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Quiz-1: Feb 13
Chap 1,2 of ①
Quiz-2: mar 24
Chap 3,4 of ①
endsem: may 5
chap 5,6 of ①

① introduction to matrix theory (A-Singh)
② introduction to linear algebra (G-Strang)

Chapter 3

$\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$ (real, complex)

[Linear Algebra done right → Sheldon Axler]

Matrix Operations

3.1 Examples of linear equations

Linear equations are everywhere, starting from mental arithmetic problems to advanced defense applications. We start with an example. The system of linear equations

$$\begin{aligned} x_1 + x_2 &= 3 \\ x_1 - x_2 &= 1 \end{aligned}$$

has a unique solution $x_1 = 2,\ x_2 = 1$. Substituting these values for the unknowns, we see that the equations are satisfied; but why are there no other solutions? Well, we have not merely guessed this solution; we have *solved* it! The details are as follows:

Suppose the pair (x_1, x_2) is a solution of the system. Subtracting the first from the second, we get another equation: $-2x_2 = -2$. It implies $x_2 = 1$. Then from either of the equations, we get $x_1 = 1$. To proceed systematically, we would like to replace the original system with the following:

$$\begin{aligned} x_1 + x_2 &= 3 \\ x_2 &= 1 \end{aligned}$$

Substituting $x_2 = 1$ in the first equation of the new system, we get $x_1 = 2$. In fact, substituting these values of x_1 and x_2, we see that the original equation is satisfied.

Convinced? The only solution of the system is $x_1 = 2, x_2 = 1$. What about the system

$$\begin{aligned} x_1 + x_2 &= 3 \\ x_1 - x_2 &= 1 \\ 2x_1 - x_2 &= 3 \end{aligned}$$

The first two equations have a unique solution and that satisfies the third. Hence this system also has a unique solution $x_1 = 2, x_2 = 1$. So the extra equation does not put any constraint on the solutions that we obtained earlier.

But what about our systematic solution method? We aim at eliminating the first unknown from all

but the first equation. We replace the second equation with the one obtained by second minus the first. We also replace the third by third minus twice the first. It results in

$$\begin{aligned} x_1 + x_2 &= 3 \\ -x_2 &= -1 \\ -3x_2 &= 3 \end{aligned}$$

Notice that the second and the third equations *coincide*, hence the conclusion. We give another twist. Consider the system

$$\begin{aligned} x_1 + x_2 &= 3 \\ x_1 - x_2 &= 1 \\ 2x_1 + x_2 &= 3 \end{aligned}$$

The first two equations again have the same solution $x_1 = 2, x_2 = 1$. But this time, the third is not satisfied by these values of the unknowns. So, the system has no solution. Also, by using our elimination method, we obtain the equations as:

$$\begin{aligned} x_1 + x_2 &= 3 \\ -x_2 &= -1 \\ -x_2 &= -3 \end{aligned}$$

The last two equations are not *consistent*. So, the original system has no solution.

Finally, instead of adding another equation, we drop one. Consider the linear equation

$$x_1 + x_2 = 3$$

having only one equation. The old solution $x_1 = 2, x_2 = 1$ is still a solution of this system. But $x_1 = 1, x_2 = 2$ is also a solution. Moreover, since $x_1 = 3 - x_2$, by assigning x_2 any real number, we get a corresponding value for x_1, which together give a solution. Thus, it has infinitely many solutions. Notice that the same conclusion holds if we have more equations, which are some multiple of the only given equation. For example,

$$\begin{aligned} x_1 + x_2 &= 3 \\ 2x_1 + 2x_2 &= 6 \\ 3x_1 + 3x_2 &= 9 \end{aligned}$$

We see that the number of equations really does not matter, but the number of *independent* equations does matter.

Warning: the notion of independent equations is not yet clear; nonetheless we have some working idea.

It is not also very clear when does a system of equations have a solution, a unique solution, infinitely many solutions, or even no solutions. And why not a system of equations has more than one but finitely many solutions? How do we use our elimination method for obtaining infinite number of solutions? To answer these questions, we will introduce matrices. Matrices will help us in representing the problem is a compact way and also will lead to a definitive answer. We will also study the eigenvalue problem for matrices which come up often in applications.

3.2 Basic matrix operations

A **matrix** is a rectangular array of symbols. For us these symbols are real numbers or, in general, complex numbers. The individual numbers in the array are called the **entries** of the matrix. The number of rows and the number of columns in any matrix are necessarily positive integers. A matrix with m rows and n columns is called an $m \times n$ matrix and it may be written as

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix},$$

or as $A = [a_{ij}]$ for short with $a_{ij} \in \mathbb{F}$ for $i = 1, \ldots, m]$ $j = 1, \ldots, n$. The number a_{ij} which occurs at the entry in ith row and jth column is referred to as the (ij)th entry (sometimes as (i, j)-th entry) of the matrix $[a_{ij}]$.

As usual, $\mathbb{R}$ denotes the set of all real numbers and $\mathbb{C}$ denotes the set of all complex numbers. We will write $\mathbb{F}$ for either $\mathbb{R}$ or $\mathbb{C}$. The numbers in $\mathbb{F}$ will also be referred to as **scalars**. Thus each entry of a matrix is a scalar.

Any matrix with m rows and n columns will be referred as an $m \times n$ matrix. The set of all $m \times n$ matrices with entries from $\mathbb{F}$ will be denoted by $\mathbb{F}^{m \times n}$.

A **row vector** of size n is a matrix in $\mathbb{F}^{1 \times n}$. Similarly, a **column vector** of size m is a matrix in $\mathbb{F}^{m \times 1}$. Sometimes we will write $\mathbb{F}^{1 \times n}$ as $\mathbb{F}^n$. The vectors in $\mathbb{F}^n$ will be written as

$$(a_1, \ldots, a_n) \quad \text{or as} \quad [a_1, \ \cdots, \ a_n] \quad \text{or as} \quad [a_1 \ \cdots \ a_n].$$

We will sometimes write a column vector as $[b_1 \ \cdots \ b_n]^t$, for saving vertical space.

Any matrix in $\mathbb{F}^{m \times n}$ is said to have its **size** as $m \times n$. If $m = n$, the rectangular array becomes a square array with m rows and m columns; and the matrix is called a square matrix of **order** m.

Naturally, two matrices of the same size are considered **equal** when their corresponding entries coincide, i.e., if $A = [a_{ij}]$ and $B = [b_{ij}]$ are in $\in \mathbb{F}^{m \times n}$, then

$$A = B \quad \text{iff} \quad a_{ij} = b_{ij}$$

for each $i \in \{1, \ldots, m\}$ and for each $j \in \{1, \ldots, n\}$. Thus matrices of different sizes are unequal.

The **zero matrix** is a matrix each entry of which is 0. We write 0 for all zero matrices of all sizes. The size is to be understood from the context.

Let $A = [a_{ij}] \in \mathbb{F}^{n \times n}$ be a square matrix of order n. The entries a_{ii} are called as the **diagonal entries** of A. The **diagonal** of A consists of all diagonal entries; the first entry on the diagonal is a_{11}, and the last diagonal entry is a_{nn}. The entries of A, which are not on the diagonal, are called as **off diagonal entries** of A; they are a_{ij} for $i \neq j$. The diagonal of the following matrix is shown in red:

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 0 \end{bmatrix}.$$

upper Δ: lower Δ is 0
lower Δ: upper Δ is 0

Here, 1 is the first diagonal entry, 3 is the second diagonal entry and 5 is the third and the last diagonal entry.

If all off-diagonal entries of A are 0, then A is said to be a **diagonal matrix**. Only a square matrix can be a diagonal matrix. There is a way to generalize this notion to any matrix, but we do not require it. Notice that the diagonal entries in a diagonal matrix need not all be nonzero. For example, the zero matrix of order n is also a diagonal matrix. The following is a diagonal matrix. We follow the convention of not showing the 0 entries in a matrix.

$$\begin{bmatrix} 1 & & \\ & 3 & \\ & & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

We also write a diagonal matrix with diagonal entries $d_1, \dots, d_n$ as $\operatorname{diag}(d_1, \dots, d_n)$. Thus the above diagonal matrix is also written as

$$\operatorname{diag}(1, 3, 0).$$

The **identity matrix** is a square matrix of which each diagonal entry is 1 and each off-diagonal entry is 0.

$$I = \operatorname{diag}(1, \dots, 1).$$

When identity matrices of different orders are used in a context, we will use the notation I_m for the identity matrix of order m.

We write e_i for a column vector whose ith component is 1 and all other components 0. When we consider e_i as a column vector in $\mathbb{F}^{n\times 1}$, the jth component of e_i is δ_{ij}. Here,

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

is the Kroneker's delta. Notice that the identity matrix $I = [\delta_{ij}]$.

There are then n distinct column vectors $e_1, \dots, e_n$. The list of column vectors $e_1, \dots, e_n$ is called the **standard basis** for $\mathbb{F}^{n\times 1}$. Accordingly, the e_is are referred to as the **standard basis vectors**. These are the columns of the identity matrix of order n, in that order; that is, e_i is the ith column of I. The transposes of these e_is are the rows of I. That is, the ith row of I is e_i^t. Thus

$$I = \begin{bmatrix} e_1 & \cdots & e_n \end{bmatrix} = \begin{bmatrix} e_1^t \\ \vdots \\ e_n^t \end{bmatrix}.$$

A **scalar matrix** is a matrix of which each diagonal entry is a scalar, the same scalar, and each off-diagonal entry is 0. Each scalar matrix is a diagonal matrix with same scalar on the diagonal. The following is a scalar matrix:

$$\begin{bmatrix} 3 & & & \\ & 3 & & \\ & & 3 & \\ & & & 3 \end{bmatrix}.$$

$e_i \to i^{th}$ column of $n \times n$ identity matrix

$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & & & \end{bmatrix} \begin{bmatrix} 0 \\ 1 \to j \\ 0 \end{bmatrix} \Rightarrow A \cdot e_j = j^{th}$ column of A

$\begin{bmatrix} \cdots & \cdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

It is also written as $\operatorname{diag}(3,3,3,3)$. If $A, B \in \mathbb{F}^{m\times m}$ and A is a scalar matrix, then $AB = BA$. Conversely, if $A \in \mathbb{F}^{m\times m}$ is such that $AB = BA$ for all $B \in \mathbb{F}^{m\times m}$, then A must be a scalar matrix. This fact is not obvious, and its proof will require much more than discussed until now.

A matrix $A \in \mathbb{F}^{m\times n}$ is said to be **upper triangular** iff all entries above the diagonal are zero. That is, $A = [a_{ij}]$ is upper triangular when $a_{ij} = 0$ for $i > j$. In writing such a matrix, we simply do not show the zero entries below the diagonal. Similarly, a matrix is called **lower triangular** iff all its entries above the diagonal are zero. Both upper triangular and lower triangular matrices are referred to as **triangular** matrices. A diagonal matrix is both upper triangular and lower triangular. The following are examples of lower triangular matrix L and upper triangular matrix U, both of order 3.

$$L = \begin{bmatrix} 1 & & \\ 2 & 3 & \\ 3 & 4 & 5 \end{bmatrix}, \quad U = \begin{bmatrix} 1 & 2 & 3 \\ & 3 & 4 \\ & & 5 \end{bmatrix}.$$

$A^* \to$ adjoint A

$\overline{A} \to$ conjugate of A

Sum of two matrices of the same size is a matrix whose entries are obtained by adding the corresponding entries in the given two matrices. That is, if $A = [a_{ij}]$ and $B = [b_{ij}]$ are in $\in \mathbb{F}^{m\times n}$, then

$$A + B = [a_{ij} + b_{ij}] \in \mathbb{F}^{m\times n}.$$

For example,

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix} + \begin{bmatrix} 3 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 4 & 3 & 5 \\ 4 & 4 & 4 \end{bmatrix}.$$

We informally say that matrices are added entry-wise. Matrices of different sizes can never be added.

It then follows that

$$A + B = B + A.$$

Similarly, matrices can be **multiplied by a scalar** entry-wise. If $A = [a_{ij}] \in \mathbb{F}^{m\times n}$, and $\alpha \in \mathbb{F}$, then

$$\alpha A = [\alpha\, a_{ij}] \in \mathbb{F}^{m\times n}.$$

Therefore, a scalar matrix with α on the diagonal is written as αI. Notice that

$$A + 0 = 0 + A = A$$

for all matrices $A \in \mathbb{F}^{m\times n}$, with an implicit understanding that $0 \in \mathbb{F}^{m\times n}$. For $A = [a_{ij}]$, the matrix $-A \in \mathbb{F}^{m\times n}$ is taken as one whose (ij)th entry is $-a_{ij}$. Thus

$$-A = (-1)A \quad \text{and} \quad A + (-A) = -A + A = 0.$$

We also abbreviate $A + (-B)$ to $A - B$, as usual.

For example,

$$3\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix} - \begin{bmatrix} 3 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 0 & 5 & 7 \\ 4 & 8 & 0 \end{bmatrix}.$$

The addition and scalar multiplication as defined above satisfy the following properties:

Let $A, B, C \in \mathbb{F}^{m\times n}$. Let $\alpha, \beta \in \mathbb{F}$.

1. $A + B = B + A$.
2. $(A + B) + C = A + (B + C)$.
3. $A + 0 = 0 + A = A$.
4. $A + (-A) = (-A) + A = 0$.
5. $\alpha(\beta A) = (\alpha\beta)A$.
6. $\alpha(A + B) = \alpha A + \alpha B$.
7. $(\alpha + \beta)A = \alpha A + \beta A$.
8. $1\,A = A$.

Notice that whatever we discuss here for matrices apply to row vectors and column vectors, in particular. But remember that a row vector cannot be added to a column vector unless both are of size 1×1, when both become numbers in $\mathbb{F}$.

Another operation that we have on matrices is **multiplication of matrices**, which is a bit involved. Let $A = [a_{ik}] \in \mathbb{F}^{m\times n}$ and $B = [b_{kj}] \in \mathbb{F}^{n\times r}$. Then their **product** AB is a matrix $[c_{ij}] \in \mathbb{F}^{m\times r}$, where the entries are

$$c_{ij} = a_{i1}b_{1j} + \cdots + a_{in}b_{nj} = \sum_{k=1}^{n} a_{ik}b_{kj}.$$

Notice that the matrix product AB is defined only when the number of columns in A is equal to the number of rows in B.

A particular case might be helpful. Suppose A is a row vector in $\mathbb{F}^{1\times n}$ and B is a column vector in $\mathbb{F}^{n\times 1}$. Then their product $AB \in \mathbb{F}^{1\times 1}$; it is a matrix of size 1×1. Often we will identify such matrices with numbers. The product now looks like:

$$\begin{bmatrix} a_1 & \cdots & a_n \end{bmatrix} \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} = \begin{bmatrix} a_1b_1 + \cdots + a_nb_n \end{bmatrix}$$

This is helpful in visualizing the general case, which looks like

$$\begin{bmatrix} a_{11} & & a_{1k} & & a_{1n} \\ & & & & \\ a_{i1} & \cdots & a_{ik} & \cdots & a_{in} \\ & & & & \\ a_{m1} & & a_{mk} & & a_{mn} \end{bmatrix} \begin{bmatrix} b_{11} & b_{1j} & b_{1r} \\ & \vdots & \\ b_{\ell 1} & b_{\ell j} & b_{\ell r} \\ & \vdots & \\ b_{n1} & b_{nj} & b_{nr} \end{bmatrix} = \begin{bmatrix} c_{11} & c_{1j} & c_{1r} \\ & & \\ c_{i1} & c_{ij} & c_{ir} \\ & & \\ c_{m1} & c_{mj} & c_{mr} \end{bmatrix}$$

The ith row of A multiplied with the jth column of B gives the (ij)th entry in AB. Thus to get AB, you have to multiply all m rows of A with all r columns of B. Besides writing a linear system in compact form, we will see later why matrix multiplication is defined this way.

For example,

$$\begin{bmatrix} 3 & 5 & -1 \\ 4 & 0 & 2 \\ -6 & -3 & 2 \end{bmatrix} \begin{bmatrix} 2 & -2 & 3 & 1 \\ 5 & 0 & 7 & 8 \\ 9 & -4 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 22 & -2 & 43 & 42 \\ 26 & -16 & 14 & 6 \\ -9 & 4 & -37 & -28 \end{bmatrix}.$$

If $u \in \mathbb{F}^{1\times n}$ and $v \in \mathbb{F}^{n\times 1}$, then $uv \in \mathbb{F}$; but $vu \in \mathbb{F}^{n\times n}$.

$$\begin{bmatrix} 3 & 6 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 19 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} \begin{bmatrix} 3 & 6 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 6 & 1 \\ 6 & 12 & 2 \\ 12 & 24 & 4 \end{bmatrix}.$$

It shows clearly that matrix multiplication is not commutative. Commutativity can break down due to various reasons. First of all when AB is defined, BA may not be defined. Secondly, even when both AB and BA are defined, they may not be of the same size; and thirdly, even when they are of the same size, they need not be equal. For example,

$$\begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 4 & 7 \\ 6 & 11 \end{bmatrix} \quad \text{but} \quad \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 8 & 13 \end{bmatrix}.$$

It does not mean that AB is never equal to BA. There can be some particular matrices A and B both in $\mathbb{F}^{n\times n}$ such that $AB = BA$.

If $A \in \mathbb{F}^{m\times n}$, then $AI_n = A$ and $I_m A = A$.

Unlike numbers, product of two nonzero matrices can be a zero matrix. For example,

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

It is easy to verify the following properties of matrix multiplication:

1. If $A \in \mathbb{F}^{m\times n}$, $B \in \mathbb{F}^{n\times r}$ and $C \in \mathbb{F}^{r\times p}$, then $(AB)C = A(BC)$.
2. If $A, B \in \mathbb{F}^{m\times n}$ and $C \in \mathbb{F}^{n\times r}$, then $(A+B)C = AC + BC$
3. If $A \in \mathbb{F}^{m\times n}$ and $B, C \in \mathbb{F}^{n\times r}$, then $A(B+C) = AB + AC$.
4. If $\alpha \in \mathbb{F}$, $A \in \mathbb{F}^{m\times n}$ and $B \in \mathbb{F}^{n\times r}$, then $\alpha(AB) = (\alpha A)B = A(\alpha B)$.

You can see matrix multiplication in a block form. Suppose $A \in \mathbb{F}^{m\times n}$. Write its ith row as $A_{i\star}$ Also, write its kth column as $A_{\star k}$. Then we can write A as a row of columns and also as a column of rows in the following manner:

$$A = [a_{ik}] = \begin{bmatrix} A_{\star 1} & \cdots & A_{\star n} \end{bmatrix} = \begin{bmatrix} A_{1\star} \\ \vdots \\ A_{m\star} \end{bmatrix}.$$

$A \in \mathbb{F}^{m\times n}$ $B \in \mathbb{F}^{n\times k}$

rows of AB: $A_1B, A_2B, \ldots A_mB$

Write $B \in \mathbb{F}^{n\times r}$ similarly as

$$B = [b_{kj}] = \begin{bmatrix} B_{\star 1} & \cdots & B_{\star r} \end{bmatrix} = \begin{bmatrix} B_{1\star} \\ \vdots \\ B_{\star n} \end{bmatrix}.$$

columns of AB: $AB_1, AB_2, \ldots AB_k$

Another notation: $E_{ij} \in \mathbb{F}^{m\times n}$ ⟶ basis for $\mathbb{F}^{m\times n}$

E_{ij} means only element at coordinates i,j is 1, rest all is 0.

$A = [a_{ij}] \in \mathbb{F}^{m\times n} \Rightarrow A = \sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}E_{ij}$

Then their product AB can now be written as

$$AB = \begin{bmatrix} AB_{\star 1} & \cdots & AB_{\star r}\end{bmatrix} = \begin{bmatrix} A_{1\star}B \\ \vdots \\ A_{m\star}B\end{bmatrix}.$$

When writing this way, we ignore the extra brackets [and].

Powers of square matrices can be defined inductively by taking

$$A^0 = I \quad \text{and} \quad A^n = AA^{n-1} \text{ for } n \in \mathbb{N}.$$

A square matrix A of order m is called **invertible** iff there exists a matrix B of order m such that (non-singular)

$$AB = I = BA.$$

Such a matrix B is called an **inverse** of A. If C is another inverse of A, then

$$C = CI = C(AB) = (CA)B = IB = B.$$

Therefore, an inverse of a matrix is unique and is denoted by A^{-1}. We talk of invertibility of square matrices only; and all square matrices are not invertible. For example, I is invertible but 0 is not. If $AB = 0$ for square matrices A and B, then neither A nor B is invertible.

If both $A, B \in \mathbb{F}^{n\times n}$ are invertible, then $(AB)^{-1} = B^{-1}A^{-1}$.

Reason?

$$B^{-1}A^{-1}AB = B^{-1}IB = I = AIA^{-1} = ABB^{-1}A^{-1}.$$

Invertible matrices play a crucial role in solving linear systems uniquely. We will come back to the issue later.

Exercises for § 3.2

1. Compute $AB,\ CA,\ DC,\ DCAB,\ A^2,\ D^2$ and A^3B^2, where

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 2\end{bmatrix},\ B = \begin{bmatrix} 4 & -1 \\ 4 & 0\end{bmatrix},\ C = \begin{bmatrix} -1 & 2 \\ 2 & -1 \\ 1 & 3\end{bmatrix},\ D = \begin{bmatrix} 3 & 2 & 1 \\ 4 & -6 & 0 \\ 1 & -2 & -2\end{bmatrix}.$$

2. Let E_{ij} be the $n \times n$ matrix whose ijth entry is 1 and all other entries are 0. Show that each $A = [a_{ij}] \in \mathbb{C}^{n\times n}$ can be expressed as $A = \sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij}E_{ij}$. Also show that $E_{ij}E_{km} = 0$ if $j \neq k$, and $E_{ij}E_{jm} = E_{im}$.

3. Let $A \in \mathbb{C}^{m\times n}$, $B \in \mathbb{C}^{n\times p}$. Let $B_1, \ldots, B_p$ be the columns of B. Show that $AB_1, \ldots, AB_p$ are the columns of AB.

4. Let $A \in \mathbb{C}^{m\times n}$, $B \in \mathbb{C}^{n\times p}$. Let $A_1, \ldots, A_m$ be the rows of A. Show that $A_1B, \ldots, A_mB$ are the rows of AB.

5. Let $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1\end{bmatrix}$. Show that $A^n = \begin{bmatrix} 1 & n & n(n-1) \\ 0 & 1 & 2n \\ 0 & 0 & 1\end{bmatrix}$ for $n \in \mathbb{N}$. induction

6. Construct two 3×3 matrices A and B such that $AB = 0$ but $BA \neq 0$.

$B \leftarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0\end{bmatrix}\begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0\end{bmatrix} \rightarrow A$

3.3 Transpose and adjoint

We consider another operation on matrices. Given a matrix $A \in \mathbb{F}^{m\times n}$, its **transpose** is a matrix in $\mathbb{F}^{n\times m}$, which is denoted by A^t, and is defined by

$$\text{the } (ij)\text{th entry of } A^t = \text{the } (ji)\text{th entry of } A.$$

That is, the ith column of A^t is the column vector $[a_{i1}, \cdots, a_{in}]^t$. The rows of A are the columns of A^t and the columns of A become the rows of A^t. In particular, if $u = \begin{bmatrix} a_1 & \cdots & a_m \end{bmatrix}$ is a row vector, then its transpose is

$$u^t = \begin{bmatrix} a_1 \\ \vdots \\ a_m \end{bmatrix},$$

which is a column vector, as mentioned earlier. Similarly, the transpose of a column vector is a row vector. If you write A as a row of column vectors, then you can express A^t as a column of row vectors, as in the following:

$$A = \begin{bmatrix} A_{\star 1} & \cdots & A_{\star n} \end{bmatrix} \Rightarrow A^t = \begin{bmatrix} A_{\star 1}^t \\ \vdots \\ A_{\star n}^t \end{bmatrix}.$$

$$A = \begin{bmatrix} A_{1\star} \\ \vdots \\ A_{m\star} \end{bmatrix} \Rightarrow A^t = \begin{bmatrix} A_{1\star}^t & \cdots & A_{m\star}^t \end{bmatrix}.$$

For example,

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix} \Rightarrow A^t = \begin{bmatrix} 1 & 2 \\ 2 & 3 \\ 3 & 1 \end{bmatrix}.$$

It then follows that transpose of the transpose is the original matrix. The following are some of the properties of this operation of transpose.

1. $(A^t)^t = A$.
2. $(A + B)^t = A^t + B^t$.
3. $(\alpha A)^t = \alpha A^t$.
4. $(AB)^t = B^t A^t$.
5. If A is invertible, then A^t is invertible, and $(A^t)^{-1} = (A^{-1})^t$.

In the above properties, we assume that the operations are allowed, that is, in (2), A and B must be of the same size. Similarly, in (4), the number of columns in A must be equal to the number of rows in B; and in (5), A must be a square matrix.

It is easy to see all the above properties, except perhaps the fourth one. For this, let $A \in \mathbb{F}^{m\times n}$ and $B \in \mathbb{F}^{n\times r}$. Now, the (ji)th entry in $(AB)^t$ is the (ij)th entry in AB; and it is given by

$$a_{i1}b_{j1} + \cdots + a_{in}b_{jn}.$$

On the other side, the (ji)th entry in B^tA^t is obtained by multiplying the jth row of B^t with the ith column of A^t. This is same as multiplying the entries in the jth column of B with the corresponding entries in the ith row of A, and then taking the sum. Thus it is

$$b_{j1}a_{i1} + \cdots + b_{jn}a_{in}.$$

This is the same as computed earlier.

The fifth one follows from the fourth one and the fact that $(AB)^{-1} = B^{-1}A^{-1}$.

Observe that transpose of a lower triangular matrix is an upper triangular matrix, and vice versa.

Close to the operations of transpose of a matrix is the adjoint. Let $A = [a_{ij}] \in \mathbb{F}^{m\times n}$. The **adjoint** of A is denoted as A^*, and is defined by

$$\text{the } (ij)\text{th entry of } A^* = \text{the complex conjugate of } (ji)\text{th entry of } A.$$

We write $\overline{\alpha}$ for the **complex conjugate** of a scalar α. That is, $\overline{\alpha + i\beta} = \alpha - i\beta$. Thus, if $a_{ij} \in \mathbb{R}$, then $\overline{a_{ij}} = a_{ij}$. Thus, when A has only real entries, $A^* = A^t$. Also, the ith column of A^t is the column vector $(\overline{a}_{i1}, \cdots, \overline{a}_{in})^t$. For example,

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix} \Rightarrow A^* = \begin{bmatrix} 1 & 2 \\ 2 & 3 \\ 3 & 1 \end{bmatrix}.$$

$$A = \begin{bmatrix} 1+i & 2 & 3 \\ 2 & 3 & 1-i \end{bmatrix} \Rightarrow A^* = \begin{bmatrix} 1-i & 2 \\ 2 & 3 \\ 3 & 1+i \end{bmatrix}.$$

Similar to the transpose, the adjoint satisfies the following properties:

1. $(A^*)^* = A$.
2. $(A + B)^* = A^* + B^*$.
3. $(\alpha A)^* = \overline{\alpha} A^*$.
4. $(AB)^* = B^*A^*$.
5. If A is invertible, then A^* is invertible, and $(A^*)^{-1} = (A^{-1})^*$.

Here also, in (2), the matrices A and B must be of the same size, and in (4), the number of columns in A must be equal to the number of rows in B. The adjoint of A is also called the **conjugate transpose** of A. Notice that if $A \in \mathbb{R}^{m\times n}$, then $A^* = A^t$.

Occasionally, we will use $\overline{A}$ for the matrix obtained from A by taking complex conjugate of each

entry. That is, the (ij)th entry of $\overline{A}$ is the complex conjugate of the (ij)th entry of A. Hence $A^* = (\overline{A})^t$.

Further, the familiar dot product in $\mathbb{R}^{1\times 3}$ can be generalized to $\mathbb{F}^{1\times n}$ or to $\mathbb{F}^{n\times 1}$. For vectors $u, v \in \mathbb{F}^{1\times n}$, we define their **inner product** as

$$\langle u, v\rangle = uv^*.$$

For example, $u = \begin{bmatrix}1 & 2 & 3\end{bmatrix}$, $v = \begin{bmatrix}2 & 1 & 3\end{bmatrix}$ $\Rightarrow$ $\langle u, v\rangle = 1\times 2 + 2\times 1 + 3\times 3 = 13$.

Similarly, for $x, y \in \mathbb{F}^{n\times 1}$, we define their inner product as

$$\langle x, y\rangle = y^*x.$$

In case, $\mathbb{F} = \mathbb{R}$, the x^* becomes x^t. The inner product satisfies the following properties:

Let $x, y, z \in \mathbb{F}^{n\times 1}$ (or in $\mathbb{F}^{1\times n}$); $\alpha, \beta \in \mathbb{F}$.

1. $\langle x, x\rangle \geq 0$.
2. $\langle x, x\rangle = 0$ iff $x = 0$.
3. $\langle x, y\rangle = \overline{\langle y, x\rangle}$.
4. $\langle x + y, z\rangle = \langle x, z\rangle + \langle y, z\rangle$.
5. $\langle z, x + y\rangle = \langle z, x\rangle + \langle z, y\rangle$.
6. $\langle \alpha x, y\rangle = \alpha\langle x, y\rangle$.
7. $\langle x, \beta y\rangle = \overline{\beta}\langle x, y\rangle$.

The inner product gives rise to the length of a vector as in the familiar case of $\mathbb{R}^{1\times 3}$. We now call the generalized version of length as the *norm*. If u is in $\mathbb{F}^{1\times n}$ or in $\mathbb{F}^{n\times 1}$), we define its **norm**, denoted by $\|u\|$ as the nonnegative square root of $\langle u, u\rangle$. That is,

$$\|u\| = \sqrt{\langle u, u\rangle}.$$

The norm satisfies the following properties:

Let $x, y \in \mathbb{F}^{1\times n}$ (or in $\mathbb{F}^{n\times 1}$); $\alpha \in \mathbb{F}$.

1. $\|x\| \geq 0$.
2. $\|x\| = 0$ iff $x = 0$.
3. $\|\alpha x\| = |\alpha|\,\|x\|$.
4. $|\langle x, y\rangle| \leq \|x\|\,\|y\|$. (*Cauchy-Schwartz inequality*)
5. $\|x + y\| = \|x\| + \|y\|$. (*Triangle inequality*)

Using these properties, the acute angle between any two nonzero vectors can be defined. Let $x, y \in \mathbb{F}^{1\times n}$ (or in $\mathbb{F}^{n\times 1}$). The **angle θ between** x and y, denoted by $\theta(x, y)$ is defined by

$$\cos\theta(x, y) = \frac{|\langle x, y\rangle|}{\|x\|\,\|y\|}.$$

In particular, when $\theta(x, y) = \pi/2$, we say that the vectors x and y are **orthogonal**, and we write this as $x \perp y$. That is,

$$x \perp y \quad \text{iff} \quad \langle x, y\rangle = 0.$$

Notice that this definition allows x and y to be zero vectors. Also, the zero vector is orthogonal to every vector.

It follows that if $x \perp y$, then $\|x\|^2 + \|y\|^2 = \|x + y\|^2$. This is referred to as **Pythagoras law**. The converse of Pythagoras law holds when $\mathbb{F} = \mathbb{R}$. For $\mathbb{F} = \mathbb{C}$, it does not hold, in general.

Exercises for § 3.3

1. Determine A^t, $\overline{A}$, A^*, A^*A and AA^*, where

   (a) $A = \begin{bmatrix} -1 & 2 & 3 & 1 \\ 2 & -1 & 0 & 3 \\ 0 & -1 & -3 & 1 \end{bmatrix}$ (b) $A = \begin{bmatrix} 1 & -2+i & 3-i \\ i & -1-i & 2i \\ 1+3i & -i & -3 \\ -2 & 0 & -i \end{bmatrix}$

2. Let u and v be the first and second rows of the matrix A in Exercise 1(a), and let x and y be the first and second columns of A in Exercise 1(b), respectively. Compute the inner products $\langle u, v\rangle$ and $\langle x, y\rangle$.

3. In $\mathbb{C}$, consider the inner product $\langle x, y\rangle = x\overline{y}$. Let $x = 1$ and $y = i$ be two vectors in $\mathbb{C}$. Show that $\|x\|^2 + \|y\|^2 = \|x + y\|^2$ but $\langle x, y\rangle \neq 0$.

4. In $\mathbb{F}^{n\times 1}$, show that the parallelogram law holds. That is, for all $x, y \in \mathbb{F}^{n\times 1}$, we have $\|x + y\|^2 + \|x - y\|^2 = 2(\|x\|^2 + \|y\|^2)$.

5. In $\mathbb{F}^{1\times n}$, derive the triangle inequality from Cauchy-Schwartz inequality.

6. Let $A \in \mathbb{C}^{m\times n}$. Suppose $AA^* = I_m$. Does it follow that $A^*A = I_n$?

3.4 Elementary row operations

Recall that while solving linear equations in two or three variables, you try to eliminate a variable from all but one equation by adding an equation to the other, or even adding a constant times one equation to another. We do similar operations on the rows of a matrix. These are achieved by multiplying a given matrix with some special matrices, called elementary matrices.

Let $e_1, \ldots, e_m \in \mathbb{F}^{m\times 1}$ be the standard basis vectors. Let $1 \le i, j \le m$. The product $e_i e_j^t$ is an $m \times m$ matrix whose (i, j)th entry is 1 and all other entries are 0. We write such a matrix as E_{ij}.

① exchange of rows: $E[i,j]A$ (i, j rows swapped)

② multiplication of row by a non zero const: $E_\alpha[i]A$ (i^{th} row multiplied by α)

③ replacing a row by a sum of that row with a non-zero constant multiple of another row: $E_\alpha[3,1]\,A$ i^{th} row → i^{th} row + $\alpha \times j^{th}$ row

For instance, when $m = 3$, we have

$$e_2 e_3^t = \begin{bmatrix}0\\1\\0\end{bmatrix}\begin{bmatrix}0 & 0 & 1\end{bmatrix} = \begin{bmatrix}0 & 0 & 0\\0 & 0 & 1\\0 & 0 & 0\end{bmatrix} = E_{23}.$$

- $E[i,j]^{-1} = E[i,j]$
- $E_\alpha[i]^{-1} = E_{1/\alpha}[i]$

An **elementary matrix** of order m is one of the following three types:

- $E_\alpha[2] = \begin{bmatrix}1&0&0\\0&\alpha&0\\0&0&1\end{bmatrix}$ 3×3
- $E_\alpha[i,j] \in \mathbb{F}^{m\times m} = I_m + \alpha E_{ij}$
- $E_\alpha[i,j]^{-1} = E_{-\alpha}[i,j]$

1. $E[i,j] := I - E_{ii} - E_{jj} + E_{ij} + E_{ji}$ with $i \neq j$.
2. $E_\alpha[i] := I - E_{ii} + \alpha E_{ii}$, where α is a nonzero scalar.
3. $E_\alpha[i,j] := I + \alpha E_{ij}$, where α is a nonzero scalar and $i \neq j$.

Here, I is the identity matrix of order m. Similarly, the order of the elementary matrices will be understood from the context; we will not show that in our symbolism.

Example 3.1. The following are instances of elementary matrices of order 3.

$$E[1,2] = \begin{bmatrix}0&1&0\\1&0&0\\0&0&1\end{bmatrix},\quad E_{-1}[2] = \begin{bmatrix}1&0&0\\0&-1&0\\0&0&1\end{bmatrix},\quad E_2[3,1] = \begin{bmatrix}1&0&0\\0&1&0\\2&0&1\end{bmatrix}.$$

We observe that for a matrix $A \in \mathbb{F}^{m\times n}$, the following are true:

1. $E[i,j]\,A$ is the matrix obtained from A by exchanging its ith and jth rows.
2. $E_\alpha[i]\,A$ is the matrix obtained from A by replacing its ith row with α times the ith row.
3. $E_\alpha[i,j]\,A$ is the matrix obtained from A by replacing its ith row with the ith row plus α times the jth row.

We call these operations of pre-multiplying a matrix with an elementary matrix as **elementary row operations**. Thus there are three kinds of elementary row operations as listed above. Sometimes, we will refer to them as of Type-1, 2, or 3, respectively. Also, in computations, we will write

$$A \xrightarrow{E} B$$

to mean that the matrix B has been obtained by an elementary row operation E, that is, $B = EA$.

Example 3.2. See the following applications of elementary row operations:

$$\begin{bmatrix}1&1&1\\2&2&2\\3&3&3\end{bmatrix} \xrightarrow{E_{-3}[3,1]} \begin{bmatrix}1&1&1\\2&2&2\\0&0&0\end{bmatrix} \xrightarrow{E_{-2}[2,1]} \begin{bmatrix}1&1&1\\0&0&0\\0&0&0\end{bmatrix}$$

Often we will apply elementary row operations in a sequence. In this way, the above operations could be shown in one step as $E_{-3}[3,1]$, $E_{-2}[2,1]$. However, remember that the result of application of this sequence of elementary row operations on a matrix A is $E_{-2}[2,1]\,E_{-3}[3,1]\,A$; the products are in reverse order.

Exercises for § 3.4

1. Compute $E[2,3]A$, $E_i[2]A$, $E_{-1/2}[1,3]A$ and $E_i[1,2]A$, where A is given by

$$A = \begin{bmatrix} -1 & 2 & 3 & 1 \\ 2 & -1 & 0 & 3 \\ 0 & -1 & -3 & 1 \end{bmatrix} \quad \text{(b) } A = \begin{bmatrix} 1 & -2+i & 3-i \\ i & -1-i & 2i \\ 1+3i & -i & -3 \\ -2 & 0 & -i \end{bmatrix}$$

2. Argue in general terms why the following are true:

   (a) $E[i,j]\,A$ is the matrix obtained from A by exchanging its ith and jth rows.

   (b) $E_\alpha[i]\,A$ is the matrix obtained from A by replacing its ith row with α times the ith row.

   (c) $E_\alpha[i,j]\,A$ is the matrix obtained from A by replacing its ith row with the ith row plus α times the jth row.

3.5 Row reduced echelon form

Elementary operations can be used to reduce a matrix to a nice form, bringing in many zero entries. Recall that this corresponds to eliminating a variable from an equation of a linear system. The first, from left, nonzero entry in a nonzero row of a matrix is called a **pivot**. We denote a pivot in a row by putting a box around it. A column where a pivot occurs is called a **pivotal column**.

A matrix $A \in \mathbb{F}^{m\times n}$ is said to be in **row reduced echelon form** (RREF) iff the following conditions are satisfied:

(1) Each pivot is equal to 1.

(2) The column index of the pivot in the $(i+1)$th row is greater than the column index of the pivot in the ith row, for all i from 1 to $m-1$.

(3) In a pivotal column, all entries other than the pivot are zero.

(4) All zero rows are at the bottom.

Example 3.3. The matrix $\begin{bmatrix} \boxed{1} & 2 & 0 & 0 \\ 0 & 0 & \boxed{1} & 0 \\ 0 & 0 & 0 & \boxed{1} \end{bmatrix}$ is in row reduced echelon form whereas the matrices

$$\begin{bmatrix} 0 & \boxed{1} & 3 & 0 \\ 0 & 0 & 0 & \boxed{2} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & \boxed{1} & 3 & \boxed{1} \\ 0 & 0 & 0 & \boxed{1} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & \boxed{1} & 3 & 0 \\ 0 & 0 & 0 & \boxed{1} \\ 0 & 0 & 0 & \boxed{1} \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & \boxed{1} & 0 & 0 \\ 0 & 0 & \boxed{1} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \boxed{1} \end{bmatrix}$$

are not in row reduced echelon form.

Any matrix can be brought to a row reduced echelon form by using elementary row operations. We give an algorithm to achieve this.

Reduction to Row Reduced Echelon Form

1. Set the work region R as the whole matrix A.
2. If all entries in R are 0, then stop.
3. If there are nonzero entries in R, then find the leftmost nonzero column. Mark it as the pivotal column.
4. Find the topmost nonzero entry in the pivotal column. Box it; it is a pivot.
5. If the pivot is not on the top row of R, then exchange the row of A which contains the top row of R with the row where the pivot is.
6. If the pivot, say, α is not equal to 1, then replace the top row of R in A by $1/\alpha$ times that row.
7. Make all entries, except the pivot, in the pivotal column as zero by replacing each row above and below the top row of R using elementary row operations in A with that row and the top row of R.
8. Find the sub-matrix to the right and below the pivot. If no such sub-matrix exists, then stop. Else, reset the work region R to this sub-matrix, and go to 2.

We will refer to the output of the above reduction algorithm as *the row reduced echelon form* (the RREF) of a given matrix.

Example 3.4.

$$A = \begin{bmatrix} \boxed{1} & 1 & 2 & 0 \\ 3 & 5 & 7 & 1 \\ 1 & 5 & 4 & 5 \\ 2 & 8 & 7 & 9 \end{bmatrix} \xrightarrow{R1} \begin{bmatrix} \boxed{1} & 1 & 2 & 0 \\ 0 & 2 & 1 & 1 \\ 0 & 4 & 2 & 5 \\ 0 & 6 & 3 & 9 \end{bmatrix} \xrightarrow{E_{1/2}[2]} \begin{bmatrix} \boxed{1} & 1 & 2 & 0 \\ 0 & \boxed{1} & \frac{1}{2} & \frac{1}{2} \\ 0 & 4 & 2 & 5 \\ 0 & 6 & 3 & 9 \end{bmatrix}$$

$$\xrightarrow{R2} \begin{bmatrix} \boxed{1} & 0 & \frac{3}{2} & -\frac{1}{2} \\ 0 & \boxed{1} & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 6 \end{bmatrix} \xrightarrow{E_{1/3}[3]} \begin{bmatrix} \boxed{1} & 0 & \frac{3}{2} & -\frac{1}{2} \\ 0 & \boxed{1} & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & \boxed{1} \\ 0 & 0 & 0 & 6 \end{bmatrix} \xrightarrow{R3} \begin{bmatrix} \boxed{1} & 0 & \frac{3}{2} & 0 \\ 0 & \boxed{1} & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \boxed{1} \\ 0 & 0 & 0 & 0 \end{bmatrix} = B$$

Here, $R1 = E_{-3}[2,1],\ E_{-1}[3,1],\ E_{-2}[4,1]$; $R2 = E_{-1}[2,1],\ E_{-4}[3,2],\ E_{-6}[4,2]$; and $R3 = E_{1/2}[1,3],\ E_{-1/2}[2,3],\ E_{-6}[4,3]$. The matrix B is the RREF of A. Notice that

$$B = E_{-6}[4,3]\, E_{-1/2}[2,3]\, E_{1/2}[1,3]\, E_{-1/3}[3]\, E_{-6}[4,2]\, E_{-4}[3,2]\, E_{-1}[2,1] E_{-1/2}[2]$$
$$E_{-2}[4,1]\, E_{-1}[3,1]\, E_{-3}[2,1]\, A.$$

The products are in reverse order.

$\Rightarrow EU_{k_1} = e_1,\ EU_{k_2} = e_2, \ldots\ EU_{k_r} = e_r$

let $\theta_1, \theta_2, \ldots \theta_{n-r}$ be non pivotal columns in order $\Rightarrow$

if θ_i occurs b/w e_j & e_{j+1}, then θ_i should be of the form $\theta_i = [a_1\ a_2 \ldots a_j\ 0\ 0 \ldots 0]^T$

$= a_1 e_1 + a_2 e_2 + \cdots a_j e_j$

column of A $= a_1 EU_{k_1} + a_2 EU_{k_2} + \cdots a_j EU_{k_j}$

$\Rightarrow E^{-1}\theta_i = a_1 u_{k_1} + a_2 u_{k_2} + \cdots + a_j u_{k_j}$

above example: $\theta_1 = 3/2\, u_1 + 1/2\, u_2$

Theorm: RREF of a matrix is unique (2 matrices can have same RREF)

$$A = \begin{bmatrix} 0 & 4 & -6 \\ \boxed{4} & 0 & 10 \\ -6 & 10 & 0 \end{bmatrix} \xrightarrow[R_2 \leftrightarrow R_1]{E[1,2]} \begin{bmatrix} \boxed{4} & 0 & 10 \\ 0 & 4 & -6 \\ -6 & 10 & 0 \end{bmatrix} \xrightarrow[R_1 = R_1/4]{E_{1/4}[1]} \begin{bmatrix} \boxed{1} & 0 & 2.5 \\ 0 & 4 & -6 \\ -6 & 10 & 0 \end{bmatrix}$$

$$\xrightarrow{E_6[3,1]\ \ R_3 = R_3 + 6R_1} \begin{bmatrix} \boxed{1} & 0 & 2.5 \\ 0 & 4 & -6 \\ 0 & 10 & 15 \end{bmatrix} \xrightarrow[R_2 = R_2/4]{E_{1/4}[2]} \begin{bmatrix} \boxed{1} & 0 & 2.5 \\ 0 & \boxed{4} & -6 \\ 0 & 10 & 15 \end{bmatrix} \xrightarrow[R_3 = R_3 - 10R_2]{E_{-10}[3,2]} \begin{bmatrix} \boxed{1} & 0 & 2.5 \\ 0 & \boxed{1} & -1.5 \\ 0 & 0 & \boxed{30} \end{bmatrix}$$

$$\xrightarrow{R_3 = R_3/30\ \ E_{1/30}[3]} \begin{bmatrix} \boxed{1} & 0 & 2.5 \\ 0 & \boxed{1} & -1.5 \\ 0 & 0 & \boxed{1} \end{bmatrix} \xrightarrow[E_{3/2}[2,3]]{E_{-5/2}[1,3]} \begin{bmatrix} \boxed{1} & 0 & 0 \\ 0 & \boxed{1} & 0 \\ 0 & 0 & \boxed{1} \end{bmatrix}$$

number of pivots in the RREF of a matrix $A \in \mathbb{F}^{m\times n}$ is called rank of A and denoted by rank(A)

rank $\le \min(m,n)$

Exercises for § 3.5

1. Compute row reduced echelon forms of the following matrices:

   (a) $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 2 & -1 & -1 & 0 \\ 1 & 0 & 1 & -4 \\ 0 & 1 & -1 & -4 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 2 & 1 & -1 \\ 0 & 2 & 3 & 3 \\ 1 & -1 & -3 & -4 \\ 1 & 1 & 5 & -2 \end{bmatrix}$

2. Suppose that a matrix is already in RREF with r number of pivots. Show that the pivotal columns are the basis vectors $e_1, \dots, e_r$ in that order, from left to right.

3. Suppose that a matrix is already in RREF with the r pivotal columns as $C_1, \dots, C_r$. Is it true that each non-pivotal column can be expressed as $\alpha_1 C_1 + \cdots + \alpha_r C_r$ for some suitable scalars $\alpha_1, \dots, \alpha_r$?

4. Argue why our algorithm for reducing a matrix to its RREF gives a unique output.

3.6 Determinant

There are two important quantities associated with a square matrix. One is the trace and the other is the determinant.

The sum of all diagonal entries of a square matrix is called the **trace** of the matrix. That is, if $A = [a_{ij}] \in \mathbb{F}^{m\times m}$, then

$$\operatorname{tr}(A) = a_{11} + \cdots + a_{nn} = \sum_{k=1}^{n} a_{kk}.$$

In addition to $\operatorname{tr}(I_m) = m,\ \operatorname{tr}(0) = 0$, the trace satisfies the following properties:

Let $A \in \mathbb{F}^{n\times n}$.

1. $\operatorname{tr}(\alpha A) = \alpha \operatorname{tr}(A)$ for each $\alpha \in \mathbb{F}$.
2. $\operatorname{tr}(A^t) = \operatorname{tr}(A)$ and $\operatorname{tr}(A^*) = \overline{\operatorname{tr}(A)}$.
3. $\operatorname{tr}(A + B) = \operatorname{tr}(A) + \operatorname{tr}(B)$ and $\operatorname{tr}(AB) = \operatorname{tr}(BA)$.
4. $\operatorname{tr}(A^*A) = 0$ iff $\operatorname{tr}(AA^*) = 0$ iff $A = 0$.

tr(ABC) = tr(BCA) = tr(CAB)

Observe that $\operatorname{tr}(A^*A) = \sum_{i=1}^{m}\sum_{j=1}^{m} |a_{ij}|^2 = \operatorname{tr}(AA^*)$. Form this (4) follows.

The second quantity, called the **determinant** of a square matrix $A = [a_{ij}] \in \mathbb{F}^{n\times n}$, written as $\det(A)$, is defined inductively as follows:

If $n = 1$, then $\det(A) = a_{11}$.
If $n > 1$, then $\det(A) = \sum_{j=1}^{n} (-1)^{1+j} a_{1j} \det(A_{1j})$

where the matrix $A_{1j} \in \mathbb{F}^{(n-1)\times(n-1)}$ is obtained from A by deleting the first row and the jth column of A.

A_{ij} → remove ith row, jth col

minor ← $M_{ij}(A)$ → $\det(A_{ij})$

cofactor ← $C_{ij}(A)$ → $(-1)^{i+j} M_{ij}(A)$

When $A = [a_{ij}]$ is written showing all its entries, we also write $\det(A)$ by replacing the two big closing brackets [and] by two vertical bars | and |. For a 2×2 matrix, its determinant is seen as follows:

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = (-1)^{1+1} a_{11} \det[a_{22}] + (-1)^{1+2} a_{12} \det[a_{21}] = a_{11}a_{22} - a_{12}a_{21}.$$

Similarly, for a 3×3 matrix, we need to compute three 2×2 determinants. For example,

$$\begin{aligned} \det \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & 2 \end{bmatrix} &= \begin{vmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & 2 \end{vmatrix} \\ &= (-1)^{1+1} \times 1 \times \begin{vmatrix} 3 & 1 \\ 1 & 2 \end{vmatrix} + (-1)^{1+2} \times 2 \times \begin{vmatrix} 2 & 1 \\ 3 & 2 \end{vmatrix} + (-1)^{1+3} \times 3 \times \begin{vmatrix} 2 & 3 \\ 3 & 1 \end{vmatrix} \\ &= 1 \times \begin{vmatrix} 3 & 1 \\ 1 & 2 \end{vmatrix} - 2 \times \begin{vmatrix} 2 & 1 \\ 3 & 2 \end{vmatrix} + 3 \times \begin{vmatrix} 2 & 3 \\ 3 & 1 \end{vmatrix} \\ &= (3 \times 2 - 1 \times 1) - 2 \times (2 \times 2 - 1 \times 3) + 3 \times (2 \times 1 - 3 \times 3) \\ &= 5 - 2 \times 1 + 3 \times (-7) = -18. \end{aligned}$$

For a lower triangular matrix, we see that

$$\begin{vmatrix} a_{11} & & & & \\ a_{12} & a_{22} & & & \\ a_{13} & a_{23} & a_{33} & & \\ & & & \ddots & \\ a_{n1} & & \cdots & & a_{nn} \end{vmatrix} = a_{11} \begin{vmatrix} a_{22} & & & \\ a_{23} & a_{33} & & \\ & & \ddots & \\ a_{n1} & & \cdots & a_{nn} \end{vmatrix} = \cdots = a_{11}a_{22} \cdots a_{nn}.$$

In general, the determinant of any triangular matrix (upper or lower), is the product of its diagonal entries. In particular, the determinant of a diagonal matrix is also the product of its diagonal entries. Thus, if I is the identity matrix of order n, then $\det(I) = 1$ and $\det(-I) = (-1)^n$.

Our definition of determinant *expands* the determinant in the first row. In fact, the same result may be obtained by expanding it in any other row, or even any other column. Along with this, some more properties of the determinant are listed in the following.

Let $A \in \mathbb{F}^{n \times n}$. The sub-matrix of A obtained by deleting the ith row and the jth column is called the (ij)th **minor** of A, and is denoted by A_{ij}. The (ij)th **co-factor** of A is $(-1)^{i+j} \det(A_{ij})$; it is denoted by $C_{ij}(A)$. Sometimes, when the matrix A is fixed in a context, we write $C_{ij}(A)$ as C_{ij}. The **adjugate** of A is the $n \times n$ matrix obtained by taking transpose of the matrix whose (ij)th entry is $C_{ij}(A)$; it is denoted by $\operatorname{adj}(A)$. That is, $\operatorname{adj}(A) \in \mathbb{F}^{n \times n}$ is the matrix whose (ij)th entry is the (ji)th co-factor $C_{ji}(A)$. Also, we write $A_i(x)$ for the matrix obtained from A by replacing its ith row by a row vector x of appropriate size.

Let $A \in \mathbb{F}^{n \times n}$. Let $i, j, k \in \{1, \ldots, n\}$. Let $E[i, j],\ E_\alpha[i]$ and $E_\alpha[i, j]$ be the elementary matrices of order n with $1 \leq i \neq j \leq n$ and $\alpha \neq 0$, a scalar. Then the following statements are true.

1. $\det(E[i,j]\,A) = -\det(A)$.
2. $\det(E_\alpha[i]\,A) = \alpha\det(A)$.
3. $\det(E_\alpha[i,j]\,A) = \det(A)$.
4. If some row of A is the zero vector, then $\det(A) = 0$.
5. If one row of A is a scalar multiple of another row, then $\det(A) = 0$.
6. For any $i \in \{1,\dots,n\}$, $\det(\,A_i(x+y)\,) = \det(\,A_i(x)\,) + \det(\,A_i(y)\,)$.
7. $\det(A^t) = \det(A)$.
8. If A is a triangular matrix, then $\det(A)$ is equal to the product of the diagonal entries of A.
9. $\det(AB) = \det(A)\det(B)$ for any matrix $B \in \mathbb{F}^{n\times n}$.
10. $\det(A^t) = \det(A)$.
11. $A\,\mathrm{adj}(A) = \mathrm{adj}(A)A = \det(A)\,I$.
12. A is invertible iff $\det(A) \neq 0$.

Elementary column operations are operations similar to row operations, but with columns instead of rows. Notice that since $\det(A^t) = \det(A)$, the facts concerning elementary row operations also hold true if elementary column operations are used. Using elementary operations, the *computational complexity* for evaluating a determinant can be reduced drastically. The trick is to bring a matrix to a triangular form by using elementary row operations, so that the determinant of the triangular matrix can be computed easily.

Example 3.5.

$$\begin{vmatrix} 1 & 0 & 0 & 1 \\ -1 & 1 & 0 & 1 \\ -1 & -1 & 1 & 1 \\ -1 & -1 & -1 & 1 \end{vmatrix} \overset{R1}{=} \begin{vmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & -1 & 1 & 2 \\ 0 & -1 & -1 & 2 \end{vmatrix} \overset{R2}{=} \begin{vmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & -1 & 4 \end{vmatrix} \overset{R3}{=} \begin{vmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 8 \end{vmatrix} = 8.$$

Here, $R1 = E_1[2,1];\ E_1[3,1];\ E_1[4,1]$, $R2 = E_1[3,2];\ E_1[4,2]$, and $R3 = E_1[4,3]$.

Finally, the upper triangular matrix has the required determinant.

Example 3.6. See that the following is true, for verifying Property (6) as mentioned above:

$$\begin{vmatrix} 3 & 1 & 2 & 4 \\ -1 & 1 & 0 & 1 \\ -1 & -1 & 1 & 1 \\ -1 & -1 & -1 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 & 1 \\ -1 & 1 & 0 & 1 \\ -1 & -1 & 1 & 1 \\ -1 & -1 & -1 & 1 \end{vmatrix} + \begin{vmatrix} 2 & 1 & 2 & 3 \\ -1 & 1 & 0 & 1 \\ -1 & -1 & 1 & 1 \\ -1 & -1 & -1 & 1 \end{vmatrix}.$$

Exercises for § 3.6

1. Construct an $n \times n$ nonzero matrix, where no row is a scalar multiple of another row but its determinant is 0.

2. Let $A \in \mathbb{C}^{n\times n}$. Show that if $\operatorname{tr}(A^*A) = 0$, then $A = 0$.

3. Let $a_1, \dots, a_n \in \mathbb{C}$. Let A be the $n \times n$ matrix whose first row has all entries as 1 and whose kth row has entries $a_1^{k-1}, \dots, a_n^{k-1}$ in that order. Show that $\det(A) = \Pi_{i<j}(a_i - a_j)$.

4. Let A be an $n \times n$ matrix with integer entries. Prove that if $\det(A) = \pm 1$, then A^{-1} has only integer entries.

5. Determine $\operatorname{adj}(A)$ and A^{-1} using the adjugate, where $A = \begin{bmatrix} 1 & 0 & 0 & 1 \\ -1 & 1 & 0 & 1 \\ -1 & -1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$.

A is invertible ; $\det(A) \neq 0$; RREF of A is I ; rank of A is n ;

3.7 Computing inverse of a matrix

The adjugate property of the determinant provides a way to compute the inverse of a matrix, provided it is invertible. However, it is very inefficient. We may use elementary row operations to compute the inverse. Our computation of the inverse bases on the following fact.

Theorem 3.1. *A square matrix is invertible iff it is a product of elementary matrices.*

Proof: Each elementary matrix is invertible since $E[i, j]$ is its own inverse, $E_{1/\alpha}[i]$ is the inverse of $E_\alpha[i]$, and $E_{-\alpha}[i, j]$ is the inverse of $E_\alpha[i, j]$. Therefore, any product of elementary matrices is invertible.

Conversely, suppose that A is an invertible matrix. Let EA^{-1} be the RREF of A^{-1}. If EA^{-1} has a zero row, then $EA^{-1}A$ also has a zero row. That is, E has a zero row. But E is a product of elementary matrices, which is invertible; it does not have a zero row. Therefore, EA^{-1} does not have a zero row. Then each row in the square matrix EA^{-1} has a pivot. But the only square matrix in RREF having a pivot at each row is the identity matrix. Therefore, $EA^{-1} = I$. That is, $A = E$, a product of elementary matrices. □

The computation of inverse will be easier if we write the matrix A and the identity matrix I side by side and apply the elementary operations on both of them simultaneously. For this purpose, we introduce the notion of an augmented matrix.

If $A \in \mathbb{F}^{m\times n}$ and $B \in \mathbb{F}^{m\times k}$, then the matrix $[A|B] \in \mathbb{F}^{m\times(n+k)}$ obtained from A and B by writing first all the columns of A and then the columns of B, in that order, is called an **augmented matrix**.

The vertical bar shows the separation of columns of A and of B, though, conceptually unnecessary.

For computing the inverse of a matrix, start with the augmented matrix $[A|I]$. Apply elementary row operations for reducing A to its row reduced echelon form, while simultaneously applying the

same operations on the entries of I. This means we pre-multiply the matrix $[A|I]$ with a product B of elementary matrices. In block form, our result is the augmented matrix $[BA|BI]$. If $BA = I$, then $BI = A^{-1}$. That is, the part that contained I originally will give the matrix A^{-1} after the elementary row operations have been applied. If after row reduction, it turns out that $B \neq I$, then A is not invertible; this information is a bonus.

Example 3.7. For illustration, consider the following square matrices:

$$A = \begin{bmatrix} 1 & -1 & 2 & 0 \\ -1 & 0 & 0 & 2 \\ 2 & 1 & -1 & -2 \\ 1 & -2 & 4 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -1 & 2 & 0 \\ -1 & 0 & 0 & 2 \\ 2 & 1 & -1 & -2 \\ 0 & -2 & 0 & 2 \end{bmatrix}.$$

We want to find the inverses of the matrices, if at all they are invertible.

Augment A with an identity matrix to get

$$\left[\begin{array}{cccc|cccc} \boxed{1} & -1 & 2 & 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 2 & 0 & 1 & 0 & 0 \\ 2 & 1 & -1 & -2 & 0 & 0 & 1 & 0 \\ 1 & -2 & 4 & 2 & 0 & 0 & 0 & 1 \end{array}\right].$$

Use elementary row operations. Since $a_{11} = 1$, we leave $row(1)$ untouched. To zero-out the other entries in the first column, we use the sequence of elementary row operations $E_1[2,1]$, $E_{-2}[3,1]$, $E_{-1}[4,1]$ to obtain

$$\left[\begin{array}{cccc|cccc} \boxed{1} & -1 & 2 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 2 & 2 & 1 & 1 & 0 & 0 \\ 0 & 3 & -5 & -2 & -2 & 0 & 1 & 0 \\ 0 & -1 & 2 & 2 & -1 & 0 & 0 & 1 \end{array}\right].$$

The pivot is -1 in $(2,2)$ position. Use $E_{-1}[2]$ to make the pivot 1.

$$\left[\begin{array}{cccc|cccc} 1 & -1 & 2 & 0 & 1 & 0 & 0 & 0 \\ 0 & \boxed{1} & -2 & -2 & -1 & -1 & 0 & 0 \\ 0 & 3 & -5 & -2 & -2 & 0 & 1 & 0 \\ 0 & -1 & 2 & 2 & -1 & 0 & 0 & 1 \end{array}\right].$$

Use $E_1[1,2]$, $E_{-3}[3,2]$, $E_1[4,2]$ to zero-out all non-pivot entries in the pivotal column to 0:

$$\left[\begin{array}{cccc|cccc} \boxed{1} & 0 & 0 & -2 & 0 & -1 & 0 & 0 \\ 0 & \boxed{1} & -2 & -2 & -1 & -1 & 0 & 0 \\ 0 & 0 & \boxed{1} & 4 & 1 & 3 & 1 & 0 \\ 0 & 0 & 0 & 0 & -2 & -1 & 0 & 1 \end{array}\right].$$

Since a zero row has appeared in the A portion of the augmented matrix, we conclude that A is not invertible. You see that the second portion of the augmented matrix has no meaning now. However,

it records the elementary row operations which were carried out in the reduction process. Verify that this matrix is equal to

$$E_1[4,2]\,E_{-3}[3,2]\,E_1[1,2]\,E_{-1}[2]\,E_{-1}[4,1]\,E_{-2}[3,1]\,E_1[2,1]$$

and that the first portion is equal to this matrix times A.

For B, we proceed similarly. The augmented matrix $[B|I]$ with the first pivot looks like:

$$\left[\begin{array}{cccc|cccc} \boxed{1} & -1 & 2 & 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 2 & 0 & 1 & 0 & 0 \\ 2 & 1 & -1 & -2 & 0 & 0 & 1 & 0 \\ 0 & -2 & 0 & 2 & 0 & 0 & 0 & 1 \end{array}\right].$$

The sequence of elementary row operations $E_1[2,1];\ E_{-2}[3,1]$ yields

$$\left[\begin{array}{cccc|cccc} \boxed{1} & -1 & 2 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 2 & 2 & 1 & 1 & 0 & 0 \\ 0 & 3 & -5 & -2 & -2 & 0 & 1 & 0 \\ 0 & -2 & 0 & 2 & 0 & 0 & 0 & 1 \end{array}\right].$$

Next, the pivot is -1 in $(2,2)$ position. Use $E_{-1}[2]$ to get the pivot as 1.

$$\left[\begin{array}{cccc|cccc} \boxed{1} & -1 & 2 & 0 & 1 & 0 & 0 & 0 \\ 0 & \boxed{1} & -2 & -2 & -1 & -1 & 0 & 0 \\ 0 & 3 & -5 & -2 & -2 & 0 & 1 & 0 \\ 0 & -2 & 0 & 2 & 0 & 0 & 0 & 1 \end{array}\right].$$

And then $E_1[1,2];\ E_{-3}[3,2];\ E_2[4,2]$ gives

$$\left[\begin{array}{cccc|cccc} \boxed{1} & 0 & 0 & -2 & 0 & -1 & 0 & 0 \\ 0 & \boxed{1} & -2 & -2 & -1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 4 & 1 & 3 & 1 & 0 \\ 0 & 0 & -4 & -2 & -2 & -2 & 0 & 1 \end{array}\right].$$

Next pivot is 1 in $(3,3)$ position. Now, $E_2[2,3];\ E_4[4,3]$ produces

$$\left[\begin{array}{cccc|cccc} \boxed{1} & 0 & 0 & -2 & 0 & -1 & 0 & 0 \\ 0 & \boxed{1} & 0 & 6 & 1 & 5 & 2 & 0 \\ 0 & 0 & \boxed{1} & 4 & 1 & 3 & 1 & 0 \\ 0 & 0 & 0 & 14 & 2 & 10 & 4 & 1 \end{array}\right].$$

Next pivot is 14 in $(4,4)$ position. Use $[4;1/14]$ to get the pivot as 1:

$$\left[\begin{array}{cccc|cccc} \boxed{1} & 0 & 0 & -2 & 0 & -1 & 0 & 0 \\ 0 & \boxed{1} & 0 & 6 & 1 & 5 & 2 & 0 \\ 0 & 0 & \boxed{1} & 4 & 1 & 3 & 1 & 0 \\ 0 & 0 & 0 & \boxed{1} & 1/7 & 5/7 & 2/7 & 1/14 \end{array}\right].$$

Use $E_2[1,4];\ E_{-6}[2,4];\ E_{-4}[3,4]$ to zero-out the entries in the pivotal column:

$$\left[\begin{array}{cccc|cccc} \boxed{1} & 0 & 0 & 0 & 2/7 & 3/7 & 4/7 & 1/7 \\ 0 & \boxed{1} & 0 & 0 & 1/7 & 5/7 & 2/7 & -3/7 \\ 0 & 0 & \boxed{1} & 0 & 3/7 & 1/7 & -1/7 & -2/7 \\ 0 & 0 & 0 & \boxed{1} & 1/7 & 5/7 & 2/7 & 1/14 \end{array}\right].$$

Thus $B^{-1} = \frac{1}{7}\begin{bmatrix} 2 & 3 & 4 & 1 \\ 1 & 5 & 2 & -3 \\ 3 & 1 & -1 & -2 \\ 1 & 5 & 2 & \frac{1}{2} \end{bmatrix}$. Verify that $B^{-1}B = BB^{-1} = I$.

Observe that if a matrix is not invertible, then our algorithm for reduction to RREF produces a pivot in the I portion of the augmented matrix.

Exercises for § 3.7

1. Compute the inverses of the following matrices, if possible:

   (a) $\begin{bmatrix} 2 & 1 & 2 \\ 1 & 3 & 1 \\ -1 & 1 & 2 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 4 & -6 \\ -1 & -1 & 3 \\ 1 & -2 & 3 \end{bmatrix}$ (c) $\begin{bmatrix} 3 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 \\ 1 & 1 & 2 & -1 \\ -2 & 1 & -1 & 3 \end{bmatrix}$

2. Let $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & -b & -c \end{bmatrix}$, where $b, c \in \mathbb{C}$. Show that $A^{-1} = bI + cA$.

3. Show that if a matrix A is upper triangular and invertible, then so is A^{-1}.

4. Show that if a matrix A is lower triangular and invertible, then so is A^{-1}.

5. Show that every $n \times n$ matrix can be written as a sum of two invertible matrices.

6. Show that every $n \times n$ invertible matrix can be written as a sum of two non-invertible matrices.

$A \xrightarrow{RREF} B = \begin{bmatrix} 0 & 1 & a_1 & 0 & b_1 & c_1 & 0 & d_1 \\ 0 & 0 & 0 & 1 & b_2 & c_2 & 0 & d_2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\vartheta_1\ e_1\ \vartheta_2\ e_2\ \vartheta_3\ \vartheta_4\ e_3\ \vartheta_5$

$u_1\ u_2\ u_3\ u_4\ u_5\ u_6\ u_7\ u_8$

$e_1, e_2, e_3 \in F^{5\times 1}$

$\vartheta_2 = \begin{pmatrix} a_1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad \vartheta_3 = \begin{bmatrix} b_1 \\ b_2 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad \vartheta_4 = \begin{bmatrix} c_1 \\ c_2 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad \vartheta_5 = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \\ 0 \\ 0 \end{bmatrix}$

$b_1e_1 + b_2e_2 \qquad c_1e_1 + c_2e_2 \qquad d_1e_1 + d_2e_2 + d_3e_3$

$A = [u_1 \quad u_2 \quad u_3 \quad \dots \quad u_8]$

$B = EA = [EU_1 \,|Eu_2|\, Eu_3 \dots |Eu_8]$

$Eu_1 = \vartheta_1 \quad Eu_2 = e_1 \quad Eu_3 = \vartheta_2 \quad Eu_4 = e_2 \quad Eu_5 = \vartheta_3$

$Eu_6 = \vartheta_4 \quad Eu_7 = e_3 \quad Eu_8 = \vartheta_5$

$\Rightarrow u_1 = E^{-1}\vartheta_1 \quad u_2 = E^{-1}e_1 \quad u_3 = E^{-1}\vartheta_2 \quad u_4 = E^{-1}e_2 \quad u_5 = E^{-1}\vartheta_3$

$u_6 = E^{-1}\vartheta_4 \quad u_7 = E^{-1}e_3 \quad u_8 = E^{-1}\vartheta_5$

$\vartheta_5 = d_1 Eu_2 + d_2 Eu_4 + d_3 Eu_7 \Rightarrow E^{-1}\vartheta_5 = d_1u_2 + d_2u_4 + d_3u_7$

$\Rightarrow \boxed{u_8 = d_1u_2 + d_2u_4 + d_3u_7}$

$$v_4 = c_1 E u_2 + c_2 E u_4 \Rightarrow \vec{E} v_4 = c_1 u_2 + c_2 u_4$$

$$\Rightarrow \boxed{u_6 = c_1 u_2 + c_2 u_4}$$

$$v_3 = b_1 E u_2 + b_2 E u_4 \Rightarrow \vec{E} v_3 = b_1 u_2 + b_2 u_4$$

$$\Rightarrow \boxed{u_5 = b_1 u_2 + b_2 u_4}$$

$$v_2 = a_1 E u_2 \Rightarrow \vec{E} v_2 = a_1 u_2$$

$$\Rightarrow \boxed{u_3 = a_1 u_2}$$

If rank(A)= r < m, then E(−1)e(r +1), . . . , E(−1)e(m) are not expressible in the form α1u1 + ·· · + αnun for any α1, . . . , αn ∈ F.
Given a matrix A, our algorithm produces a unique matrix B in RREF. It raises the question whether by following another algorithm that employs elementary row operations on A, we would obtain a matrix in RREF different from B? The following result shows that this is not possible; the row reduced echelon form of a matrix is canonical.

Let $a_1, \ldots, a_n \in \mathbb{C}$. Let A be the $n \times n$ matrix whose first row has all entries as 1 and whose kth row has entries $a_1^{k-1}, \ldots, a_n^{k-1}$ in that order. Show that $\det(A) = (-1)^{n(n-1)/2} \Pi_{1 \le i < j \le n}(a_i - a_j)$.

Chapter 4

Rank and Linear Equations

4.1 Matrices as linear maps

Let $A \in \mathbb{F}^{m\times n}$. We may view the matrix A as a function from $\mathbb{F}^{n\times 1}$ to $\mathbb{F}^{m\times 1}$. It goes as follows.

Let $x \in \mathbb{F}^{n\times 1}$. Then define the matrix A as a function $A : \mathbb{F}^{n\times 1} \to \mathbb{F}^{m\times 1}$ by

$$A(x) = Ax.$$

That is, the value of the function A at any vector $x \in \mathbb{F}^{n\times 1}$ is the vector Ax in $\mathbb{F}^{m\times 1}$.

Since the matrix product Ax is well defined, such a function is meaningful. We see that due to the properties of matrix product, the following are true:

1. $A(u+v) = A(u) + A(v)$ for all $u, v \in \mathbb{F}^{n\times 1}$.
2. $A(\alpha v) = \alpha A(v)$ for all $v \in \mathbb{F}^{n\times 1}$ and for all $\alpha \in \mathbb{F}$.

In this manner a matrix is considered as a linear map. A **linear map** is any function from $\mathbb{F}^{n\times 1}$ to $\mathbb{F}^{m\times 1}$ satisfying the above two properties. To see the connection between the matrix as a rectangular array and as a function, consider the values of the matrix A at the standard basis vectors $e_1, \dots, e_n$ in $\mathbb{F}^{n\times 1}$. Recall that e_j is a column vector in $\mathbb{F}^{n\times 1}$ where the jth entry is 1 and all other entries are 0. Let $A = [a_{ij}] \in \mathbb{F}^{m\times n}$. We see that

$$Ae_j = \begin{bmatrix} a_{11} & \cdots & a_{1j} & \cdots & a_{1n} \\ & & \vdots & & \\ a_{i1} & \cdots & a_{ij} & \cdots & a_{in} \\ & & \vdots & & \\ a_{m1} & \cdots & a_{mj} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} = \begin{bmatrix} a_{1j} \\ \vdots \\ a_{ij} \\ \vdots \\ a_{mj} \end{bmatrix} = j\text{th column of } A.$$

A matrix $A \in \mathbb{F}^{m\times n}$ is viewed as the linear map $A : \mathbb{F}^{n\times 1} \to \mathbb{F}^{m\times 1}$, where $A(e_j)$ is the jth column of A, and $A(v) = Av$ for each $v \in \mathbb{F}^{n\times 1}$.

The **range** of the matrix A (of the linear map A) is the set $R(A) = \{Ax : x \in \mathbb{F}^{n\times 1}\}$. Now, each vector $x = [\alpha_1, \dots, \alpha_n]^t \in \mathbb{F}^{n\times 1}$ can be written as

$$x = \alpha_1 e_1 + \cdots + \alpha_n e_n.$$

If $y \in R(T)$, then there exists an $x \in \mathbb{F}^{n\times 1}$ such that $y = Ax$. Then such a y can be written as

$$y = Ax = \alpha_1 Ae_1 + \cdots + \alpha_n Ae_n.$$

Conversely we see that each vector $\alpha_1 Ae_1 + \cdots + \alpha_n Ae_n$ is in $R(A)$. Since Ae_j is the jth column of A, we find that

$$R(A) = \{\alpha_1 A_1 + \cdots + \alpha_n A_n \ : a_1, \ldots, a_n \in \mathbb{F}\},$$

where $A_1, \ldots, A_n$ are the n columns of A.

Exercises for § 4.1

1. Let $A \in \mathbb{C}^{m\times n}$. Define $T : \mathbb{C}^{1\times m} \to \mathbb{C}^{1\times n}$ by $T(x) = xA$ for $x \in \mathbb{C}^{1\times m}$. Show that T is a linear map. Identify $T(e_j)$, where e_j is a vector in $\mathbb{C}^{1\times m}$ whose jth component is 1 and all other components are 0.

2. Define $T : \mathbb{R}^{3\times 1} \to \mathbb{R}^{2\times 1}$ by $T([a, b, c]^t) = [c, b + a]^t$. Show that T is a linear map. Find a matrix $A \in \mathbb{R}^{2\times 3}$ such that $T([a, b, c]^t) = A[a, b, c]^t$.

4.2 Linear independence

We give a name to an expression of the type we have seen in the last section.

If $v_1, \ldots, v_m \in \mathbb{F}^{1\times n}$, (or in $\mathbb{F}^{n\times 1}$) then the vector

$$\alpha_1 v_1 + \cdots \alpha_m v_m$$

α_i can also be 0

is called a **linear combination** of $v_1, \ldots, v_m$, where $\alpha_1, \ldots, \alpha_m \in \mathbb{F}$ are some scalars.

For example, in $\mathbb{F}^{1\times 2}$, one linear combination of $v_1 = [1,\ 1]$ and $v_2 = [1,\ -1]$ is as follows:

$$2[1,\ 1] + 1[1,\ -1].$$

This linear combination evaluates to $[3,\ 1]$. Thus $[3,\ 1]$ is a linear combination of v_1, v_2.

Is $[4,\ -2]$ a linear combination of v_1 and v_2? Yes, since

$$[4,\ -1] = 1[1,\ 1] + 3[1,\ -1].$$

In fact, every vector in $\mathbb{F}^{1\times 2}$ is a linear combination of v_1 and v_2. Reason:

$$[a,\ b] = \tfrac{a+b}{2}\,[1,\ 1] + \tfrac{a-b}{2}\,[1,\ -1].$$

However, every vector in $\mathbb{F}^{1\times 2}$ is not a linear combination of $[1,\ 1]$ and $[2,\ 2]$. Reason? Any linear combination of these two vectors is a multiple of $[1,\ 1]$. Then $[1,\ 0]$ is not a linear combination of these two vectors.

The vectors $v_1, \ldots, v_m$ in $\mathbb{F}^{n\times 1}$ are called **linearly dependent** iff at least one of them is a linear combination of others. The vectors are called **linearly independent** iff none of them is a linear combination of others.

$\text{span}\{\theta_1, \theta_2, \ldots, \theta_k\} = \{\alpha_1\theta_1 + \alpha_2\theta_2 + \cdots \alpha_k\theta_k : \alpha_j \in F, j \in [1{:}k]\}$

span $\{(1,1),(1,-1)\} = \{(\alpha_1+\alpha_2),(\alpha_1-\alpha_2) : \alpha_1,\alpha_2 \in \mathbb{R}\}$

$(4,2) \to$ in the span

For example, $[1,\ 1]$, $[1,\ -1]$, $[4,\ -1]$ are linearly dependent vectors whereas $[1,\ 1]$, $[1,\ -1]$ are linearly independent vectors.

Notice that if $\alpha_1 = \cdots = \alpha_m = 0$, then obviously, the linear combination $\alpha_1 v_1 + \cdots + \alpha_m v_m$ evaluates to 0. That is, the zero vector can always be written as a *trivial* linear combination.

Suppose the vectors $v_1, \ldots, v_m$ are linearly dependent. Then one of them, say, v_i is a linear combination of others. That is,

$$v_i = \alpha_1 v_1 + \cdots + \alpha_{i-1} v_{i-1} + \alpha_{i+1} v_{i+1} + \cdots + \alpha_m v_m.$$

Then

$$\alpha_1 v_1 + \cdots + \alpha_{i-1} v_{i-1} + (-1) v_i + \alpha_{i+1} v_{i+1} + \cdots + \alpha_m v_m = 0.$$

Here, we see that a linear combination becomes zero, where at least one of the coefficients, that is, the ith one is nonzero.

Conversely, suppose that we have scalars $\beta_1, \ldots, \beta_m$ not all zero such that

$$\beta_1 v_1 + \cdots + \beta_m v_m = 0.$$

Suppose that the kth scalar β_k is nonzero. Then

$$v_k = -\frac{1}{\beta_k}\big(\beta_1 v_1 + \cdots + \beta_{k-1} v_{k-1} + \beta_{k+1} v_{k+1} + \cdots + \beta_m v_m\big).$$

That is, the vectors $v_1, \ldots, v_m$ are linearly dependent.

Thus we have proved the following:

$v_1, \ldots, v_m$ are linearly dependent iff $\alpha_1 v_1 + \cdots + \alpha_m v_m = 0$ for scalars $\alpha_1, \ldots, \alpha_m$ not all zero.

The same may be written in terms of linear independence.

Theorem 4.1. *The vectors $v_1, \ldots, v_m \in \mathbb{F}^{1\times n}$ are linearly independent iff for all $\alpha_1, \ldots, \alpha_m \in \mathbb{F}$,*

$$\alpha_1 v_1 + \cdots \alpha_m v_m = 0 \text{ implies that } \alpha_1 = \cdots = \alpha_m = 0.$$

The same is true when $v_1, \ldots, v_n \in \mathbb{F}^{n\times 1}$.

Theorem 4.1 provides a way to determine whether a finite number of vectors are linearly independent or not. You start with a linear combination of the given vectors; and equate it to 0. Then you must be able to derive that each coefficient in that linear combination is 0. If this is the case, then the given vectors are linearly independent. If it is not possible, then from its proof you must be able to find a way of expressing one of the vectors as a linear combination of the others, showing that the vectors are linearly dependent.

Example 4.1. Are the vectors $[1,\ 1,\ 1]$, $[2,\ 1,\ 1]$, $[3,\ 1,\ 0]$ linearly independent?

We start with an arbitrary linear combination and equate it to the zero vector. Solve the resulting linear equations to determine whether all the coefficients are necessarily 0 or not.

So, let

$$a[1,\ 1,\ 1] + b[2,\ 1,\ 1] + c[3,\ 1,\ 0] = [0,\ 0,\ 0].$$

Comparing the components, we have

$$a + 2b + 3c = 0,\ a + b + c = 0,\ a + b = 0.$$

The last two equations imply that $c = 0$. Substituting in the first, we see that

$$a + 2b = 0.$$

This and the equation $a + b = 0$ give $b = 0$. Then it follows that $a = 0$.

We conclude that the given vectors are linearly independent.

Example 4.2. Are the vectors $[1,\ 1,\ 1],\ [2,\ 1,\ 1],\ [3,\ 2,\ 2]$ linearly independent?

Clearly, the third one is the sum of the first two. So, the given vectors are linearly dependent. To illustrate our method, we start with an arbitrary linear combination and equate it to the zero vector. Solve the resulting linear equations to determine whether all the coefficients are necessarily 0 or not.

So, as earlier, let

$$a[1,\ 1,\ 1] + b[2,\ 1,\ 1] + c[3,\ 2,\ 2] = [0,\ 0,\ 0].$$

Comparing the components, we have

$$a + 2b + 3c = 0,\ a + b + 2c = 0,\ a + b + 2c = 0.$$

The last equation is redundant. Subtracting the second from the first, we have

$$b + c = 0.$$

We may choose $b = 1,\ c = -1$ to satisfy this equation. Then from the second equation, we have $a = 1$. Then our starting equation says that

$$1[1,\ 1,\ 1] + 1[2,\ 1,\ 1] + (-1)[3,\ 2,\ 2] = [0,\ 0,\ 0].$$

That is, the third vector is the sum of the first two.

Be careful with the direction of implication here. Your work-out must be in the form

$$\alpha_1 v_1 + \cdots + \alpha_m v_m = 0 \Rightarrow \cdots \Rightarrow \alpha_1 = \cdots = \alpha_m = 0.$$

And that would prove linear independence.

To see how linear independence is helpful, consider the following system of linear equations:

$$\begin{array}{rrrl} x_1 & +2x_2 & -3x_3 & = 2 \\ 2x_1 & -x_2 & +2x_3 & = 3 \\ 4x_1 & +3x_2 & -4x_3 & = 7 \end{array}$$

Here, we find that the third equation is redundant, since 2 times the first plus the second gives the third. That is, the third one linearly depends on the first two. (You can of course choose any other

• A superset of a linearly dependent set is linearly dependent

- $e_1, e_2, e_3, \ldots e_m \rightarrow$ linearly independent
- a subset of a linearly independent set is linearly independant

equation here as linearly depending on other two, but that is not important.) Now, take the row vectors of coefficients of the unknowns as in the following:

$$v_1 = [1,\ 2,\ -3,\ 2], \quad v_2 = [2,\ -1,\ 2,\ 3], \quad v_3 = [4,\ 3,\ -4,\ 7].$$

We see that $v_3 = 2v_1 + v_2$, as it should be. We see that the vectors v_1, v_2, v_3 are linearly dependent. But the vectors v_1, v_2 are linearly independent. Thus, solving the given system of linear equations is the same thing as solving the system with only first two equations. For solving linear systems, it is of primary importance to find out which equations linearly depend on others. Once determined, such equations can be thrown away, and the rest can be solved.

Exercises for § 4.2

1. Determine whether the given vectors are linearly independent, in each case:
   (a) $[1,\ 2,\ 6]$, $[-1,\ 3,\ 4]$, $[-1,\ -4,\ 2]$ in $\mathbb{R}^{1\times 3}$.
   (b) $[1,\ 0,\ 2,\ 1]$, $[1,\ 3,\ 2,\ 1]$, $[4,\ 1,\ 2,\ 2]$ in $\mathbb{C}^{1\times 4}$.
2. Suppose that u, v, w are linearly independent in $\mathbb{C}^{1\times 5}$. Are the following lists of vectors linearly independent?
   (a) $u, v + \alpha w, w$, where α is a nonzero complex number. → independent
   (b) $u + v,\ v + w,\ w + u$. → independent
   (c) $u - v,\ v - w,\ w - u$. → independent

- singleton sets are linearly independent

4.3 Gram-Schmidt orthogonalization

It is easy to see that if the nonzero vectors $v_1, \ldots, v_n$ are orthogonal, then they are also linearly independent. How?

Suppose $v_1, \ldots, v_n$ are nonzero orthogonal vectors. Assume that

$$\alpha_1 v_1 + \cdots + \alpha_n v_n = 0.$$

Take inner product with v_j for $j \in \{1, \ldots, n\}$. For $i \neq j$, $\langle v_i, v_j \rangle = 0$. So,

$$\alpha_j \langle v_j, v_j \rangle = 0.$$

But $v_j \neq 0$ implies that $\langle v_j, v_j \rangle \neq 0$. Therefore, $\alpha_j = 0$. That is,

$$\alpha_1 = \cdots = \alpha_n = 0.$$

Therefore, the vectors $v_1, \ldots, v_n$ are linearly independent.

Conversely, given n linearly independent vectors $v_1, \ldots, v_n$ (necessarily all nonzero), we can orthogonalize them. If $v_1, \ldots, v_k$ are linearly independent but $v_1, \ldots, v_k, v_{k+1}$ are linearly dependent, then we will see that our orthogonalization process will yield the $(k+1)$th vector as the zero vector. We now discuss this method, called *Gram-Schmidt orthogonalization*.

Given two linearly independent vectors u_1, u_2 on the plane how do we construct two orthogonal vectors?

Keep $v_1 = u_1$. Take out the projection of u_2 on u_1 to get v_2. Now, $v_2 \perp v_1$.

What is the projection of u_2 on u_1?

Its length is $\langle u_2, u_1\rangle$. Its direction is that of u_1. Thus taking $v_1 = u_1$ and $v_2 = u_2 - \dfrac{\langle u_2, v_1\rangle}{\langle v_1, v_1\rangle} v_1$ does the job. You can now verify that $\langle v_2, v_1\rangle = 0$.

We may continue this process of taking away projections in $\mathbb{F}^{n\times 1}$, or in $\mathbb{F}^{1\times n}$. It results in the following process.

Theorem 4.2. (Gram-Schmidt orthogonalization) *Let $u_1, u_2, \ldots, u_n$ be linearly independent vectors. Define*

$$\begin{aligned} v_1 &= u_1 \\ v_2 &= u_2 - \frac{\langle u_2, v_1\rangle}{\langle v_1, v_1\rangle} v_1 \\ &\vdots \\ v_{n+1} &= u_{n+1} - \frac{\langle u_{n+1}, v_1\rangle}{\langle v_1, v_1\rangle} v_1 - \cdots - \frac{\langle u_{n+1}, v_n\rangle}{\langle v_n, v_n\rangle} v_n \end{aligned}$$

Then $v_1, v_2, \ldots, v_n$ are orthogonal and the set of linear combinations of $v_1, v_2, \ldots, v_n$ is equal to the set of linear combinations of $u_1, u_2, \ldots, u_n$.

Proof: $\langle v_2, v_1\rangle = \langle u_2 - \dfrac{\langle u_2, v_1\rangle}{\langle v_1, v_1\rangle} v_1, v_1\rangle = \langle u_2, v_1\rangle - \dfrac{\langle u_2, v_1\rangle}{\langle v_1, v_1\rangle}\langle v_1, v_1\rangle = 0$. Use Induction. □

Observe that if $u_1, \ldots, u_k$ are linearly independent but $u_1, \ldots, u_{k+1}$ are linearly dependent, then Gram-Schmidt process will compute nonzero orthogonal vectors $v_1, \ldots, v_k$ and it will give v_{k+1} as the zero vector.

Example 4.3. The vectors $u_1 = [1,\ 0,\ 0], u_2 = [1,\ 1,\ 0], u_3 = [1,\ 1,\ 1]$ are linearly independent in $\mathbb{R}^{1\times 3}$. Apply Gram-Schmidt Orthogonalization.

$v_1 = [1,\ 0,\ 0]$.

$v_2 = u_2 - \dfrac{\langle u_2, v_1\rangle}{\langle v_1, v_1\rangle} v_1 = [1,\ 1,\ 0] - 1\,[1,\ 0,\ 0] = [0,\ 1,\ 0]$.

$v_3 = u_3 - \dfrac{\langle u_3, v_1\rangle}{\langle v_1, v_1\rangle} v_1 - \dfrac{\langle u_3, v_2\rangle}{\langle v_2, v_2\rangle} v_2 = [1,\ 1,\ 1] - [1,\ 0,\ 0] - [0,\ 1,\ 0] = [0,\ 0,\ 1]$.

The vectors $[1,\ 0,\ 0]$, $[0,\ 1,\ 0]$, $[0,\ 0,\ 1]$ are orthogonal.

Example 4.4. Apply Gram-Schmidt orthogonalization process on the vectors $u_1 = [1,\ 1,\ 0,\ 1]$, $u_2 = [0,\ 1,\ 1,\ -1]$ and $u_3 = [1,\ 3,\ 2,\ -1]$.

$v_1 = [1,\ 1,\ 0,\ 1]$.

$v_2 = u_2 - \dfrac{\langle u_2, v_1\rangle}{\langle v_1, v_1\rangle} v_1 = [0,\ 1,\ 1,\ -1] - 0\,[1,\ 1,\ 0,\ 1] = [0,\ 1,\ 1,\ -1]$.

Given $\{v_1, v_2, \dots v_n\}$, by removing some of them we can always get a linearly independent set. How to find maximum linearly independent set.

$A = [u_1\ u_2 \dots u_n] \in F^{m \times n}$ $\quad B = EA$

$B = \begin{bmatrix} 0 & 1 & 0 & * & * & 0 & \dots & 0 & \dots \\ 0 & 0 & 1 & * & * & 0 & & \vdots & \\ 0 & 0 & 0 & & & 1 & & & \\ \vdots & \vdots & & & & \vdots & & 1 & \\ 0 & 0 & & & & 0 & & 0 & \end{bmatrix}$

any pivotal column $= \sum_{i=1}^{j} a_i e_i$

to find linearly independent set, remove non pivotal colums first.

$j_1, j_2, \dots j_r \rightarrow$ index of pivotal colums

$u_{j_1}, u_{j_2}, \dots u_{j_r} \rightarrow$ linearly independent $\quad r = \text{rank}(A)$

Theorem

Let $v_1, v_2, \cdots, v_n \in \mathbb{F}^m$. Then there is a **unique** $r \leq n$ such that some r of these vectors are linearly independent and the remaining $n - r$ vectors are linear combinations of these vectors.

- In the above theorem $r = \text{rank}(A)$, where $A = [v_1\ v_2\ \cdots\ v_n] \in \mathbb{F}^{m \times n}$.

Corollary

Any set containing more than m vectors from $\mathbb{F}^m$ is linearly dependent.

Corollary

Let $v_1, v_2, \cdots, v_n \in \mathbb{F}^{m \times 1}$, where $n \leq m$, and $A = [v_1\ v_2\ \cdots\ v_n] \in \mathbb{F}^{m \times n}$. Then the vectors $v_1, \cdots, v_n$ is linearly independent **iff** $\text{rank}(A) = n$.

- **Thus**, a square matrix $A \in \mathbb{F}^{m \times m}$ is invertible iff $\text{rank}(A) = m$ iff the columns of A are linearly independent.

$$v_3 = u_3 - \frac{\langle u_3, v_1\rangle}{\langle v_1, v_1\rangle} v_1 - \frac{\langle u_3, v_2\rangle}{\langle v_2, v_2\rangle} v_2$$
$$= [1,\ 3,\ 2,\ -1] - [1,\ 1,\ 0,\ 1] - 2[0,\ 1,\ 1,\ -1] = [0,\ 0,\ 0,\ 0].$$

Discarding v_3, which is the zero vector, we have only two linearly independent vectors out of u_1, u_2, u_3. They are u_1 and u_2; and u_3 is a linear combination of these two. In fact, the process also revealed that $u_3 = u_1 - 2u_2$.

An orthogonal set can be made orthonormal by dividing each vector by its norm. Also you can modify Gram-Schmidt orthogonalization process to directly output orthonormal vectors.

Exercises for § 4.3

1. Find $u \in \mathbb{R}^{1\times 3}$ so that $[1/\sqrt{3},\ 1/\sqrt{3},\ 1/\sqrt{3}]$, $[1/\sqrt{2},\ 0,\ -1/\sqrt{2}]$, u are orhtonormal. Form a matrix with the vectors as rows, in that order. Verify that the columns of the matrix are also orthonormal.

2. Using Gram-Schmidt process, orthonormalize the vectors $[1,\ 1,\ 1]$, $[1,\ 0,\ 1]$, $[0,\ 1,\ 2]$.

3. Show that the cross product $u \times v$ of two linearly independent vectors u, v in $\mathbb{R}^{1\times 3}$ is orthogonal to both u and v. How to obtain this third vector as $u \times v$ by Gram-Schmidt process?

4.4 Determining linear independence

Gram-Schmidt process has the main goal of orthogonalizing a list of vectors. We could use it to determine liner independence. The elementary row operations provide a more efficient way. Given m row vectors $v_1, \dots, v_m \in \mathbb{F}^{1\times n}$, we form a matrix A with its ith row as v_i. Then using elementary row operations, we bring it to its RREF.

Observe that exchanging v_i with v_j in the list of vectors does not change linear independence of the vectors. Multiplying a nonzero scalar with v_i does not affect linear independence. Also, replacing v_i with $v_i + \alpha v_j$ does not alter linear independence.

To see the last one, suppose $v_1, \dots, v_m$ are linearly independent. Let $w_i = v_i + \alpha v_j$, $i \neq j$. To show the linear independence of $v_1, \dots, v_{i-1}, w_i, v_{i+1}, \dots, v_n$, suppose that

$$\beta_1 v_1 + \cdots + \beta_{i-1} v_{i-1} + \beta_i w_i + \beta_{i+1} v_{i+1} + \cdots + \beta_m v_m = 0.$$

Then

$$\beta_1 v_1 + \cdots + \beta_{i-1} v_{i-1} + \beta_i (v_i + \alpha v_j) + \beta_{i+1} v_{i+1} + \cdots + \beta_m v_m = 0.$$

Simplifying, we have

$$\beta_1 v_1 + \cdots + \beta_i v_i + \cdots + (\beta_j + \alpha\beta_i) v_j + \cdots + \beta_m v_m = 0.$$

Using linear independence of $v_1, \dots, v_m$, we obtain

$$\beta_1 = \cdots \beta_i = \cdots \beta_j + \alpha b_i = \cdots = \beta_m = 0.$$

This gives $\beta_j = -\alpha\beta_i = 0$ and all other βs are zero. Thus $v_1, \ldots, w_i, \ldots, v_m$ are linearly independent. Similarly, the converse also holds.

Thus, we take these vectors as the rows of a matrix and apply our reduction to RREF algorithm. From the RREF, we know that all rows where a pivot occurs are linearly independent. If you want to determine exactly which vectors among these are linearly independent, you must keep track of the row exchanges. A summary of the discussion in terms of a matrix is as follows.

Theorem 4.3. *Let $A \in \mathbb{F}^{m\times n}$. The rows of A are linearly independent iff the rows of the RREF of A are linearly independent.*

Example 4.5. To determine whether the vectors $[1, 1, 0, 1]$, $[0, 1, 1, -1]$ and $[1, 3, 2, -1]$ are linearly independent or not, we proceed as follows.

$$\begin{bmatrix} \boxed{1} & 1 & 0 & 1 \\ 0 & 1 & 1 & -1 \\ 1 & 3 & 2 & -1 \end{bmatrix} \xrightarrow{E_{-1}[3,1]} \begin{bmatrix} \boxed{1} & 1 & 0 & 1 \\ 0 & \boxed{1} & 1 & -1 \\ 0 & 2 & 2 & -2 \end{bmatrix} \xrightarrow{R1} \begin{bmatrix} \boxed{1} & 0 & -1 & 2 \\ 0 & \boxed{1} & 1 & -1 \\ 0 & 0 & 0 & -4 \end{bmatrix}$$

$$\xrightarrow{R2} \begin{bmatrix} \boxed{1} & 0 & -1 & 0 \\ 0 & \boxed{1} & 1 & 0 \\ 0 & 0 & 0 & \boxed{1} \end{bmatrix}$$

Here, $R1 = E_{-1}[1, 2],\ E_{-2}[3, 2]$ and $R2 = E_{-1/4}[3],\ E_{-2}[1, 3],\ E_1[2, 3]$.

The last matrix is in RREF in which each row has a pivot. Thus all the rows in the RREF are linearly independent. Therefore, the original vectors are linearly independent.

Example 4.6. Are the vectors $[1, 1, 0, 1]^t$, $[0, 1, 1, -1]^t$ and $[2, -1, -3, 5]^t$ linearly independent?

The vectors are in $\mathbb{F}^{4\times 1}$. These are linearly independent iff their transposes are. Forming a matrix with the transpose of the given vectors as rows, and reducing it to its RREF, we see that

$$\begin{bmatrix} \boxed{1} & 1 & 0 & 1 \\ 0 & 1 & 1 & -1 \\ 2 & -1 & -3 & 5 \end{bmatrix} \xrightarrow{E_{-2}[3,1]} \begin{bmatrix} \boxed{1} & 1 & 0 & 1 \\ 0 & \boxed{1} & 1 & -1 \\ 0 & -3 & -3 & 3 \end{bmatrix} \xrightarrow{R1} \begin{bmatrix} \boxed{1} & 0 & -1 & -2 \\ 0 & \boxed{1} & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Here, $R1 = E_{-1}[1, 2],\ E_3[3, 2]$. Since a zero row has appeared, the original vectors are linearly dependent. Also, notice that no row exchanges were carried out in the reduction process. Therefore, the third vector is a linear combination of the first two vectors; and the first two vectors are linearly independent.

Exercises for § 4.4

1. Using elementary row operations determine whether the given vectors are linearly dependent or independent in each of the following cases.
   (a) $[1,\ 0,\ -1,\ 2,\ -3]$, $[-2,\ 1,\ 2,\ 4,\ -1]$, $[3,\ 0,\ -1,\ 1,\ 1]$, $[-2,\ 1,\ 1,\ -1,\ -2]$.
   (b) $[1,\ 0,\ -1,\ 2,\ -3]$, $[-2,\ 1,\ 2,\ 4,\ -1]$, $[3,\ 0,\ -1,\ 1,\ 1]$, $[-2,\ -1,\ 0,\ -7,\ 3]$.
   (c) $[1,\ i,\ -1,\ 1-i]$, $[i,\ -1,\ -i,\ 1+i]$, $[2,\ 0,\ 1,\ i]$, $[1+i,\ 1-i,\ -1,\ -i]$.
2. Suppose $A \in \mathbb{F}^{n\times n}$ is an invertible matrix, and $v_1, \ldots, v_m \in \mathbb{F}^{n\times 1}$. Prove that $v_1, \ldots, v_m$ are linearly independent iff $Av_1, \ldots, Av_n$ are linearly independent.

4.5 Rank

Consider an $m \times n$ matrix A. It may very well happen that all its rows are linearly independent. It may also happen that only some r of the m rows are linearly independent and other $m - r$ rows are linear combinations of those r rows. This number r, which is the maximum number of linearly independent rows in A, is called the **row rank** of A. The row rank is equal to 0 when all entries of the matrix are 0. If all the rows of an $m \times n$ matrix are linearly independent, its row rank is m.

Similarly, the maximum number of linearly independent columns of a matrix is called its **column rank**. We take the column rank of a zero matrix as 0; and if all columns are linearly independent, the column rank is the number of columns. Recall that for a matrix A, its range $R(A)$ is the set of all linear combinations of its columns. Therefore, the column rank of A is k means that there are exactly k columns of A such that each vector in $R(A)$ is a linear combination of these k columns.

Example 4.7. Find the row rank and the column rank of the matrix

$$A = \begin{bmatrix} 1 & 1 & 1 & 2 & 1 \\ 1 & 2 & 1 & 1 & 1 \\ 3 & 5 & 3 & 4 & 3 \\ -1 & 0 & -1 & -3 & -1 \end{bmatrix}.$$

Here, we see that $row(3) = row(1) + 2row(2), \quad row(4) = row(2) - 2row(1)$.
But $row(2)$ is not a scalar multiple of $row(1)$, that is, $row(1),\ row(2)$ are linearly independent. Therefore, the row rank of A is 2.

For the linear independence of columns, we see that

$$col(3) = col(5) = col(1),\ col(4) = 3col(1) - col(2).$$

And columns one and two are linearly independent. So, the column rank of A is also 2.

It can be proved that an elementary row operation neither alters the row rank nor the column rank of a matrix. Therefore, the row rank of a matrix is the same as the row rank of its RREF. Similarly, the column rank of a matrix is same as the column rank of its RREF. However, the row rank and the column rank of a matrix in RREF is equal to the number of pivots. Therefore, the row rank and the column rank of a matrix are equal.

We thus define the **rank** of a matrix A as its row rank, which is also equal to its column rank, and denote it by $\text{rank}(A)$. Since the row rank of A^t is the column rank of A, it follows that $\text{rank}(A^t) = \text{rank}(A)$. The number $n - \text{rank}(A)$ is called the **nullity** of A. We will connect the nullity to the number of linearly independent solutions of a linear homogeneous system $Ax = 0$ later.

For instance, look at Examples 4.5-4.6. There, the rank of

$$\begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & -1 \\ 1 & 3 & 2 & -1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & -1 \\ 2 & -1 & -3 & 5 \end{bmatrix}$$

$A \in F^{m \times n}$, for any invertible matrix $P \in F^{m \times n}$ and

$Q \in F^{n \times n}$

$\text{rank}(PAQ) = \text{rank}(AQ) = \text{rank}(A) = \text{rank}$

are, respectively, 3 and 2. Their nullity are 0 and 1, respectively.

It thus follows that if A is an $m \times n$ matrix, then $\text{rank}(A)$ must be less than or equal to $\min\{m, n\}$.

Also, if we take more than n number of vectors in $\mathbb{F}^{1\times n}$ (or in $\mathbb{F}^{n\times 1}$), then they are bound to be linearly dependent. Hint: look at the RREF of the matrix whose rows are the given vectors.

Further, we see that a square matrix is invertible iff its rank is equal to its order.

Exercises for § 4.5

1. Reduce the following matrix to its RREF and determine its rank:
$$\begin{bmatrix} 1 & 1 & 1 & 2 & 1 \\ 1 & 2 & 1 & 1 & 1 \\ 3 & 5 & 3 & 4 & 3 \\ -1 & 0 & -1 & -3 & -1 \end{bmatrix}.$$
2. Let $T : \mathbb{R}^{3\times 1} \to \mathbb{R}^{3\times 1}$ be defined by $T([a,\ b,\ c]^t) = [a + b,\ 2a - b - c,\ a + b + c]^t$. Let $A \in \mathbb{R}^{3\times 3}$ be the matrix such that $T(x) = Ax$ for $x \in \mathbb{R}^{3\times 1}$. Find $\text{rank}(A)$.
3. If $E \in \mathbb{F}^{m\times m}$ is an elementary matrix and $A \in \mathbb{F}^{m\times n}$, then show that the row rank of EA is equal to the row rank of A.
4. If $B \in \mathbb{F}^{m\times m}$ is an invertible matrix and $A \in \mathbb{F}^{m\times n}$, then show that the column rank of BA is equal to the column rank of A.
5. From previous two exercises, conclude that an elementary row operation neither alters the row rank nor the column rank of a matrix.
6. Let $A \in \mathbb{C}^{m\times k};\ B \in \mathbb{R}^{m\times k},\ C \in \mathbb{C}^{k\times m}$. Show that $\text{rank}(A^t) = \text{rank}(A)$.
7. Show that if $v_1, \ldots, v_n \in \mathbb{F}^{1\times m}$ and $n > m$, then these vectors are linearly dependent.
8. Prove that if $A \in \mathbb{F}^{n\times n}$, then $\text{rank}(A) = n$ iff $\det(A) \neq 0$.

4.6 Solvability of linear equations

We can now use our knowledge about matrices to settle some issues regarding solvability of linear systems. A **linear system** with m equations in n unknowns looks like:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots a_{mn}x_n &= b_m \end{aligned}$$

It is the set of all possible outputs of the linear transformation defined by A, meaning:R(A)={Ax|x is in the domain of A}, R(A)={Ax|x is in the domain of A} This is the column space (or image) of A, consisting of all linear combinations of the columns of A. The system Ax=b has a solution if and only if b lies in this range R(A).

consistent: has a solution
inconsistent: 0 solutions

Solving such a linear system amounts to determining the unknowns $x_1, \ldots, x_n$ with known scalars a_{ij} and b_i. Using the abbreviation $x = [x_1, \ldots, x_n]^t$, $b = [b_1, \ldots, b_m]^t$ and $A = [a_{ij}]$, the system can be written in the compact form:

$$Ax = b.$$

Here, $A \in \mathbb{F}^{m\times n}$, $x \in \mathbb{F}^{n\times 1}$ and $b \in \mathbb{F}^{m\times 1}$. We also say that the matrix A is the **system matrix** of the linear system $Ax = b$. Observe that the matrix A is a linear transformation from $\mathbb{F}^{n\times 1}$ to $\mathbb{F}^{m\times 1}$, where m is the *number of equations* and n is the *number of unknowns* in the system.

There is a slight deviation from our accepted symbolism. In case of linear systems, we write b as a column vector and x_i are unknown scalars.

$Sol(A|b) = \{y : Ay = b\}$ → solution space

Let $A \in \mathbb{F}^{m\times n}$ and $b \in \mathbb{F}^{m\times 1}$. A **solution** of the system $Ax = b$ is any vector $y \in \mathbb{F}^{n\times 1}$ such that $Ay = b$. In such a case, if $y = [a_1, \ldots, a_n]^t$, then a_i is called as the **value of the unknown** x_i in the solution y. In this language a solution of the system is also written informally as

$$x_1 = a_1, \ \cdots, \ x_n = a_n.$$

If $Ax=b$ has more than 1 soln ⇒ it has ∞'ly many solutions
y, z → solns ⇒ $ty + (1-t)z$ ✓

The system $Ax = b$ has a solution iff $b \in R(A)$; and it has a unique solution iff $b \in R(A)$ and A is a one-one map. Corresponding to the linear system $Ax = b$ is the **homogeneous system**

$$Ax = 0.$$

The homogeneous system always has a solution since $y := 0$ is a solution. It has infinitely many solutions when it has a nonzero solution. For, if y is a solution of $Ax = 0$, then so is αx for any scalar α.

To study the non-homogeneous system, we use the *augmented matrix* $[A|b] \in \mathbb{F}^{m\times(n+1)}$ which has its first n columns as those of A in the same order, and the $(n+1)$th column is b. For example,

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix}, \ b = \begin{bmatrix} 4 \\ 5 \end{bmatrix} \quad \Rightarrow \quad [A|b] = \left[\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 2 & 3 & 1 & 5 \end{array}\right].$$

Theorem 4.4. *Let $A \in \mathbb{F}^{m\times n}$ and $b \in \mathbb{F}^{m\times 1}$. Then the following statements are true.*

(1) *$Ax = b$ has a solution iff* $\operatorname{rank}([A|b]) = \operatorname{rank}(A)$. → consistent

(2) *If u is a particular solution of $Ax = b$, then each solution of $Ax = b$ is given by $u + y$, where y is a solution of the homogeneous system $Ax = 0$.*

(3) *If $[A'|b']$ is obtained from $[A|b]$ by a finite sequence of elementary row operations, then each solution of $Ax = b$ is a solution of $A'x = b'$, and vice versa.*

(4) *If $r = \operatorname{rank}([A|b]) = \operatorname{rank}(A) < n$, then there are $n - r$ unknowns which can take arbitrary values and other r unknowns be determined from the values of these $n - r$ unknowns.*

↳ infinite solutions

(5) *$Ax = b$ has a unique solution iff* $\operatorname{rank}([A|b]) = \operatorname{rank}(A) = n$. → 1 soln

(6) *If $m = n$, then $Ax = b$ has a unique solution iff* $\det(A) \neq 0$.

If z is a solution of $Ax=b$ ⇒ $Sol(A|b) = \{z+y : y \in Sol(A|0)\}$
$Ax=0$ → associated homogeneous system

Proof: (1) $Ax = b$ has a solution iff $b \in R(A)$ iff b is a linear combination of columns of A iff $\text{rank}([A|b]) = \text{rank}(A)$.

(2) Let u be a *particular solution* of $Ax = b$. Then $Au = b$. Now, y is a solution of $Ax = b$ iff $Ay = b$ iff $Ay = Au$ iff $A(y - u) = 0$ iff $y - u$ is a solution of $Ax = 0$.

(3) If $[A'|b']$ has been obtained from $[A|b]$ by a finite sequence of elementary row operations, then $A' = EA$ and $b' = Eb$, where E is the product of corresponding elementary matrices. Thus E is invertible. Now, $A'x = b'$ iff $EAx = Eb$ iff $Ax = E^{-1}Eb = b$.

(4) Due to (2), consider solving the corresponding homogeneous system. Let $\text{rank}(A) = r < n$. Due to (3), assume that A is in RREF. There are r number of pivots in A and $m - r$ number of zero rows. Omit all the zero rows. It does not affect the solutions. The $n - r$ unknowns which do not correspond to pivots can take arbitrary values, and the unknowns corresponding to pivots can be expressed in terms of these $n - r$ unknowns.

(5) It follows from (1) and (4).

(6) Notice that for a matrix $A \in \mathbb{F}^{n\times n}$, it is invertible iff $\text{rank}(A) = n$ iff $\det(A) \neq 0$. Then the statement follows from (5). □

A system of linear equations $Ax = b$ is said to be **consistent** iff $\text{rank}([A|b]) = \text{rank}(A)$.

Theorem 4.4(1) says that only consistent systems have solutions. Conversely, if a system has a solution, then the system must be consistent. The statement in Theorem 4.4(5) is sometimes informally stated as follows:

A consistent system has $n - \text{rank}(A)$ number of linearly independent solutions.

Exercises for § 4.6

1. Show that a linear system $Ax = b$ is solvable iff b is a linear combination of columns of A.
2. Let $A \in \mathbb{C}^{n\times n}$. Show that A is invertible iff for each $B \in \mathbb{C}^{n\times n}$, $AB = 0$ implies that $B = 0$.
3. Consider the linear system $Ax = b$, where $A \in \mathbb{F}^{m\times n}$ and $\text{rank}(A) = r$. Write explicit conditions on m, n, r so that the system has
   (a) no solution (b) unique solution (c) infinite number of solutions

4.7 Gauss-Jordan elimination

Gauss-Jordan elimination is an application of converting the augmented matrix to its row reduced echelon form for solving linear systems.

To determine whether a system of linear equations is consistent or not, we convert the augmented matrix $[A|b]$ to its RREF. In the RREF, if an entry in the b portion has become a pivot, then the system is inconsistent; otherwise, the system is consistent.

Example 4.8. Is the following system of linear equations consistent?

$$\begin{aligned} 5x_1 + 2x_2 - 3x_3 + x_4 &= 7 \\ x_1 - 3x_2 + 2x_3 - 2x_4 &= 11 \\ 3x_1 + 8x_2 - 7x_3 + 5x_4 &= 8 \end{aligned}$$

We take the augmented matrix and reduce it to its row reduced echelon form by elementary row operations.

$$\left[\begin{array}{cccc|c} 5 & 2 & -3 & 1 & 7 \\ 1 & -3 & 2 & -2 & 11 \\ 3 & 8 & -7 & 5 & 8 \end{array}\right] \xrightarrow{R1} \left[\begin{array}{cccc|c} \boxed{1} & 2/5 & -3/5 & 1/5 & 7/5 \\ 0 & -17/5 & 13/5 & -11/5 & 48/5 \\ 0 & 34/5 & -26/5 & 22/5 & -19/5 \end{array}\right]$$

$$\xrightarrow{R2} \left[\begin{array}{cccc|c} \boxed{1} & 0 & -5/17 & -1/17 & 43/17 \\ 0 & \boxed{1} & -13/17 & 11/17 & -48/17 \\ 0 & 0 & 0 & 0 & \boxed{77/5} \end{array}\right]$$

Here, $R_1 = E_{1/5}[1]$, $E_{-1}[2,1]$, $E_{-3}[3,1]$ and $R2 = E_{-5/17}[2]$, $E_{-2/5}[1,2]$, $E_{-34/5}[3,2]$. Since an entry in the b portion has become a pivot, the system is inconsistent. In fact, you can verify that the third row in A is simply first row minus twice the second row, whereas the third entry in b is not the first entry minus twice the second entry. Therefore, the system is inconsistent.

Example 4.9. We change the last equation in the previous example to make it consistent. We consider the new system

$$\begin{aligned} 5x_1 + 2x_2 - 3x_3 + x_4 &= 7 \\ x_1 - 3x_2 + 2x_3 - 2x_4 &= 11 \\ 3x_1 + 8x_2 - 7x_3 + 5x_4 &= -15 \end{aligned}$$

The reduction to echelon form will change the changed entry as follows:

$$\left[\begin{array}{cccc|c} 5 & 2 & -3 & 1 & 7 \\ 1 & -3 & 2 & -2 & 11 \\ 3 & 8 & -7 & 5 & -15 \end{array}\right] \xrightarrow{R1} \left[\begin{array}{cccc|c} \boxed{1} & 2/5 & -3/5 & 1/5 & 7/5 \\ 0 & -17/5 & 13/5 & -11/5 & 48/5 \\ 0 & 34/5 & -26/5 & 22/5 & -96/5 \end{array}\right]$$

$$\xrightarrow{R2} \left[\begin{array}{cccc|c} \boxed{1} & 0 & -5/17 & -1/17 & 43/17 \\ 0 & \boxed{1} & -13/17 & 11/17 & -48/17 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

with $R_1 = E_{1/5}[1]$, $E_{-1}[2,1]$, $E_{-3}[3,1]$ and $R2 = E_{-5/17}[2]$, $E_{-2/5}[1,2]$, $E_{-34/5}[3,2]$ as the row operations. This expresses the fact that the third equation is redundant. Now, solving the new system in row reduced echelon form is easier. Writing as linear equations, we have

$$\begin{aligned} \boxed{1}\, x_1 \qquad & -\frac{5}{17}x_3 - \frac{1}{17}x_4 = \frac{43}{17} \\ \boxed{1}\, x_2 \quad & -\frac{13}{17}x_3 + \frac{11}{17}x_4 = -\frac{48}{17} \end{aligned}$$

The unknowns corresponding to the pivots are called the **basic variables** and the other unknowns are called the **free variable**. The number of basic variables is equal to the number of pivots, which is the rank of the system matrix. By assigning the free variables x_i to any arbitrary values, say, α_i, the basic variables can be evaluated in terms of α_i.

In the above reduced system, the basic variables are x_1 and x_2; and the unknowns x_3, x_4 are free variables. We assign x_3 to α and x_4 to β. Then we have

$$x_1 = \frac{43}{17} + \frac{5}{17}\alpha + \frac{1}{17}\beta, \quad x_2 = -\frac{48}{17} + \frac{13}{17}\alpha - \frac{11}{17}\beta.$$

Therefore, any vector $y \in \mathbb{F}^{4\times 1}$ in the form

$$y := \begin{bmatrix} \frac{43}{17} + \frac{5}{17}\alpha + \frac{1}{17}\beta \\ -\frac{48}{17} + \frac{13}{17}\alpha - \frac{11}{17}\beta \\ \alpha \\ \beta \end{bmatrix} \quad \text{for } \alpha, \beta \in \mathbb{F}$$

is a solution of the linear system. Observe that

$$y = \begin{bmatrix} 43/17 \\ -48/17 \\ 0 \\ 0 \end{bmatrix} + \alpha \begin{bmatrix} 5/17 \\ 13/17 \\ 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 1/17 \\ -11/17 \\ 0 \\ 1 \end{bmatrix}.$$

Here, the first vector is a particular solution of the original system. The two vectors

$$\begin{bmatrix} 5/17 \\ 13/17 \\ 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1/17 \\ -11/17 \\ 0 \\ 1 \end{bmatrix}$$

are linearly independent solutions of the corresponding homogeneous system. There should be exactly two such linearly independent solutions of the homogeneous system, because the nullity of the system matrix is the number of unknowns minus its rank, which is $4 - 2 = 2$.

There are variations of Gauss-Jordan elimination. Instead of reducing the augmented matrix to its row reduced echelon form, if we reduce it to another intermediary form, called *the row echelon form*, then we obtain the method of **Gaussian elimination**. In the row echelon form, we do not require the entries above a pivot to be 0; also the pivots need not be equal to 1. In that case, we will require back-substitution in solving a linear system. To illustrate this process, we redo Example 4.9 starting with the augmented matrix, as follows:

$$\left[\begin{array}{cccc|c} \boxed{5} & 2 & -3 & 1 & 7 \\ 1 & -3 & 2 & -2 & 11 \\ 3 & 8 & -7 & 5 & -15 \end{array}\right] \xrightarrow{R1} \left[\begin{array}{cccc|c} \boxed{5} & 2 & -3 & 1 & 7 \\ 0 & \boxed{-17/5} & 13/5 & -11/5 & 48/5 \\ 0 & 34/5 & -26/5 & 22/5 & -96/5 \end{array}\right]$$

$$\xrightarrow{E_2[3,2]} \left[\begin{array}{cccc|c} \boxed{5} & 2 & -3 & 1 & 7 \\ 0 & \boxed{-17/5} & 13/5 & -11/5 & 48/5 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

$$
\begin{aligned}
x_1+x_2+x_3+x_4-3x_5&=6\\
2x_1+3x_2+x_3+4x_4-9x_5&=17\\
x_1+x_2+x_3+2x_4-5x_5&=8\\
2x_1+2x_2+2x_3+3x_4-8x_5&=14
\end{aligned}
\Rightarrow
\left[\begin{array}{ccccc|c}
1&1&1&1&-3&6\\
2&3&1&4&-9&17\\
1&1&1&2&-5&8\\
2&2&2&3&-8&14
\end{array}\right]
$$

consistent

$\text{Rank}(A|B) = \text{Rank}(A) = 3 < (n=5)$

$\Downarrow$

infinite solutions

RREF (A|B)

$$
\left[\begin{array}{ccccc|c}
\boxed{1}&0&2&0&-2&3\\
0&\boxed{1}&-1&0&1&1\\
0&0&0&\boxed{1}&-2&2\\
0&0&0&0&0&0
\end{array}\right]
$$

RREF (A)

RREF corresponds to linear system:

$$
\begin{aligned}
x_1+2x_3-2x_5&=3\\
x_2-x_3+x_5&=1\\
x_4-2x_5&=2
\end{aligned}
\qquad
\begin{aligned}
x_3&=\alpha\\
x_5&=\beta
\end{aligned}
\Big\}\ \text{not pivotal columns}
$$

$$
\begin{aligned}
x_1&=3-2\alpha+2\beta\\
x_2&=1+\alpha-\beta\\
x_4&=2+2\beta
\end{aligned}
\Rightarrow
x=\begin{bmatrix}3-2\alpha+2\beta\\1+\alpha-\beta\\\alpha\\2+2\beta\\\beta\end{bmatrix}
=\begin{bmatrix}3\\1\\0\\2\\0\end{bmatrix}
+\alpha\begin{bmatrix}-2\\1\\1\\0\\0\end{bmatrix}
+\beta\begin{bmatrix}2\\-1\\0\\2\\1\end{bmatrix}
$$

particular solution

linear independent solutions of homogeneous system

$$x_1 - x_2 + 2x_3 - 2x_4 = 7$$
$$4x_1 + 3x_3 + x_4 = 9$$
$$2x_1 - 5x_2 + x_3 = -2$$
$$3x_1 - x_2 - x_3 + 2x_4 = -2$$

$$\Rightarrow \begin{bmatrix} 1 & -1 & 2 & -2 & 7 \\ 4 & 0 & 3 & 1 & 9 \\ 2 & -5 & 1 & 0 & -2 \\ 3 & -1 & 1 & 2 & -2 \end{bmatrix}$$

$\text{Rank}(A) = \text{Rank}(A|B) = 4 = n$

$\Rightarrow$ unique solution

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & \frac{10}{9} \\ 0 & 1 & 0 & 0 & 11/9 \\ 0 & 0 & 1 & 0 & 17/9 \\ 0 & 0 & 0 & 1 & -10/9 \end{array}\right]$$

$x_1 = 10/9 \quad x_2 = 11/9$

$x_3 = 17/9 \quad x_4 = -10/9$

Here, $R_1 = E_{-1/5}[2,1]$, $E_{-3/5}[3,1]$. The augmented matrix is now in row echelon form. It is a consistent system, since no entry in the b portion is a pivot. The pivots say that x_1, x_2 are basic variables and x_3, x_4 are free variables. We assign x_3 to α and x_4 to β. Writing in equations form, we have

$$x_1 = 7 - 2\,x_2 + 3\,\alpha - \beta, \quad x_2 = -\frac{5}{17}\Big(\frac{48}{5} - \frac{13}{5}\alpha + \frac{11}{5}\beta\Big).$$

First we determine x_2 and then back-substitute. We obtain

$$x_1 = \quad \frac{43}{17} + \frac{5}{17}\alpha + \frac{1}{17}, \quad x_2 = -\frac{48}{17} + \frac{13}{17}\alpha - \frac{11}{17}\beta, \quad x_3 = \alpha, \quad x_4 = \beta.$$

As you see we end up with the same set of solutions as in Gauss-Jordan elimination.

Exercises for § 4.7

1. Using Gauss-Jordan elimination, and also by Gaussian elimination, solve the following linear systems:

   (a) $3w + 2x + 2y - z = 2,\ 2x + 3y + 4z = -2,\ y - 6z = 6.$

   (b) $w + 4x + y + 3z = 1,\ 2x + y + 3z = 0,\ w + 3x + y + 2z = 1,\ 2x + y + 6z = 0.$

   (c) $w - x + y - z = 1,\ w + x - y - z = 1,\ w - x - y + z = 2,\ 4w - 2x - 2y = 1.$

2. Show that the following linear system $x + y + kz = 1,\ x - y - z = 2,\ 2x + y - 2z = 3$ has no solution for $k = 1$, and has a unique solution for each $k \neq 1$.

Chapter 3
Matrix as a Linear Map



3.1 Subspace and Span

Recall that $\mathbb{F}$ stands for either $\mathbb{R}$ or $\mathbb{C}$; and $\mathbb{F}^n$ is either $\mathbb{F}^{1\times n}$ or $\mathbb{F}^{n\times 1}$. Also, recall that a typical row vector in $\mathbb{F}^{1\times n}$ is written as $[a_1, \ldots, a_n]$ and a column vector in $\mathbb{F}^{n\times 1}$ is written as $[a_1, \ldots, a_n]^t$. Both the row and column vectors are written uniformly as $(a_1, \ldots, a_n)$; these constitute the vectors in $\mathbb{F}^n$. In $\mathbb{F}^n$, we have a special vector, called the zero vector, which we denote by 0; that is, $0 = (0, \ldots, 0)$. And if $x = (a_1, \ldots, a_n) \in \mathbb{F}^n$, then its additive inverse is $-x = (-a_1, \ldots, -a_n)$.

The operations of addition and scalar multiplication in $\mathbb{F}^n$ enjoy the following properties:
For all $u, v, w \in \mathbb{F}^n$, and for all $\alpha, \beta \in \mathbb{F}$,

1. $u + v = v + u$.
2. $(u + v) + w = u + (v + w)$.
3. $u + 0 = 0 + u = u$.
4. $u + (-u) = -u + u = 0$.
5. $\alpha(\beta u) = (\alpha\beta)u$.
6. $\alpha(u + v) = \alpha u + \alpha v$.
7. $(\alpha + \beta)u = \alpha u + \beta u$.
8. $1\,u = u$.
9. $(-1)u = -u$.
10. If $u + v = u + w$, then $v = w$.
11. If $\alpha u = 0$, then $\alpha = 0$ or $u = 0$.

It so happens that the last three properties follow from the earlier ones. Any nonempty set where the two operations of addition and scalar multiplication are defined, and which enjoy the first eight properties above, is called a *vector space* over $\mathbb{F}$. In this sense, $\mathbb{F}^n$, that is, both $\mathbb{F}^{1\times n}$ and $\mathbb{F}^{n\times 1}$, are vector spaces over $\mathbb{F}$. In such a general setting if a nonempty subset of a vector space is closed under both

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For real numbers, only 2 proper subspaces are there i.e {0} and $\mathbb{F}^n$

the operations, then it is called a *subspace*. We may not need these general notions. However, we define a subspace of our specific vector spaces.

Let V be a nonempty subset of $\mathbb{F}^n$. We say that V is a **subspace of** $\mathbb{F}^n$ iff for each scalar $\alpha \in \mathbb{F}$ and for each pair of vectors $u, v \in V$, we have both $u + v \in V$ and $\alpha u \in V$.

This is the meaning of the informal phrase: V is closed under addition and scalar multiplication. It is easy to see that a nonempty subset V of $\mathbb{F}^n$ is a subspace of $\mathbb{F}^n$ iff the following single condition is satisfied:

for each $\alpha \in \mathbb{F}$ and for all vectors u, v from V, $\alpha u + v \in V$.

Z (integers) → not a Q (rational) → subspace

Subspace examples

Example 3.1 1. $\{0\}$ and $\mathbb{F}^n$ are subspaces of $\mathbb{F}^n$ → trivial subspaces

2. Let $V = \{(a, b, c) : 2a + 3b + 5c = 0, a, b, c \in \mathbb{F}\}$. Clearly, $(0, 0, 0) \in V$. So, $V \neq \varnothing$. If $(a_1, b_1, c_1),\ (a_2, b_2, c_2) \in V$, then

$$2a_1 + 3b_1 + 5c_1 = 0, \quad 2a_2 + 3b_2 + 5c_2 = 0.$$

Adding them, we have $2(a_1 + a_2) + 3(b_1 + b_2) + 5(c_1 + c_2) = 0$. Therefore, $(a_1, b_1, c_1) + (a_2, b_2, c_2) \in V$.
If $\alpha \in \mathbb{F}$, then $2(\alpha a_1) + 3(\alpha b_1) + 5(\alpha c_1) = 0$. So, $\alpha(a_1, b_1, c_1) \in V$.
Hence, V is a subspace of $\mathbb{F}^3$.

3. Let $V = \{(a, b, c) : 2a + 3b + 5c = 1, a, b, c \in \mathbb{F}\}$. Clearly, $(1/2, 0, 0) \in V$. So, $V \neq \varnothing$. Also, $(0, 1/3, 0) \in V$.
We see that $(1/2, 0, 0) + (0, 1/3, 0) = (1/2, 1/3, 0)$. And

$$2 \times 1/2 + 3 \times 1/3 + 5 \times 0 = 2 \neq 1.$$

That is, $(1/2, 0, 0) + (0, 1/3, 0) \notin V$. Therefore, V is not a subspace of $\mathbb{F}^3$.
Also, notice that $2\ (1/2, 0, 0) \notin V$.

4. Let $\beta_1, \ldots, \beta_n \in \mathbb{F}$. Let $V = \{[a_1, \ldots, a_n] : \beta_1 a_1 + \cdots + \beta_n a_n = 0\}$. It is easy to check that V is a subspace of $\mathbb{F}^{1\times n}$.

5. Let $\alpha, \beta_1, \ldots, \beta_n \in \mathbb{F}$ with $\alpha \neq 0$. Then, the subset $V = \{[a_1, \ldots, a_n] : \beta_1 a_1 + \cdots + \beta_n a_n = \alpha\}$ is not a subspace of $\mathbb{F}^{1\times n}$. Why? □

Verify that in a subspace V, all the properties (1–8) above hold true. This is the reason we call such a nonempty subset as a subspace. Further, if U and V are subspaces of $\mathbb{F}^n$ and $U \subseteq V$, then we say that U is a subspace of V.

A singleton with a nonzero vector is not a subspace. For example, $\{(1, 1)\}$ is not a subspace of $\mathbb{F}^2$ since $2(1, 1)$ is not an element of this set.

What about the set $\{\alpha(1, 1) : \alpha \in \mathbb{F}\}$? Take any two vectors from this set, say, $\alpha(1, 1)$ and $\beta(1, 1)$. Let $\gamma \in \mathbb{F}$. Now,

$$\gamma(\alpha(1, 1)) + \beta(1, 1) = (\gamma\alpha + \beta)(1, 1)$$

is an element of the set. Therefore, the set is a subspace of $\mathbb{F}^2$. Notice that this set is the set of all linear combinations of the vector $(1, 1)$.

V is a subspace of F^n and $S \subseteq V \Rightarrow S$ is said to be spanning subset of V / S spans V if V = span(S)

Recall that a linear combination of vectors $v_1, \ldots, v_m$ is any vector in the form

$$\alpha_1 v_1 + \cdots + \alpha_m v_m$$

for scalars $\alpha_1, \ldots, \alpha_m$. If S is any nonempty subset of $\mathbb{F}^n$, we define span(S) as the set of all linear combinations of finite number of vectors from S. That is,

$$\text{span}(S) = \{\alpha_1 v_1 + \cdots + \alpha_m v_m : \alpha_1, \ldots, \alpha_m \in \mathbb{F}, v_1, \ldots, v_m \in S \text{ for an } m \in \mathbb{N}\}.$$

Also, we define span$(\varnothing) = \{0\}$. We read span(S) as **span** of S.

When $S = \{v_1, \ldots, v_m\}$, we also write span(S) as span$\{v_1, \ldots, v_m\}$. Thus,

$$\text{span}\{v_1, \ldots, v_m\} = \{\alpha_1 v_1 + \cdots + \alpha_m v_m : \alpha_1, \ldots, \alpha_m \in \mathbb{F}\}.$$

For instance, $v_1 + \cdots + v_m$ and $v_1 + 5v_2$ are in span$\{v_1, \ldots, v_m\}$. In the first case, each α_i is equal to 1, whereas in the second case, $\alpha_1 = 1, \alpha_2 = 5$ and all other α_is are 0.

Also, for $S \neq \varnothing$, we may write span(S) in the following way:

$$\begin{aligned}
\text{span}(S) &= \cup_{m=1}^{\infty} \{\alpha_1 v_1 + \cdots + \alpha_m v_m : \alpha_1, \ldots, \alpha_m \in \mathbb{F}, v_1, \ldots, v_m \in S\} \\
&= \cup_{m=1}^{\infty} \text{span}\{v_1, \ldots, v_m\} \text{ for } v_1, \ldots, v_m \in S \\
&= \cup_{m=1}^{\infty} \text{span}(A) \text{ for } A \subseteq S \text{ with } |A| = m.
\end{aligned}$$

Here, $|A|$ means the number of elements of the set A. Further, when we speak of a set of vectors, it is implicitly assumed that the set is a subset of $\mathbb{F}^n$ for some n.

Example 3.2 In $\mathbb{R}^2$, $U = \text{span}\{(1, 1)\} = \{(a, a) : a \in \mathbb{R}\}$ is subspace of $\mathbb{R}^2$; it is the straight line that passes through the origin with slope 1.

Again, let $V = \text{span}\{(1, 1), (1, 2)\} = \{(\alpha(1, 1) + \beta(1, 2) : \alpha, \beta \in \mathbb{R}\}$, in $\mathbb{R}^2$. Any vector in V can be written as $u + v$, where u is a vector on the straight line $y = x$, and v is vector on the straight line $y = 2x$. Here, a vector on the straight line means vector directed from the origin to a point on the straight line. It seems any vector in the plane can be written as $u + v$ for some such u and v. To show that this is the case, we try writing any point $(a, b) \in \mathbb{R}^2$ as $u + v$. So, suppose $(a, b) = \alpha(1, 1) + \beta(1, 2)$. Then, $a = \alpha + \beta, b = \alpha + 2\beta$. Solving, we obtain $\beta = b - a, \alpha = 2a - b$. We may now verify that

$$(a, b) = (2a - b)(1, 1) + (b - a)(1, 2).$$

Therefore, $\mathbb{R}^2 \subseteq V$. On the other hand, $V \subseteq \mathbb{R}^2$, since each linear combination of $(1, 1)$ and $(1, 2)$ is in $\mathbb{R}^2$. Therefore, $V = \mathbb{R}^2$.

Notice that both U and V are subspaces of $\mathbb{R}^2$. Further, U is a *proper subspace* of $\mathbb{R}^2$, whereas $V = \mathbb{R}^2$. □

Theorem 3.1 *Let $S \subseteq \mathbb{F}^n$. Then, $S \subseteq$ span(S), and* span(S) *is a subspace of $\mathbb{F}^n$.*

Proof Any vector $u \in S$ is a linear combination of vectors from S, for $u = 1\,u$. Thus, $S \subseteq \operatorname{span}(S)$.

Now, if $S = \varnothing$, then $\operatorname{span}(S) = \{0\}$. And if $S \neq \varnothing$, then $S \subseteq \operatorname{span}(S)$ implies that $\operatorname{span}(S) \neq \varnothing$. In any case, $\operatorname{span}(S)$ is nonempty.

If $u, v \in \operatorname{span}(S)$, then both of them are linear combinations of vectors from S. So, $\alpha u + v$ is also a linear combination of vectors from S for any scalar α. Therefore, $\operatorname{span}(S)$ is a subspace of $\mathbb{F}^n$. ■

We see that $\operatorname{span}(\operatorname{span}(S)) = \operatorname{span}(S)$. (How?) In general, span of any subspace is the subspace itself.

Let V be a subspace of $\mathbb{F}^n$, and let $S \subseteq V$. We say that S is a **spanning subset of** V, or that S **spans** V iff $V = \operatorname{span}(S)$. In this case, each vector in V can be expressed as a linear combination of vectors from S; we informally say that the vectors in S span V.

Notice that $(2, 2) \in \operatorname{span}\{(1, 1)\}$. So, $\operatorname{span}\{(1, 1), (2, 2)\} = \operatorname{span}\{(1, 1)\}$. As in Example 3.2, we have in $\mathbb{F}^2$,

$$\operatorname{span}\{(1, 1), (2, 2)\} = \operatorname{span}\{(1, 1)\} \neq \mathbb{F}^2, \quad \operatorname{span}\{(1, 1), (1, 2)\} = \mathbb{F}^2.$$

Further, the vectors $(1, 1), (1, 2), (1, -1)$ also span $\mathbb{F}^2$. In fact, since the first two vectors span $\mathbb{F}^2$, any list of vectors from $\mathbb{F}^2$ containing these two will also span $\mathbb{F}^2$.

Similarly, the vectors $e_1, \ldots, e_n$ in $\mathbb{F}^{n\times 1}$ span $\mathbb{F}^{n\times 1}$, where e_i is the column vector in $\mathbb{F}^{n\times 1}$ whose ith component is 1 and all other components are 0.

In this terminology, vectors $v_1, \ldots, v_n$ are linearly dependent iff one of the vectors in this list is in the span of the rest. If no vector in the list is in the span of the rest, then the vectors are linearly independent.

Exercises for Sect. 3.1

1. Let V be a nonempty subset of $\mathbb{F}^n$. Prove the following:
   (a) V is a subspace of $\mathbb{F}^n$ iff for all $\alpha, \beta \in \mathbb{F}$ and for all vectors $u, v \in V$, $\alpha u + \beta v \in V$.
   (b) V is a subspace of $\mathbb{F}^n$ iff V with addition as in $\mathbb{F}^n$ and scalar multiplication as in $\mathbb{F}^n$ satisfies the first eight properties of a vector space, as mentioned in the text.
2. Let U be a subspace of V, and let V be a subspace of W, where W is a subspace of $\mathbb{F}^n$. Is U a subspace of W?
3. Let $u, v_1, v_2, \ldots, v_n$ be $n + 1$ distinct vectors in $\mathbb{F}^n$, $S_1 = \{v_1, v_2, \ldots, v_n\}$, and let $S_2 = \{u, v_1, v_2, \ldots, v_n\}$. Prove that $\operatorname{span}(S_1) = \operatorname{span}(S_2)$ iff $u \in \operatorname{span}(S_1)$.
4. Let A and B be subsets of $\mathbb{F}^n$. Prove or disprove the following:
   (a) A is a subspace of $\mathbb{F}^n$ if and only if $\operatorname{span}(A) = A$.
   (b) If $A \subseteq B$, then $\operatorname{span}(A) \subseteq \operatorname{span}(B)$.
   (c) $\operatorname{span}(A \cup B) = \{u + v : u \in \operatorname{span}(A),\ v \in \operatorname{span}(B)\}$.
   (d) $\operatorname{span}(A \cap B) \subseteq \operatorname{span}(A) \cap \operatorname{span}(B)$.
   (e) $\operatorname{span}(A) \cap \operatorname{span}(B) \subseteq \operatorname{span}(A \cap B)$.

5. Let $S \subseteq U \subseteq \mathbb{F}^n$, where U is a subspace of $\mathbb{F}^n$. Suppose for any subspace V of $\mathbb{F}^n$, if $S \subseteq V$ then $U \subseteq V$. Prove that $U = \text{span}(S)$.
6. Let A and B be subsets of $\mathbb{F}^n$. Prove or disprove: $A \cup B$ is linearly independent iff $\text{span}(A) \cap \text{span}(B) = \{0\}$.

3.2 Basis and Dimension

Suppose we are given with a subset S of a subspace V of $\mathbb{F}^n$. The subset S may or may not span V. If it spans V, it is possible that it has a proper subset which also spans V. For instance,

$$S = \{[1, 2, -3], [1, 0, -1], [2, -4, 2], [0, -2, 2]\}$$

spans the subspace $V = \{[a, b, c] : a + b + c = 0\}$ of $\mathbb{F}^{1\times 3}$. Also, the subset

$$\{[1, 2, -3], [1, 0, -1], [2, -4, 2]\}$$

of S spans the same subspace V. Notice that S is linearly dependent. Reason:

$$[0, -2, 2] = (-1)[1, 2, -3] + [1, 0, -1].$$

On the other hand, the linearly independent subset $\{[1, 2, -3]\}$ of S does not span V. For instance,

$$[1, 0, 1] \in V, \quad [1, 0, -1] \neq \alpha[1, 2, -3], \quad \text{for any } \alpha \in \mathbb{F}.$$

That is, a spanning subset may be superfluous and a linearly independent set may be deficient. A linearly independent set which also spans a subspace may be just adequate in spanning the subspace.

Let V be a subspace of $\mathbb{F}^n$. Let B be a set of vectors from V. We say that the set B is a **basis** of V iff B is linearly independent and B spans V. Also, we define $\varnothing$ as the basis for the zero subspace $\{0\}$.

In what follows, we consider ordered sets, and the ordering of vectors in a set is shown by the way they are written. For instance, in the ordered set $\{v_1, v_2, v_3\}$, the vector v_1 is the first vector, v_2 is the second, and v_3 is the third, whereas in $\{v_2, v_3, v_1\}$, the vector v_2 is the first, v_3 is the second, and v_1 is the third. We assume implicitly that *each basis is an ordered set*.

It follows from Theorem 2.5 that each basis for a subspace of $\mathbb{F}^n$ has at most n vectors.

Example 3.3 1. It is easy to check that $B = \{e_1, \ldots, e_n\}$ is a basis of $\mathbb{F}^{n\times 1}$. Similarly, $E = \{e_1^t, \ldots, e_n^t\}$ is a basis of $\mathbb{F}^{1\times n}$. These are the *standard bases* of the spaces.

$V \subseteq \mathbb{F}^n$ be a subspace; $B \subseteq V$ is basis of V iff (i) B is linearly independent (ii) $\text{span}(B) = V$

2. We show that $B = \{[1, 2, -3], [1, 0, -1]\}$ is a basis of

$$V = \{[a, b, c] : a + b + c = 0, a, b, c \in \mathbb{F}\}.$$

First, $B \subseteq V$. Second, any vector in V is of the form $[a, b, -a - b]$ for $a, b \in \mathbb{F}$. Now,

$$[a, b, -a - b] = \tfrac{b}{2}[1, 2, -3] + \left(a - \tfrac{b}{2}\right)[1, 0, -1]$$

shows that $\text{span}(B) = V$. For linear independence, suppose

$$\alpha[1, 2, -3] + \beta[1, 0, -1] = [0, 0, 0].$$

Then, $\alpha + \beta = 0, 2\alpha = 0, -3\alpha - \beta = 0$. It implies that $\alpha = \beta = 0$.

3. Also, $E = \{[1, -1, 0], [0, 1, -1]\}$ is another basis for the subspace V of $\mathbb{F}^{1\times 3}$ in (2). □

Let B be a basis of a subspace V of $\mathbb{F}^n$. If $C \subseteq V$ is any proper superset of B, then any vector in $C \setminus B$ is a linear combination of vectors from B. So, C is linearly dependent. On the other hand, if D is any proper subset of B, then each vector in $B \setminus D$ fails to be a linear combination of vectors from D. Otherwise, B would be linearly dependent. We thus say that

A basis is a maximal linearly independent set.
A basis is a minimal spanning set.

Of course, you can prove that a maximal linearly independent set is a basis, and a minimal spanning set is a basis. This would guarantee that each subspace of $\mathbb{F}^n$ has a basis. We take a more direct approach.

Theorem 3.2 *Each subspace of $\mathbb{F}^n$ has a basis with at most n vectors.*

Proof Let V be a subspace of $\mathbb{F}^n$. If $V = \{0\}$, then $\varnothing$ is a basis of V. Otherwise, choose a nonzero vector v_1 from V. Take $B = \{v_1\}$. If $V = \text{span}(B)$, then B is a basis of V. Else, there exists a nonzero vector, say, $v_2 \in V \setminus \text{span}(B)$. Update B to $B \cup \{v_2\}$. Notice that B is linearly independent. Continue this process to obtain larger and larger linearly independent sets in V. By Theorem 2.5, a linearly independent set in V cannot have more than n vectors. Thus, the process terminates with a basis of V having at most n vectors. ■

The zero subspace $\{0\}$ has a single basis $\varnothing$. But other subspaces do not have a unique basis. For instance, the subspace V in Example 3.3 has at least two bases. However, something remains same in all these bases. In that example, both the bases have exactly two vectors.

Theorem 3.3 *Let V be a subspace of $\mathbb{F}^n$. All bases of V have the same number of vectors.*

Proof Without loss of generality, let V be a subspace of $\mathbb{F}^{1\times n}$. By Theorem 2.5, a basis of V can have at most n vectors. So, let $\{u_1, \ldots, u_k\}$ and $\{v_1, \ldots, v_m\}$ be bases for V, where $k \le n$ and $m \le n$. Construct the matrix $A \in \mathbb{F}^{(k+m)\times n}$ by taking its rows as $u_1, \ldots, u_k, v_1, \ldots, v_m$ in that order.

The first k rows of A are linearly independent, and other rows are linear combinations of these k rows. Also, the last m rows of A are linearly independent and other rows are linear combinations of these m rows. By Theorem 2.6, $k = \text{rank}(A) = m$. ■

In view of Theorem 3.3, there exists a unique non-negative number associated with each subspace of $\mathbb{F}^n$, which is the number of vectors in any basis of the subspace.

Let V be a subspace of $\mathbb{F}^n$. The number of vectors in some (or any) basis for V is called the **dimension** of V. We write this number as $\dim(V)$ and also as $\dim V$.

Since $\{e_1, \ldots, e_n\}$ is a basis for $\mathbb{F}^{n\times 1}$, $\dim(\mathbb{F}^{n\times 1}) = n$. Similarly, $\dim(\mathbb{F}^{1\times n}) = n$. Remember that when we consider $\mathbb{C}^{n\times 1}$ or $\mathbb{C}^{1\times n}$, the scalars in any linear combination are complex numbers, and for $\mathbb{R}^{n\times 1}$ or $\mathbb{R}^{1\times n}$, the scalars are real numbers. Notice that $\dim(\{0\}) = \dim(\text{span}(\varnothing)) = 0$; the dimension of any subspace of $\mathbb{F}^n$ is at most n.

Example 3.4 The subspace $U := \{[a, b, c, d] : a - 2b + 3c = 0 = d + a, a, b, c, d \in \mathbb{F}\}$ may be written as

$2b-3c, b, c, -2b+3c$

$b[2,1,0,-2] + c[-3,0,1,3]$

$$\begin{aligned} U &= \{[a, b, c, d] : [2b - 3c, b, c, -2b + 3c] : b, c \in \mathbb{F}\} \\ &= \{b\,[2, 1, 0, -2] + c\,[-3, 0, 1, 3] : b, c \in \mathbb{F}\}. \end{aligned}$$

The vectors $[2, 1, 0, -2]$ and $[-3, 0, 1, 3]$ are linearly independent. Therefore, U has a basis $\{[2, 1, 0, -2], [-3, 0, 1, 3]\}$. So, $\dim(U) = 2$. □

Recall that $|B|$ stands for the number of elements in a set B. For any subspace V of $\mathbb{F}^n$, and any subset B of $\mathbb{F}^n$, the following statements should be obvious:

1. If $|B| < \dim(V)$, then $\text{span}(B)$ is a proper subspace of V.
2. If $|B| > \dim(V)$, then B is linearly dependent.
3. If $|B| = \dim(V)$ and $\text{span}(B) = V$, then B is a basis of V.
4. If $|B| = \dim(V)$ and B is linearly independent, then B is a basis of V.
5. If U is a subspace of V, then $\dim(U) \le \dim(V) \le n$.
6. If B is a superset of a spanning set of V, then B is linearly dependent.
7. If B is a proper subset of a linearly independent subset of V, then B is linearly independent, and $\text{span}(B)$ is a proper subspace of V.
8. Each spanning set of V contains a basis of V.
9. Each linearly independent subset of V can be extended to a basis of V.

For (8)–(9), we may employ the same construction procedure as in the proof of Theorem 3.2. A statement equivalent to (9) is proved below.

Theorem 3.4 (Basis Extension Theorem) *Let V be a subspace of $\mathbb{F}^n$. Each basis of a subspace of V can be extended to a basis of V.*

Proof Let B be a basis for a subspace U of V. If $U = V$, then B is a basis of V. Else, let $v_1 \in V \backslash U$. Now, $B \cup \{v_1\}$ is linearly independent. If this set spans V, then it is a basis of V. Otherwise, let $v_2 \in V \backslash \text{span}(B \cup \{v_1\})$. If $B \cup \{v_1, v_2\}$ is not a basis of V, then let $v_3 \in V \backslash \text{span}(B \cup \{v_1, v_2\})$. Continue this process. The process terminates since $\dim(V) \le n$. Upon termination, we obtain a basis for V. ■

We can use elementary row operations to extract a basis for a subspace which is given in the form of span of some finite number of vectors. We write the vectors as row vectors, form a matrix A, and convert it to its RREF. Then, the pivotal rows of the RREF form the required basis. Also, those rows of A which have become the pivotal rows (monitoring row exchanges) form a basis.

Example 3.5 Let $U = \text{span}\{(1, 1, 1, 1), (2, 1, 0, 3), (-1, 0, 1, -2), (0, 3, 2, 1)\}$. Find a basis for the subspace U of $\mathbb{F}^4$.

We start with the matrix with these vectors as its rows and convert it to its RREF as follows:

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 1 & 0 & 3 \\ -1 & 0 & 1 & -2 \\ 0 & 3 & 2 & 1 \end{bmatrix} \xrightarrow{R1} \begin{bmatrix} \boxed{1} & 1 & 1 & 1 \\ 0 & -1 & -2 & 1 \\ 0 & 1 & 2 & -1 \\ 0 & 3 & 2 & 1 \end{bmatrix} \xrightarrow{R2} \begin{bmatrix} \boxed{1} & 0 & -1 & 2 \\ 0 & \boxed{1} & 2 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -4 & 4 \end{bmatrix}$$

$$\xrightarrow{R3} \begin{bmatrix} \boxed{1} & 0 & 0 & 1 \\ 0 & \boxed{1} & 0 & 1 \\ 0 & 0 & \boxed{1} & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Here, $R1 = E_{-2}[2, 1],\ E_1[3, 1]$; $R2 = E_{-1}[2],\ E_{-1}[1, 2],\ E_{-1}[3, 2],\ E_{-3}[4, 2]$; and $R3 = E[3, 4],\ E_{-1/4}[3],\ E_1[1, 3],\ E_{-2}[2, 3]$.

The pivotal rows show that $\{(1, 0, 0, 1), (0, 1, 0, 1), (0, 0, 1, -1)\}$ is a basis for the given subspace. Notice that only one row exchange has been done in this reduction process, which means that the third row in the RREF corresponds to the fourth vector and the fourth row corresponds to the third vector. Thus, the pivotal rows correspond to the first, second, and the fourth vector, originally. This says that a basis for the subspace is also given by $\{(1, 1, 1, 1), (2, 1, 0, 3), (0, 3, 2, 1)\}$.

The reduction process confirms that the third vector is a linear combination of the first, second, and the fourth. □

The method illustrated in the above example can be used in another situation. Suppose B is a basis for U, which is a subspace of $V \subseteq \mathbb{F}^n$. Assume that a basis E for V is also known. Then, we can extend B to a basis for V. Towards this, we may form a matrix with the vectors in B as row vectors and then place the vectors from

E as rows below those from B. Then, reduction of this matrix to RREF produces a basis for V, which is an extension of B.

Linear independence has something to do with invertibility of a square matrix. Suppose the rows of a matrix $A \in \mathbb{F}^{n\times n}$ are linearly independent. Then, the RREF of A has n number of pivots. That is, $\text{rank}(A) = n$. Consequently, A is invertible. On the other hand, if a row of A is a linear combination of other rows, then this row appears as a zero row in the RREF of A. That is, A is not invertible.

Considering A^t instead of A, we conclude that A^t is invertible iff the rows of A are linearly independent. However, A is invertible iff A^t is invertible. Therefore, A is invertible iff its columns are linearly independent.

Theorem 3.5 *A square matrix is invertible iff its rows are linearly independent iff its columns are linearly independent.*

From Theorem 3.5, it follows that an $n \times n$ matrix is invertible iff its rows form a basis for $\mathbb{F}^{1\times n}$ iff its columns form a basis for $\mathbb{F}^{n\times 1}$.

Exercises for Sect. 3.2

1. Let V be a subspace of $\mathbb{F}^n$, and let $B \subseteq V$. Prove the following:
   (a) B is a basis of V iff B is linearly independent and each proper superset of B is linearly dependent.
   (b) B is a basis of V iff B spans V and no proper subset of B spans V.
2. Prove statements in (1)–(7) listed after Example 3.4.
3. Let $\{x, y, z\}$ be a basis for a vector space V. Is $\{x + y, y + z, z + x\}$ also a basis for V?
4. Find a basis for the subspace $\{(a, b, c) \in \mathbb{R}^3 : a + b - 5c = 0\}$ of $\mathbb{R}^3$.
5. Find bases and dimensions of the following subspaces of $\mathbb{R}^5$:
   (a) $\{(a, b, c, d, e) \in \mathbb{R}^5 : a - c - d = 0\}$.
   (b) $\{(a, b, c, d, e) \in \mathbb{R}^5 : b = c = d, a + e = 0\}$.
   (c) $\text{span}\{(1, -1, 0, 2, 1), (2, 1, -2, 0, 0), (0, -3, 2, 4, 2), (3, 3, -4, -2, -1), (5, 7, -3, -2, 0)\}$.
6. Extend the set $\{(1, 0, 1, 0), (1, 0, -1, 0)\}$ to a basis of $\mathbb{R}^4$.
7. Let $u_1, v, w, x_1, x_2, x_3, x_4 \in \mathbb{R}^5$ satisfy $x_1 = u + 2v + 3w$, $x_2 = 2u - 3v + 4w$, $x_3 = -3u + 4v - 5w$, and $x_4 = -u + 6v + w$. Is $\{x_1, x_2, x_3, x_4\}$ linearly dependent?
8. Prove that the only proper subspaces of $\mathbb{R}^2$ are the straight lines passing through the origin.
9. Describe all subspaces of $\mathbb{R}^3$.
10. Let $A \in \mathbb{F}^{n\times n}$. Let $\{v_1, \ldots, v_n\}$ be any basis of $\mathbb{F}^{n\times 1}$. Prove that A is invertible iff $Ax = v_i$ has at least one solution for each $i = 1, \ldots, n$.
11. Let A be a matrix. Call a row (column) of A *redundant* if it can be expressed as a linear combination of other rows (columns). Show the following:

(a) If a redundant row of A is deleted, then the column rank of A remains unchanged.
(b) If a redundant column of A is deleted, then the row rank of A remains unchanged.
(c) The row rank of A is equal to the column rank of A.

3.3 Linear Transformations

Let $A \in \mathbb{F}^{m\times n}$. If $x \in \mathbb{F}^{n\times 1}$, then $Ax \in \mathbb{F}^{m\times 1}$. Thus, we may view the matrix A as a function from $\mathbb{F}^{n\times 1}$ to $\mathbb{F}^{m\times 1}$ which maps x to Ax. That is, the map $A : \mathbb{F}^{n\times 1} \to \mathbb{F}^{m\times 1}$ is given by

$$A(x) = Ax \quad \text{for } x \in \mathbb{F}^{n\times 1}.$$

Notice that instead of using another symbol, we write the map obtained this way from a matrix A as A itself. Due to the properties of matrix product, the following are true for the map A:

1. $A(u + v) = A(u) + A(v)$ for all $u, v \in \mathbb{F}^{n\times 1}$.
2. $A(\alpha v) = \alpha A(v)$ for all $v \in \mathbb{F}^{n\times 1}$ and for all $\alpha \in \mathbb{F}$.

In this manner, a matrix is considered as a *linear transformation*. In fact, any function A from a vector space to another (both over the same field) satisfying the above two properties is called a linear transformation or a *linear map*.

To see the connection between the matrix as a rectangular array and as a function, consider the values of the matrix A at the standard basis vectors $e_1, \ldots, e_n$ in $\mathbb{F}^{n\times 1}$. The column vector $e_j \in \mathbb{F}^{n\times 1}$ has the jth entry as 1 and all other entries 0. Let $A = [a_{ij}] \in \mathbb{F}^{m\times n}$. The product Ae_j is a vector in $\mathbb{F}^{m\times 1}$, whose ith entry is

$$a_{i1} \cdot 0 + \cdots + a_{i(j-1)} \cdot 0 + a_{ij} \cdot 1 + a_{i(j+1)} \cdot 0 + \cdots + a_{in} \cdot 0 = a_{ij}.$$

That is, $Ae_j = [a_{1j}, \ldots, a_{ij}, \ldots, a_{mj}]^t = A_{\star j}$, the jth column of A.

Observation 3.1 A matrix $A \in \mathbb{F}^{m\times n}$ is viewed as the linear transformation $A : \mathbb{F}^{n\times 1} \to \mathbb{F}^{m\times 1}$, where $A(e_j) = A_{\star j}$, the jth column of A for $1 \le j \le n$, and $A(v) = Av$ for each $v \in \mathbb{F}^{n\times 1}$.

The *range* of the linear transformation A is the set $R(A) = \{Ax : x \in \mathbb{F}^{n\times 1}\}$. If $y \in R(A)$, then there exists an $x = [\alpha_1, \ldots, \alpha_n]^t \in \mathbb{F}^{n\times 1}$ such that $y = Ax$. The vector x can be written as $x = \alpha_1 e_1 + \cdots + \alpha_n e_n$. Thus, such a $y \in R(A)$ is written as

$$y = Ax = \alpha_1 Ae_1 + \cdots + \alpha_n Ae_n.$$

Conversely, each vector $\alpha_1 Ae_1 + \cdots + \alpha_n Ae_n = A(\alpha_1 e_1 + \cdots + \alpha_n e_n) \in R(A)$. Hence,

$$\begin{aligned} R(A) &= \{\alpha_1 Ae_1 + \cdots + \alpha_n Ae_n : \alpha_1, \ldots, \alpha_n \in \mathbb{F}\} \\ &= \{\alpha_1 A_{\star 1} + \cdots + \alpha_n A_{\star n} : \alpha_1, \ldots, \alpha_n \in \mathbb{F}\} \\ &= \text{span of columns of } A. \end{aligned}$$

Therefore, $R(A)$ is a subspace of $\mathbb{F}^{m\times 1}$. We refer to $R(A)$ as the **range space** of A. Since $R(A)$ is the span of the columns of A, it is also called the *column space of* A. Its dimension is the *column rank* of A. We know that

$$\dim(R(A)) = \text{ the column rank of } A = \text{rank}(A).$$

Similarly, the subspace of $\mathbb{F}^{1\times n}$ which is spanned by the rows of A is called the *row space* of A. Notice that the nonzero rows in the RREF of A form a basis for the row space of A. The dimension of the row space is the *row rank* of A, which we know to be equal to $\text{rank}(A)$ also.

For an $m \times n$ matrix A, viewed as a linear transformation, the set of all vectors which map to the zero vector is denoted by $N(A)$. That is,

$$N(A) = \{x \in \mathbb{F}^{n\times 1} : Ax = 0\}.$$

We find that $N(A)$ is the set of all solutions of the linear system $Ax = 0$. Also,

if $u, v \in N(A)$ and $\alpha \in \mathbb{F}$, then $A(\alpha u + v) = \alpha Au + Av = 0$.

Therefore, $N(A)$ is a subspace of $\mathbb{F}^{n\times 1}$. We refer to $N(A)$ as the **null space** of A. The dimension of the null space is called the **nullity** of the matrix A and is denoted by $\text{null}(A)$. That is,

$$\text{null}(A) = \dim(N(A)).$$

Theorem 2.9 (4) implies that $\dim(R(A)) + \dim(N(A)) = n$. Since this will be used often, we mention it as a theorem. An alternate proof of this theorem is given in Problem 12.

Theorem 3.6 (Rank Nullity) *Let $A \in \mathbb{C}^{m\times n}$. Then,* $\dim(R(A)) + \dim(N(A)) = \text{rank}(A) + \text{null}(A) = n$.

Gauss–Jordan elimination process gives us the following mechanical way of construction of a basis for $N(A)$. First, we reduce A to its RREF B. Next, we throw away the zero rows at the bottom of B to obtain an $r \times n$ matrix C. If necessary, we insert zero rows in C to obtain an $n \times n$ matrix D so that the pivots become diagonal entries in D. Each diagonal entry of D which is on a non-pivotal column is now 0. We change each such 0 to -1. Call the new matrix as E. The non-pivotal columns in E form a basis for $N(A)$.

Example 3.6 Consider the system matrix in Example 2.9. We had its RREF with (boxed) pivots as shown below:

$$A = \begin{bmatrix} 5 & 2 & -3 & 1 \\ 1 & -3 & 2 & -2 \\ 3 & 8 & -7 & 5 \end{bmatrix} \longrightarrow \begin{bmatrix} \boxed{1} & 0 & -5/17 & -1/17 \\ 0 & \boxed{1} & -13/17 & 11/17 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \text{RREF}(A).$$

The first two columns in RREF(A) are the pivotal columns. So, the first two columns in A form a basis for $R(A)$. That is,

$$\text{a basis for } R(A) \text{ is } \{[5, 1, 3]^t, [2, -3, 8]^t\}.$$

For a basis of $N(A)$, notice that each pivot has the row index equal to the column index; so, we do not require to insert zero rows between pivotal rows. To make it a square matrix, we attach a zero row to the RREF at the bottom:

$$D = \begin{bmatrix} \boxed{1} & 0 & -5/17 & -1/17 \\ 0 & \boxed{1} & -13/17 & 11/17 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Then, we change the diagonal entries in the non-pivotal columns to -1. These changed non-pivotal columns form a basis for $N(A)$. That is,

$$\text{a basis for } N(A) \text{ is } \left\{\left[-\tfrac{5}{17}, -\tfrac{13}{17}, -1, 0\right]^t, \left[-\tfrac{1}{17}, \tfrac{11}{17}, 0, -1\right]^t\right\}.$$

Thus, $\dim(R(A)) + \dim(N(A)) = 4 =$ the number of columns in A. □

Example 3.7 Find bases for $R(A)$ and $N(A)$, where $A = \begin{bmatrix} 5 & 2 & -3 & 2 \\ 10 & 4 & -5 & 5 \\ -5 & -2 & 3 & -2 \end{bmatrix}$.

We reduce A to its RREF as in the following:

$$\begin{bmatrix} 5 & 2 & -3 & 2 \\ 10 & 4 & -5 & 5 \\ -5 & -2 & 3 & -2 \end{bmatrix} \longrightarrow \begin{bmatrix} \boxed{1} & 2/5 & -3/5 & 2/5 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} \boxed{1} & 2/5 & 0 & 1 \\ 0 & 0 & \boxed{1} & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

A basis for $R(A)$ is provided by the columns of A that correspond to the pivotal columns, that is, the first and the third columns of A. So,

$$\text{a basis for } R(A) = \{[5, 10, -5]^t, [-3, -5, 3]^t\}.$$

The second and fourth columns are linear combinations of these basis vectors, where the coefficients are provided by the entries in the non-pivotal columns of the RREF. That is, the second column of A is $\frac{2}{5}$ times the first column, and the fourth column is 1 times the first column plus 1 times the second column. Indeed, we may verify that

$$[2, 4, -2]^t = \tfrac{2}{5}[5, 10, -5]^t, \quad [2, 5, -2]^t = [5, 10, -5]^t + [-3, -5, 3]^t.$$

Towards computing a basis for $N(A)$, notice that the only zero row in the RREF is the third row. So, we delete it. Next, the pivot on the second row has the column index as 3. So, we insert a zero row between first and second rows. Next, we adjoin a zero row at the bottom to make it a square matrix. Finally, we change the diagonal entry of these new rows to -1. We then obtain the matrix

$$\begin{bmatrix} \boxed{1} & 2/5 & 0 & 1 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & \boxed{1} & 1 \\ 0 & 0 & 0 & -1 \end{bmatrix}.$$

A basis for $N(A)$ is provided by the non-pivotal columns. That is,

$$\text{a basis for } N(A) = \left\{[\tfrac{2}{5}, -1, 0, 0]^t, [1, 0, 1, -1]^t\right\}. \qquad \square$$

Exercises for Sect. 3.3

1. Let $A = \begin{bmatrix} 1 & 2 & 1 & 1 & 1 \\ 3 & 5 & 3 & 4 & 3 \\ 1 & 1 & 1 & 2 & 1 \\ 5 & 8 & 5 & 7 & 5 \end{bmatrix}$. Write $r = \text{rank}(A)$ and $k = \text{null}(A)$.
   (a) Find bases for $R(A)$ and $N(A)$, and then determine r and k.
   (b) Express suitable $4 - r$ rows of A as linear combinations of other r rows.
   (c) Express suitable $5 - r$ columns of A as linear combinations of other r columns.

2. Construct bases for (i) the row space, (ii) the column space, and the null space for each of the following matrices:

   (a) $\begin{bmatrix} 1 & 2 & 4 \\ 3 & 1 & 7 \\ 2 & 4 & 8 \end{bmatrix}$ (b) $\begin{bmatrix} -3 & 1 & -3 \\ 1 & 2 & 8 \\ 3 & -1 & 4 \\ 4 & -2 & 2 \end{bmatrix}$ (c) $\begin{bmatrix} 3 & 4 & 5 & 6 \\ 2 & 1 & 3 & 2 \\ 1 & 3 & -2 & 1 \end{bmatrix}$

3. Let $A \in \mathbb{F}^{m \times n}$. Consider A as the linear transformation $A : \mathbb{F}^{n \times 1} \to \mathbb{F}^{m \times 1}$ given by $A(x) = Ax$ for each $x \in \mathbb{F}^{n \times 1}$. Prove the following:
   (a) A is one-one iff $N(A) = \{0\}$.
   (b) A is one-one and onto iff A maps any basis onto a basis of $\mathbb{F}^{n \times 1}$.
   (c) A is one-one iff A is onto.

4. Let $A \in \mathbb{F}^{m \times n}$. Let $P \in \mathbb{F}^{m \times m}$ be invertible. Is it true that the RREF of PA is same as the RREF of A?

5. Let A and B be matrices having the same RREF. Then which of the following may be concluded? Explain.
   (a) $R(A) = R(B)$ (b) $N(A) = N(B)$
   (c) The row space of A is equal to the row space of B.

6. For $A \in \mathbb{F}^{m \times n}$ and $B \in \mathbb{F}^{n \times k}$, prove: $\text{rank}(AB) \le \min\{\text{rank}(A), \text{rank}(B)\}$.

3.4 Coordinate Vectors

Let $B = \{v_1, \ldots, v_m\}$ be a basis of a subspace V of $\mathbb{F}^n$. Recall that a basis is assumed to be an ordered set of vectors. Let $v \in V$. As B spans V, the vector v is a linear combination of vectors from B. Can there be two distinct linear combinations? Suppose there exist scalars $a_1, \ldots, a_m, b_1, \ldots, b_m \in \mathbb{F}$ such that

$$v = a_1 v_1 + \cdots + a_m v_m = b_1 v_1 + \cdots + b_m v_m.$$

Then, $(a_1 - b_1)v_1 + \cdots + (a_m - b_m)v_m = 0$. Since B is linearly independent, $a_1 = b_1, \ldots, a_m = b_m$. That is, such a linear combination is unique.

Conversely, suppose that each vector in V is written uniquely as a linear combination of $v_1, \ldots, v_m$. To show linear independence of these vectors, suppose that

$$\alpha_1 v_1 + \cdots + \alpha_m v_m = 0$$

for scalars $\alpha_1, \ldots, \alpha_m$. We also have

$$0\, v_1 + \cdots + 0\, v_m = 0.$$

From the uniqueness of writing the zero vector as a linear combination of $v_1, \ldots, v_m$, we conclude that $\alpha_1 = 0, \ldots, \alpha_m = 0$. Therefore, B is linearly independent.

We note down the result we have proved.

Theorem 3.7 *Let $B = \{v_1, \ldots, v_m\}$ be an ordered set of vectors from a subspace V of $\mathbb{F}^n$. B is a basis of V iff for each $v \in V$, there exists a unique vector $[\alpha_1, \ldots, \alpha_m]^t$ in $\mathbb{F}^{m\times 1}$ such that $v = \alpha_1 v_1 + \cdots + \alpha_m v_m$.*

Notice that in Theorem 3.7, $\dim(V) = m$.

Let $B = \{v_1, \ldots, v_m\}$ be a basis for a subspace V of $\mathbb{F}^n$. Let $v \in V$. For $v = \alpha_1 v_1 + \cdots + \alpha_m v_m$, the unique column vector $[\alpha_1, \ldots, \alpha_m]^t \in \mathbb{F}^{m\times 1}$ is called the **coordinate vector** of v with respect to the basis B; it is denoted by $[v]_B$. In this sense, we say that a basis provides a coordinate system in a subspace. Notice that the ordering of basis vectors is important to obtain the coordinate vector.

Example 3.8 Let $V = \{[a, b, c]^t : a + 2b + 3c = 0, a, b, c \in \mathbb{R}\}$. This subspace has a basis

$$B = \{[0, 3, -2]^t, [2, -1, 0]^t\}.$$

For $v = [-3, 0, 1]^t \in V$, we have

$$v = [-3, 0, 1]^t = -\tfrac{1}{2}[0, 3, -2]^t - \tfrac{3}{2}[2, -1, 0]^t.$$

Therefore, $[v]_B = \left[-\tfrac{1}{2}, -\tfrac{3}{2}\right]^t$.

If $w \in V$ has coordinate vector given by $[w]_B = [1, 1]^t$, then

$$w = 1\,[0, 3, -2]^t + 1\,[2, -1, 0]^t = [2, 2, -2]^t.$$

Since $\dim(V) = 2$, all the coordinate vectors are elements of $\mathbb{F}^{2\times 1}$. □

The coordinate vector induces a linear map from V onto $\mathbb{F}^{m\times 1}$. To see this, let $B = \{v_1, \ldots, v_m\}$ be a basis for a subspace V of $\mathbb{F}^n$. Suppose $u, v \in V$ have the coordinate vectors

$$[u]_B = \big[a_1, \ldots, a_m\big]^t, \quad [v]_B = \big[b_1, \ldots, b_m\big]^t.$$

It means $u = a_1 v_1 + \cdots a_m v_m$ and $v = b_1 v_1 + \cdots + b_m v_m$. Then,

$$\alpha u + v = (\alpha a_1 + b_1) + \cdots + (\alpha a_m + b_m) v_m.$$

That is, $[\alpha u + v]_B = \alpha [u]_B + [v]_B$. In a way, this is the *linear property* of the coordinate vector map. Also,

$$v_j = 0 \cdot v_1 + \cdots + 0 \cdot v_{j-1} + 1 \cdot v_j + 0 \cdot v_{j+1} + \cdots + 0 \cdot v_m.$$

Hence, $[v_j]_B = e_j$ for each $j \in \{1, \ldots, m\}$.

We summarize these simple facts about the coordinate vector.

Observation 3.2 Let $B = \{v_1, \ldots, v_m\}$ be a basis for a subspace V of $\mathbb{F}^n$. Let $\alpha_1, \ldots, \alpha_m \in \mathbb{F}$, and let $u_1, \ldots, u_m \in V$. Then,

1. $[\alpha_1 u_1 + \cdots + \alpha_m u_m]_B = \alpha_1 [u_1]_B + \cdots + \alpha_m [u_m]_B$.
2. $[v_j]_B = e_j$, the jth standard basis vector of $\mathbb{F}^{m\times 1}$ for $1 \le j \le m$.

Consider $\mathbb{F}^{n\times 1}$ as its own subspace. It has many bases. If E is the standard basis of $\mathbb{F}^{n\times 1}$ and $v = [a_1, \ldots, a_n]^t \in \mathbb{F}^{n\times 1}$, then $[v]_E = v$. When we change the basis E to another, say, B, then how does $[v]_B$ look like?

Theorem 3.8 *Let $\{e_1, \ldots, e_n\}$ be the standard basis of $\mathbb{F}^{n\times 1}$, and let $B = \{u_1, \ldots, u_n\}$ be any basis of $\mathbb{F}^{n\times 1}$. Write $P = [u_1 \cdots u_n]$. Then, $P[e_j]_B = e_j = [u_j]_B$ for $1 \le j \le n$, and for each $u \in \mathbb{F}^{n\times 1}$, $P[u]_B = u$ and $[u]_B = P^{-1}u$.*

Proof Let $j \in \{1, \ldots, n\}$. By Observation 3.2, $[u_j]_B = e_j$. Since u_j is the jth column of P, we have $u_j = Pe_j = P[u_j]_B$.

Let $u \in \mathbb{F}^{n\times 1}$. Then, $u = a_1 u_1 + \cdots + a_n u_n$ for unique scalars $a_1, \ldots, a_n$. Now,

$$\begin{aligned} P[u]_B &= P\big[a_1, \ldots, a_n\big]^t = P(a_1 e_1 + \cdots + a_n e_n) \\ &= a_1 Pe_1 + \cdots + a_n Pe_n = a_1 u_1 + \cdots + a_n u_n = u. \end{aligned}$$

As the columns of P form a basis for $\mathbb{F}^{n\times 1}$, the matrix P is invertible. Therefore, $[u]_B = P^{-1}u$. ■

Example 3.9 Consider the basis $B = \{u_1, u_2, u_3\}$ for $\mathbb{R}^{3\times 1}$, where $u_1 = [1, 1, 1]^t$, $u_2 = [1, 0, 1]^t$, and $u_3 = [1, 0, 0]^t$. The matrix P in Theorem 3.8 and its inverse are given by

$$P = [u_1 \quad u_2 \quad u_3] = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix}, \quad P^{-1} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -1 & 1 \\ 1 & 0 & -1 \end{bmatrix}.$$

We see that

$$\begin{aligned} e_1 &= [1, 0, 0]^t = 0 \cdot u_1 + 0 \cdot u_2 + u_3 \\ e_2 &= [0, 1, 0]^t = u_1 - u_2 + 0 \cdot u_3 \\ e_3 &= [0, 0, 1]^t = 0 \cdot u_1 + u_2 - u_3 \end{aligned}$$

Also, since $P[e_j]_B = e_j$, we have $[e_j]_B = P^{-1}e_j$. That is, the columns of P^{-1} are the coordinate vectors of e_j with respect to the basis B. Either way,

$$[e_1]_B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad [e_2]_B = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \quad [e_3]_B = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}.$$

It leads to

$$\begin{aligned} P[e_1]_B &= P[0, 0, 1]^t = [1, 0, 0]^t = e_1 \\ P[e_2]_B &= P[1, -1, 0]^t = [0, 1, 0]^t = e_2 \\ P[e_3]_B &= P[0, 1, -1]^t = [0, 0, 1]^t = e_3 \end{aligned}$$

That is, $P[e_j]_B = e_j$ for each j as stated in Theorem 3.8. We verify the other conclusion of the theorem for $u = [1, 2, 3]^t$. Here, $u = 2u_1 + u_2 - 2u_3$. Therefore,

$$[u]_B = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -1 & 1 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = P^{-1}u. \qquad \square$$

It raises a more general problem: if a vector v has the coordinate vector $[v]_B$ with respect to a basis B of $\mathbb{F}^{n\times 1}$, and also a coordinate vector $[v]_C$ with respect to another basis C of $\mathbb{F}^{n\times 1}$, then how are $[v]_B$ and $[v]_C$ related? We will address this question in Sect. 3.6.

Exercises for Sect. 3.4

In the following, a basis B for $\mathbb{F}^{3\times 1}$ and a vector $u \in \mathbb{F}^{3\times 1}$ are given. Compute the coordinate vector $[u]_B$.

1. $B = \{[0, 0, 1]^t, [1, 0, 0]^t, [0, 1, 0]^t\}$, $u = [1, 2, 3]^t$
2. $B = \{[0, 1, 1]^t, [1, 0, 1]^t, [1, 1, 0]^t\}$, $u = [2, 1, 3]^t$
3. $B = \{[1, 1, 1]^t, [1, 2, 1]^t, [1, 2, 3]^t\}$, $u = [3, 2, 1]^t$

3.5 Coordinate Matrices

Let $A \in \mathbb{F}^{m\times n}$. We view A as a linear transformation from $\mathbb{F}^{n\times 1}$ to $\mathbb{F}^{m\times 1}$. Let D and E be the standard bases of $\mathbb{F}^{n\times 1}$ and $\mathbb{F}^{m\times 1}$, respectively. Then, $[u]_D = u$ and $[Au]_E = Au$. We thus obtain

$$[Au]_E = A\,[u]_D.$$

If we choose a different pair of bases, that is, a basis B for $\mathbb{F}^{n\times 1}$ and a basis C for $\mathbb{F}^{m\times 1}$, then u has the coordinate vector $[u]_B$ and Au has the coordinate vector $[Au]_C$. Looking at the above equation, we ask:

Does there exist a matrix M such that $[Au]_C = M[u]_B$?
If so, is it unique?

The following theorem answers these questions.

Theorem 3.9 *Let $A \in \mathbb{F}^{m\times n}$. Let $B = \{u_1, \ldots, u_n\}$ and $C = \{v_1, \ldots, v_m\}$ be bases for $\mathbb{F}^{n\times 1}$ and $\mathbb{F}^{m\times 1}$, respectively. Then, there exists a unique matrix $M \in \mathbb{F}^{m\times n}$ such that $[Au]_C = M[u]_B$. Moreover, the entries of $M = [\alpha_{ij}]$ satisfy the following:*

$$Au_j = \alpha_{1j}v_1 + \cdots + \alpha_{mj}v_m \quad \textit{for each } j \in \{1, \ldots, n\}.$$

Proof Let $j \in \{1, \ldots, n\}$. Since $Au_j \in \mathbb{F}^{m\times 1}$ and C is a basis for $\mathbb{F}^{m\times 1}$, there exist unique scalars $\alpha_{1j}, \ldots, \alpha_{mj}$ such that

$$Au_j = \alpha_{1j}v_1 + \cdots + \alpha_{mj}v_m.$$

Thus, we have unique mn scalars α_{ij} so that the above equation is true for each j. Construct the matrix $M = [\alpha_{ij}] \in \mathbb{F}^{m\times n}$. By Observation 3.2,

$$[Au_j]_C = \begin{bmatrix}\alpha_{1j}, \ldots, \alpha_{mj}\end{bmatrix}^t = M\,e_j = M\,[u_j]_B \quad \text{for each } j \in \{1, \ldots, n\}.$$

We must verify that such an equality holds for each $u \in \mathbb{F}^{n\times 1}$. So, let $u \in \mathbb{F}^{n\times 1}$. As B is a basis of $\mathbb{F}^{n\times 1}$, there exist unique scalars $\beta_1, \ldots, \beta_n$ such that $u = \beta_1 u_1 + \cdots + \beta_n u_n$. Now,

$$\begin{aligned}[Au]_C &= [\beta_1 u_1 + \cdots + \beta_n u_n]_C = \beta_1[Au_1]_C + \cdots + \beta_n[Au_n]_C\\ &= \beta_1 M\,[u_1]_B + \cdots + \beta_n M\,[u_n]_B = M\,[\beta_1 u_1 + \cdots + \beta_n u_n]_B = M\,[u]_B.\end{aligned}$$

This proves the existence of a required matrix M.

To prove uniqueness, let P and Q be matrices such that

$$[Au]_C = P\,[u]_B = Q\,[u]_B \quad \text{for each } u \in \mathbb{F}^{n\times 1}.$$

In particular, with $u = u_j$, we have

$$[Au_j]_C = P\,[u_j]_B = Q\,[u_j]_B \quad \text{for each } j \in \{1, \ldots, n\}.$$

Since $[u_j]_B = e_j$, we obtain $Pe_j = Qe_j$. That is, the jth column of P is equal to the jth column of Q for each $j \in \{1, \ldots, n\}$. Therefore, $P = Q$. ■

In view of Theorem 3.9, we denote the matrix M as $[A]_{C,B}$ and give it a name. Let $A \in \mathbb{F}^{m\times n}$. Let B be a basis for $\mathbb{F}^{n\times 1}$, and let C be a basis for $\mathbb{F}^{m\times 1}$. The unique $m \times n$ matrix $[A]_{C,B}$ that satisfies

$$[Au]_C = [A]_{C,B}[u]_B \quad \text{for each } u \in \mathbb{F}^{n\times 1}$$

is called the **coordinate matrix** of A with respect to the bases B and C.

Theorem 3.9 provides a method to compute the coordinate matrix. Let $A \in \mathbb{F}^{m\times n}$, $B = \{u_1, \ldots, u_n\}$ be a basis for $\mathbb{F}^{n\times 1}$, and let $C = \{v_1, \ldots, v_n\}$ be a basis for $\mathbb{F}^{m\times 1}$. We express the image of each basis vector u_j in terms of the basis vector v_is as in the following:

$$\begin{aligned} Au_1 &= \alpha_{11}v_1 + \alpha_{21}v_2 + \cdots + \alpha_{m1}v_m \\ Au_2 &= \alpha_{12}v_1 + \alpha_{22}v_2 + \cdots + \alpha_{m2}v_m \\ &\vdots \\ Au_n &= \alpha_{1n}v_1 + \alpha_{2n}v_2 + \cdots + \alpha_{mn}v_m \end{aligned} \tag{3.1}$$

Then, the coordinate matrix is (Mark which coefficient α_{ij} goes where)

$$[A]_{C,B} = \begin{bmatrix} \alpha_{11} & \alpha_{12} & \cdots & \alpha_{1n} \\ \alpha_{21} & \alpha_{22} & \cdots & \alpha_{2n} \\ & \vdots & & \\ \alpha_{m1} & \alpha_{m2} & \cdots & \alpha_{mn} \end{bmatrix}. \tag{3.2}$$

Also, notice that if the coordinate matrix is given by (3.2), then the equalities in (3.1) are satisfied.

Example 3.10 Compute the coordinate matrix $[A]_{C,B}$ of $A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 1 & -1 \end{bmatrix}$ with respect to the bases $B = \{[1, 2]^t, [3, 1]^t\}$ for $\mathbb{F}^{2\times 1}$, and $C = \{[1, 0, 0]^t, [1, 1, 0]^t, [1, 1, 1]^t\}$ for $\mathbb{F}^{3\times 1}$.

Write $u_1 = [1, 2]^t, u_2 = [3, 1]^t$ and $v_1 = [1, 0, 0]^t, v_2 = [1, 1, 0]^t, v_3 = [1, 1, 1]^t$. We obtain

$$\begin{aligned} Au_1 = [2, 3, -1]^t &= -v_1 + 4v_2 - v_3 \\ Au_2 = [1, 4, 2]^t &= -3v_1 + 2v_2 + 2v_3. \end{aligned}$$

Therefore, $[A]_{C,B} = \begin{bmatrix} -1 & -3 \\ 4 & 2 \\ -1 & 2 \end{bmatrix}$. □

We use Theorem 3.8 in proving the following result, which shows another way for computing the coordinate matrix.

Theorem 3.10 *Let $A \in \mathbb{F}^{m\times n}$. Let $B = \{u_1, \ldots, u_n\}$ and $C = \{v_1, \ldots, v_m\}$ be bases for $\mathbb{F}^{n\times 1}$ and $\mathbb{F}^{m\times 1}$, respectively. Construct the matrices*

$$P = \begin{bmatrix} u_1 & \cdots & u_n \end{bmatrix}, \quad Q = \begin{bmatrix} v_1 & \cdots & v_m \end{bmatrix},$$

by taking the column vectors u_j and v_i as columns of the respective matrices. Then, $[A]_{C,B} = Q^{-1}AP$.

Proof Let $u = a_1u_1 + \cdots + a_nu_n$. By Theorem 3.8, we obtain $P\,[u]_B = u$. Also, $Q\,[Au]_C = Au$.

Since the columns of $Q \in \mathbb{F}^{m\times m}$ are linearly independent, Q is invertible. Therefore, $[Au]_C = Q^{-1}Au = Q^{-1}AP\,[u]_B$.

It then follows that $[A]_{C,B} = Q^{-1}AP$. ■

To summarize, if $A \in \mathbb{F}^{m\times n}$, B is a basis for $\mathbb{F}^{n\times 1}$, and C is a basis for $\mathbb{F}^{m\times 1}$, then the coordinate matrix $[A]_{C,B}$ can be obtained in two ways.

Due to Theorem 3.9, we express the A-image of the basis vectors in B as linear combinations of basis vectors in C. The coefficients of each such image form the corresponding columns of the matrix $[A]_{C,B}$.

Alternatively, using Theorem 3.10, we construct P as the matrix whose jth column is the jth vector in B and the matrix Q by taking its ith column as the ith vector in C. Then, we have $[A]_{C,B} = Q^{-1}AP$.

Since inverse of a matrix is computed using the reduction to RREF, the coordinate matrix may be computed the same way. If $B = \{u_1, \ldots, u_n\}$ and $C = \{v_1, \ldots, v_m\}$ are bases for $\mathbb{F}^{n\times 1}$ and $\mathbb{F}^{m\times 1}$, respectively, then

$$P = \begin{bmatrix} u_1 & \cdots & u_n \end{bmatrix}, \quad Q = \begin{bmatrix} v_1 & \cdots & v_m \end{bmatrix}.$$

Now, $AP = \begin{bmatrix} Au_1 & \cdots & Au_n \end{bmatrix}$. To compute $Q^{-1}AP$, we start with the augmented matrix $[Q \mid AP]$ and then proceed towards its RREF. Since Q^{-1} exists, the RREF will look like $[I \mid Q^{-1}AP]$. Schematically, the computation may be written as in the following:

$$\begin{bmatrix} v_1 & \cdots & v_m \mid Au_1 & \cdots & Au_n \end{bmatrix} \overset{RREF}{\longrightarrow} [I \mid [A]_{C,B}].$$

Example 3.11 Compute the coordinate matrix $[A]_{C,B}$ of $A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 1 & -1 \end{bmatrix}$ with respect to the bases $B = \{[1, 2]^t, [3, 1]^t\}$ for $\mathbb{F}^{2\times 1}$, and $C = \{[1, 0, 0]^t, [1, 1, 0]^t, [1, 1, 1]^t\}$ for $\mathbb{F}^{3\times 1}$.

Writing the vectors of B as u_1 and u_2 in that order, we have

$$Au_1 = \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}, \quad Au_2 = \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix}.$$

We then convert the augmented matrix of columns as vectors from C followed by Au_1 and Au_2 to RREF. It is as follows:

$$\left[\begin{array}{ccc|cc} 1 & 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & 3 & 4 \\ 0 & 0 & 1 & -1 & 2 \end{array}\right] \xrightarrow{R1} \left[\begin{array}{ccc|cc} 1 & 0 & 0 & -1 & -3 \\ 0 & 1 & 0 & 4 & 2 \\ 0 & 0 & 1 & -1 & 2 \end{array}\right].$$

Here, $R1 = E_{-1}[1, 2],\ E_{-1}[2, 3]$. Therefore, $[A]_{C,B} = \begin{bmatrix} -1 & -3 \\ 4 & 2 \\ -1 & 2 \end{bmatrix}$.

It is easy to verify that $Au_1 = -v_1 + 4v_2 - v_3$ and $Au_2 = -3v_1 + 2v_2 + 2v_3$ as required by (3.1)–(3.2). □

The linear property of coordinate vectors can be extended to coordinate matrices; see Exercise 2. Moreover, the product formula $[Au]_C = [A]_{C,B}[u]_B$ has a similar looking expression for product of matrices.

Theorem 3.11 *Let $A \in \mathbb{F}^{m\times k}$ and $B \in \mathbb{F}^{k\times n}$ be matrices. Let C, D and E be bases for $\mathbb{F}^{n\times 1}$, $\mathbb{F}^{k\times 1}$ and $\mathbb{F}^{m\times 1}$, respectively. Then, $[AB]_{E,C} = [A]_{E,D}[B]_{D,C}$.*

Proof For each $v \in \mathbb{F}^{n\times 1}$,

$$[A]_{E,D}[B]_{D,C}[v]_C = [A]_{E,D}[Bv]_D = [ABv]_E = [AB]_{E,C}[v]_C.$$

Therefore, $[A]_{E,D}[B]_{D,C} = [AB]_{E,C}$. ■

Notice that in Theorem 3.11, the matrices are the linear transformations

$$A : \mathbb{F}^{k\times 1} \to \mathbb{F}^{m\times 1}, \quad B : \mathbb{F}^{n\times 1} \to \mathbb{F}^{k\times 1}, \quad AB : \mathbb{F}^{n\times 1} \to \mathbb{F}^{m\times 1}.$$

Since $ABv = (AB)(v) = A(Bv)$, the linear transformation AB is the composition of the maps A and B. The composition map AB transforms a vector from $\mathbb{F}^{n\times 1}$ to one in $\mathbb{F}^{k\times 1}$ by using B first, and then it uses A to transform that vector from $\mathbb{F}^{k\times 1}$ to one in $\mathbb{F}^{m\times 1}$. Our result says that the coordinate matrix of the composition map AB is simply the product of the coordinate matrices of the maps A and B.

Exercises for Sect. 3.5

1. Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}$. Consider the bases $B = \{[1, 0, 1]^t, [0, 1, 1]^t, [1, 1, 0]^t$ and $C = \{[1, 1, 1]^t, [1, 2, 1]^t, [1, 1, 0]^t\}$ for $\mathbb{F}^{3\times 1}$. Compute $[A]_{C,B}$ and $[A]_{B,C}$ in the following ways:

(a) By expressing A-images of the basis vectors of B in terms of those of C and vice versa.
(b) By computing the inverses of two matrices constructed from the basis vectors.
(c) By using the RREF conversion as mentioned in the text.

2. Let $A, B \in \mathbb{F}^{m\times n}$, C be a basis of $\mathbb{F}^{n\times 1}$, and D be a basis for $\mathbb{F}^{m\times 1}$, and let $\alpha \in \mathbb{F}$. Show that $[\alpha A + B]_{D,C} = \alpha[A]_{D,C} + [B]_{D,C}$.
3. Take a 2×3 real matrix A and a 3×2 real matrix B. Also, take the bases $C = \{[1, 1], [0, 1]\}$, $D = \{[1, 0, 0], [0, 1, 0], [0, 0, 1]\}$ and $E = \{[1, 0], [1, 1]\}$ for the spaces $\mathbb{R}^{2\times 1}$, $\mathbb{R}^{3\times 1}$ and $\mathbb{R}^{2\times 1}$, respectively. Verify Theorem 3.11.
4. Construct a matrix $A \in \mathbb{R}^{2\times 2}$, a vector $v \in \mathbb{R}^{2\times 1}$, and a basis $B = \{u_1, u_2\}$ for $\mathbb{R}^{2\times 1}$ satisfying $[Av]_B \neq A[v]_B$.

3.6 Change of Basis Matrix

Recall that $[A]_{C,B}$ is that matrix in $\mathbb{F}^{m\times n}$ which satisfies the equation:

$$[Au]_C = [A]_{C,B}[u]_B \quad \text{for each } u \in \mathbb{F}^{n\times 1}.$$

Further, $[A]_{C,B}$ is given by $Q^{-1}AP$, where P is the matrix whose jth column is the jth basis vector of B, and Q is the matrix whose ith column is the ith basis vector of C. In particular, taking $A \in \mathbb{F}^{n\times n}$ as the identity matrix, writing B as O, and C as N, we see that

$$[u]_N = [I]_{N,O}[u]_O = Q^{-1}P[u]_O \quad \text{for each } u \in \mathbb{F}^{n\times 1}. \tag{3.3}$$

In (3.3), P is the matrix whose jth column is the jth basis vector of O, and Q is the matrix whose ith column is the ith basis vector of N for $1 \le i, j \le n$. Both O and N are bases for $\mathbb{F}^{n\times 1}$.

The formula in (3.3) shows how the coordinate vector changes when a basis changes. The matrix $I_{N,O}$, which is equal to $Q^{-1}P$, is called the **change of basis matrix** or the **transition matrix** when the basis changes from an old basis O to a new basis N.

Observe that the change of basis matrix $I_{N,O}$ can also be computed by expressing the basis vectors of O as linear combinations of basis vectors of N as stipulated by Theorem 3.9, or equivalently, in (3.1)–(3.2). Alternatively, if $O = \{u_1, \ldots, u_n\}$ and $N = \{v_1, \ldots, v_n\}$, the change of basis matrix $I_{N,O}$, which is equal to $Q^{-1}P$, may be computed by using reduction to RREF. Schematically, it is given as follows:

$$O = \{u_1, \ldots, u_n\},\ N = \{v_1, \ldots, v_n\}, \quad \begin{bmatrix} v_1 & \cdots & v_n \,|\, u_1 & \cdots & u_n \end{bmatrix} \overset{RREF}{\longrightarrow} [I \,|\, I_{N,O}].$$

Example 3.12 Consider the following bases for $\mathbb{R}^{3\times 1}$:

$$O = \{[1,0,1]^t, [1,1,0]^t, [0,1,1]^t\}, \quad N = \{[1,-1,1]^t, [1,1,-1]^t, [-1,1,1]^t\}.$$

Find the change of basis matrix $I_{N,O}$ when the basis changes from O to N. Also, find the coordinate matrix $[A]_{N,O}$ of the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ with respect to the pair of bases O and N. Verify that

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}_N = I_{N,O} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}_O, \quad \left[A \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \right]_N = [A]_{N,O} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}_O.$$

To solve this problem, we express the basis vectors of O as linear combinations of those of N, as in Theorem 3.9. The details are as follows:

$$\begin{aligned} [1,0,1]^t &= 1[1,-1,1]^t + \tfrac{1}{2}[1,1,-1]^t + \tfrac{1}{2}[-1,1,1]^t \\ [1,1,0]^t &= \tfrac{1}{2}[1,-1,1]^t + 1[1,1,-1]^t + \tfrac{1}{2}[-1,1,1]^t \\ [0,1,1]^t &= \tfrac{1}{2}[1,-1,1]^t + \tfrac{1}{2}[1,1,-1]^t + 1[-1,1,1]^t. \end{aligned}$$

The coefficients from the first equation give the first column of $I_{N,O}$ and so on. Thus, we obtain

$$I_{N,O} = \begin{bmatrix} 1 & 1/2 & 1/2 \\ 1/2 & 1 & 1/2 \\ 1/2 & 1/2 & 1 \end{bmatrix}.$$

Alternatively, by Theorem 3.10, the change of basis matrix is given by

$$I_{N,O} = \begin{bmatrix} 1 & 1 & -1 \\ -1 & 1 & 1 \\ 1 & -1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1/2 & 0 & 1/2 \\ 1/2 & 1/2 & 0 \\ 0 & 1/2 & 1/2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1/2 & 1/2 \\ 1/2 & 1 & 1/2 \\ 1/2 & 1/2 & 1 \end{bmatrix}.$$

Again, using the RREF conversion we may obtain $I_{N,O}$ as follows:

$$\left[\begin{array}{ccc|ccc} 1 & 1 & -1 & 1 & 1 & 0 \\ -1 & 1 & 1 & 0 & 1 & 1 \\ 1 & -1 & 1 & 1 & 0 & 1 \end{array}\right] \xrightarrow{R1} \left[\begin{array}{ccc|ccc} 1 & 1 & -1 & 1 & 1 & 0 \\ 0 & 2 & 0 & 1 & 2 & 1 \\ 0 & -2 & 2 & 0 & -1 & 1 \end{array}\right]$$

$$\xrightarrow{R2} \left[\begin{array}{ccc|ccc} 1 & 0 & -1 & 1/2 & 0 & -1/2 \\ 0 & 1 & 0 & 1/2 & 1 & 1/2 \\ 0 & 0 & 2 & 1 & 1 & 2 \end{array}\right] \xrightarrow{R3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 1/2 & 1/2 \\ 0 & 1 & 0 & 1/2 & 1 & 1/2 \\ 0 & 0 & 1 & 1/2 & 1/2 & 1 \end{array}\right].$$

Here, $R1 = E_1[2,1], E_{-1}[3,1]$; $R2 = E_{1/2}[2], E_{-1}[1,2], E_2[3,2]$; $R3 = E_{1/2}[3], E_1[1,3]$. The matrix in the RREF, to the right of the vertical bar, is the required change of basis matrix $I_{N,O}$.

To verify our result for $[1,2,3]^t$, notice that

$$\begin{aligned}[1,2,3]^t &= 1[1,0,1]^t + 0[1,1,0]^t + 2[0,1,1]^t \\ [1,2,3]^t &= 2[1,-1,1]^t + \tfrac{3}{2}[1,1,-1]^t + \tfrac{5}{2}[-1,1,1]^t.\end{aligned}$$

Therefore, $[[1,2,3]^t]_O = [1,0,2]^t$ and $[[1,2,3]^t]_N = [2, 3/2, 5/2]^t$. Then,

$$I_{N,O}\begin{bmatrix}1\\2\\3\end{bmatrix}_O = \begin{bmatrix}1 & 1/2 & 1/2\\ 1/2 & 1 & 1/2\\ 1/2 & 1/2 & 1\end{bmatrix}\begin{bmatrix}1\\0\\2\end{bmatrix} = \begin{bmatrix}2\\3/2\\5/2\end{bmatrix} = \begin{bmatrix}1\\2\\3\end{bmatrix}_N.$$

According to Theorem 3.10,

$$[A]_{N,O} = \begin{bmatrix}1 & 1 & -1\\ -1 & 1 & 1\\ 1 & -1 & 1\end{bmatrix}^{-1} A \begin{bmatrix}1 & 1 & 0\\ 0 & 1 & 1\\ 1 & 0 & 1\end{bmatrix} = \frac{1}{2}\begin{bmatrix}2 & 3 & 3\\ 2 & 1 & 3\\ 0 & 0 & 2\end{bmatrix}.$$

As to the verification,

$$A\begin{bmatrix}1\\2\\3\end{bmatrix} = \begin{bmatrix}6\\2\\6\end{bmatrix} = 4\begin{bmatrix}1\\-1\\1\end{bmatrix} + 4\begin{bmatrix}1\\1\\-1\end{bmatrix} + 2\begin{bmatrix}-1\\1\\1\end{bmatrix}.$$

We find that $[A]_{N,O}\begin{bmatrix}1\\2\\3\end{bmatrix}_O = [A]_{N,O}\begin{bmatrix}1\\0\\2\end{bmatrix} = \begin{bmatrix}4\\4\\2\end{bmatrix} = \left[A\begin{bmatrix}1\\2\\3\end{bmatrix}\right]_N$. □

Exercises for Sect. 3.4

Let $A = \begin{bmatrix}3 & 1 & 0\\ 0 & 1 & 3\\ -1 & -1 & -2\end{bmatrix}$, $V = \{[a,b,c]^t : a + 2b + 3c = 0,\ a,b,c \in \mathbb{R}\}$, $B = \{[0,3,-2]^t, [2,-1,0]^t\}$, and let $C = \{[1,1,-1]^t, [3,0,-1]^t\}$.

1. Show that V is a two-dimensional subspace of $\mathbb{R}^{3\times 1}$.
2. Show that $A : V \to V$ defined by $A(v) = Av$ for $v \in V$ is a well-defined function.
3. Show that B and C are bases for V.
4. Extend the basis B of V to a basis O for $\mathbb{F}^{3\times 1}$.
5. Extend the basis C of V to a basis N for $\mathbb{F}^{3\times 1}$.
6. Find the change of basis matrix $I_{N,O}$.
7. Verify that $[v]_N = I_{N,O}\,[v]_O$ for $v = [-4,-1,2]^t$.
8. Find the matrix $[A]_{N,O}$.
9. Verify that $[Av]_N = [A]_{N,O}\,[v]_O$ for $v = [-4,-1,2]^t$.

3.7 Equivalence and Similarity

In view of Theorem 3.10, two matrices $A, B \in \mathbb{F}^{m\times n}$ are called **equivalent** iff there exist invertible matrices $P \in \mathbb{F}^{n\times n}$ and $Q \in \mathbb{F}^{m\times m}$ such that $B = Q^{-1}AP$.

Notice that 'is equivalent to' is an equivalence relation on $\mathbb{F}^{m\times n}$. Equivalent matrices represent the same matrix (linear transformation) with respect to possibly different pairs of bases.

From Theorem 2.8, it follows that ranks of two equivalent matrices are equal.

We can construct a matrix of rank r relatively easily. Let $r \le \min\{m, n\}$. The matrix $R_r \in \mathbb{F}^{m\times n}$ whose first r columns are the standard basis vectors $e_1, \ldots, e_r$ of $\mathbb{F}^{m\times 1}$ and all other columns are zero columns has rank r. This matrix, in block form, looks as follows:

$$R_r = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}.$$

Such a matrix is called a **rank echelon** matrix. If R_r is an $m \times n$ matrix, then necessarily $r \le \min\{m, n\}$. For ease in writing, we do not show the size of a rank echelon matrix in the notation R_r; we rather specify it in different contexts. From Theorem 2.8, it follows that any matrix which is equivalent to R_r has also rank r.

Conversely, if a row of a matrix is a linear combination of other rows, then the rank of the matrix is same as that of the matrix obtained by deleting such a row. Similarly, deleting a column which is a linear combination of other columns does not change the rank of the matrix. It is thus possible to perform elementary row and column operations to bring a matrix of rank r to its rank echelon form R_r. We state and prove this result rigorously.

Theorem 3.12 (Rank factorization) *A matrix is of rank r iff it is equivalent to the rank echelon matrix R_r of the same size.*

Proof Let $A \in \mathbb{F}^{m\times n}$, and let R_r be the rank echelon matrix of size $m \times n$. If A is equivalent to R_r, then $\operatorname{rank}(A) = \operatorname{rank}(R_r) = r$ by Theorem 2.8.

For the converse, suppose $\operatorname{rank}(A) = r$. Convert A to its RREF $C = E_1 A$, where E_1 is a suitable product of elementary matrices. Now, each non-pivotal column of C is a linear combination of its r pivotal columns.

Consider the matrix C^t. The pivotal columns of C are now the rows e_i^t in C^t. Each other row in C^t is a linear combination of these rows. Exchange the rows of C^t so that the first r rows of the new matrix are $e_1^t, \ldots, e_r^t$ in that order. That is, we have a matrix E_2, which is a product of elementary matrices of Type 1, such that $E_2 C^t$ has the first r rows as $e_1^t, \ldots, e_r^t$ and the next $m - r$ rows are linear combinations of these r rows.

Use suitable elementary row operations to zero-out the bottom $m - r$ rows. This is possible since each non-pivotal row is a linear combination of the pivotal rows $e_1^t, \ldots, e_r^t$. Thus, we obtain the matrix $E_3 E_2 C^t$, where E_3 is the suitable product of elementary matrices used in reducing the bottom $m - r$ rows to zero rows.

Then taking transpose, we see that $C E_3^t E_2^t$ is a matrix whose first r columns are $e_1, \ldots, e_r$ and all other columns are zero columns.

To summarize, we have obtained three matrices E_1, E_2, and E_3, which are products of elementary matrices such that

$$CE_3^tE_2^t = E_1AE_3^tE_2^t = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} = R_r.$$

Since E_1, E_2^t and E_3^t are invertible, A and R_r are equivalent. ■

Theorem 3.12 asserts that given any matrix $A \in \mathbb{F}^{m\times n}$ of rank r, there exist invertible matrices $P \in \mathbb{F}^{n\times n}$ and $Q \in \mathbb{F}^{m\times m}$ such that $A = Q^{-1}R_rP$. Thus, it is named as *rank factorization*.

The rank factorization can be used to characterize equivalence of matrices. If A and B are equivalent matrices, then clearly, they have the same rank. Conversely, if two $m \times n$ matrices have the same rank r, then both of them are equivalent to R_r. Then, they are equivalent. We thus obtain the *rank theorem*:

Two matrices of the same size are equivalent iff they have the same rank.

Observe that $R_r = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} I_r \\ 0 \end{bmatrix} \begin{bmatrix} I_r & 0 \end{bmatrix}$. Due to rank factorization, there exist invertible matrices P and Q such that

$$A = Q^{-1}R_r\,P = Q^{-1}\begin{bmatrix} I_r \\ 0 \end{bmatrix}\begin{bmatrix} I_r & 0 \end{bmatrix}P = B\,C,$$

where

$$B = Q^{-1}\begin{bmatrix} I_r \\ 0 \end{bmatrix} \in \mathbb{F}^{m\times r}, \quad C = \begin{bmatrix} I_r & 0 \end{bmatrix}P \in \mathbb{F}^{r\times n}.$$

Here, $\operatorname{rank}(B) = r =$ the number of columns in B. Similarly, $\operatorname{rank}(C) = r =$ the number of rows in C. A matrix whose rank is equal to the number of rows or the number of columns is called a *full rank matrix*. The above factorization $A = BC$ shows that

each matrix can be written as a product of full rank matrices.

This result is known as the *full rank factorization*.

The notion of equivalence stems from the change of bases in both the domain and the co-domain of a matrix viewed as a linear transformation. In case the matrix is a square matrix of order n, it is considered as a linear transformation on $\mathbb{F}^{n\times 1}$. If we change the basis in $\mathbb{F}^{n\times 1}$, we would have a corresponding representation of the matrix in the new basis.

Let $A \in \mathbb{F}^{n\times n}$, a square matrix of order n. The matrix A is (viewed as) a linear transformation from $\mathbb{F}^{n\times 1}$ to $\mathbb{F}^{n\times 1}$. Let $E = \{e_1, \ldots, e_n\}$ be the standard basis of $\mathbb{F}^{n\times 1}$. The matrix A acts in the usual way: Ae_j is the jth column of A. Suppose we change the basis of $\mathbb{F}^{n\times 1}$ to $C = \{v_1, \ldots, v_n\}$. That is, in both the domain and the co-domain space, we take the new basis as C. If the equivalent matrix of A is M, then for each $v \in \mathbb{F}^{n\times 1}$,

$$[Av]_C = P^{-1}AP\,[v]_C, \quad P = \begin{bmatrix} v_1 & \cdots & v_n \end{bmatrix}.$$

That is, the coordinate matrix of A is $P^{-1}AP$. This leads to similarity of two matrices.

We say that two matrices $A, B \in \mathbb{F}^{n\times n}$ are **similar** iff $B = P^{-1}AP$ for some invertible matrix $P \in \mathbb{F}^{n\times n}$. Observe that in this case, B is the coordinate matrix of A with respect to the basis that comprises the columns of P.

Example 3.13 Consider the basis $N = \{[1,-1,1]^t, [1,1,-1]^t, [-1,1,1]^t\}$ for $\mathbb{R}^{3\times 1}$. To determine the matrix similar to $A = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ when the basis has changed from the standard basis to N, we construct the matrix P by taking the basis vectors of N as in the following:

$$P = \begin{bmatrix} 1 & 1 & -1 \\ -1 & 1 & 1 \\ 1 & -1 & 1 \end{bmatrix}.$$

Then, the matrix similar to A when the basis changes to N is

$$B = P^{-1}AP = \tfrac{1}{2}\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} 1 & 1 & -1 \\ -1 & 1 & 1 \\ 1 & -1 & 1 \end{bmatrix} = \tfrac{1}{2}\begin{bmatrix} 0 & 2 & 2 \\ 1 & -1 & 3 \\ -1 & -1 & 3 \end{bmatrix}.$$

From Example 3.12, we know that for $u = [1,2,3]^t$,

$$[u]_N = \begin{bmatrix} 2 \\ 3/2 \\ 5/2 \end{bmatrix}, \quad [Au]_N = \begin{bmatrix} 4 \\ 4 \\ 2 \end{bmatrix}, \quad B\,[u]_N = \tfrac{1}{2}\begin{bmatrix} 0 & 2 & 2 \\ 1 & -1 & 3 \\ -1 & -1 & 3 \end{bmatrix}\begin{bmatrix} 2 \\ 3/2 \\ 5/2 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ 2 \end{bmatrix}.$$

This verifies the condition $[Au]_N = B\,[u]_N$ for the vector $u = [1,2,3]^t$. □

We emphasize that if $B = P^{-1}AP$ is a matrix similar to A, then the matrix A as a linear transformation on $\mathbb{F}^{n\times 1}$ with standard basis and the matrix B as a linear transformation on $\mathbb{F}^{n\times 1}$ with the basis consisting of columns of P are the same linear transformation. Moreover, if C is the basis whose jth element is the jth column of P, then for each vector $v \in \mathbb{F}^{n\times 1}$, $[Av]_C = P^{-1}AP\,[v]_C$.

Though equivalence is easy to characterize by the rank, similarity is much more difficult. And we postpone this to a later chapter.

Exercises for Sect. 3.7

1. Let $A \in \mathbb{C}^{m\times n}$. Define $T : \mathbb{C}^{1\times m} \to \mathbb{C}^{1\times n}$ by $T(x) = xA$ for $x \in \mathbb{C}^{1\times m}$. Show that T is a linear transformation. Identify $T(e_j^t)$.
2. In each of the following cases, show that T is a linear transformation. Find the matrix A so that $T(x) = Ax$. Determine $\operatorname{rank}(A)$. And then, construct the full rank factorization of A.

(a) $T : \mathbb{R}^{3\times 1} \to \mathbb{R}^{2\times 1},\ T([a,b,c]^t) = [c, b+a]^t$
(b) Let $T : \mathbb{R}^{3\times 1} \to \mathbb{R}^{3\times 1},\ T([a,b,c]^t) = [a+b, 2a-b-c, a+b+c]^t$

3. Which matrices are equivalent to, and which are similar to
(a) the zero matrix (b) the identity matrix?
4. Consider the bases $O = \{u_1, u_2\}$ and $N = \{v_1, v_2\}$ for $\mathbb{F}^{2\times 1}$, where $u_1 = [1,1]^t$, $u_2 = [-1,1]^t$, $v_1 = [2,1]^t$, and $v_2 = [1,0]^t$. Let $A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$.

(a) Compute $Q = [A]_{O,O}$ and $R = [A]_{N,N}$.
(b) Find the change of basis matrix $P = I_{N,O}$.
(c) Compute $S = PQP^{-1}$.
(d) Is it true that $R = S$? Why?
(e) If $S = [s_{ij}]$, verify that $Av_1 = s_{11}v_1 + s_{21}v_2$, $Av_2 = s_{12}v_1 + s_{22}v_2$.

3.8 Problems

1. Let $A \in \mathbb{F}^{n\times n}$. Show that the following are equivalent to $\det(A) \neq 0$:

(a) The columns of A are linearly independent vectors in $\mathbb{F}^{n\times 1}$.
(b) The rows of A are linearly independent vectors in $\mathbb{F}^{1\times n}$.

2. Let $x, y, z \in \mathbb{F}^n$ be linearly independent. Determine whether the following are linearly independent:

(a) $x+y, y+z, z+x$ (b) $x-y, y-z, z-x$

3. Let U and V be subspaces of $\mathbb{F}^n$. Show the following:

(a) $U \cap V$ is a subspace of $\mathbb{F}^n$.
(b) $U \cup V$ need not be a subspace of $\mathbb{F}^n$.
(c) $U + V = \{u + v : u \in U,\ v \in V\}$ is a subspace of $\mathbb{F}^n$.
(d) If W is any subspace of $\mathbb{F}^n$, $U \subseteq W$, and $V \subseteq W$, then $U + V \subseteq W$.
(e) $\dim(U+V) + \dim(U \cap V) = \dim(U) + \dim(V)$.

4. Let $A \in \mathbb{F}^{m\times n}$. Show that if the columns of A are linearly independent, then $N(A) = \{0\}$. What if the rows of A are linearly independent?
5. Let $C = AB$, where $A \in \mathbb{F}^{m\times k}$ and $B \in \mathbb{F}^{k\times n}$. Show the following:

(a) If both A and B have linearly independent columns, then the columns of C are linearly independent.
(b) If both A and B have linearly independent rows, then the rows of C are linearly independent.
(c) If B has linearly dependent columns, then the columns of C are linearly dependent.
(d) If A has linearly dependent rows, then the rows of C are linearly dependent.

6. Let $A, B \in \mathbb{F}^{n\times n}$ be such that $N(A-B) = \mathbb{F}^{n\times 1}$. Show that $A = B$.

7. Let $A, B \in \mathbb{F}^{n\times n}$ be such that $AB = 0$. Show the following:
   (a) $R(B) \subseteq N(A)$ (b) $\text{rank}(A) + \text{rank}(B) \le n$
8. Let $A = xy^t$ for nonzero vectors $x \in \mathbb{F}^{m\times 1}$ and $y \in \mathbb{F}^{n\times 1}$. Show that
   (a) $\text{span}\{x\} = R(A)$ (b) $\text{span}\{y^t\} =$ the row space of A
9. Is it true that if $A \in \mathbb{F}^{m\times n}$, then both A and A^t have the same nullity?
10. Let B be the RREF of an $m \times n$ matrix A. Are the following true?
    (a) The row spaces of A and B are equal.
    (b) The column spaces of A and B are equal.
11. Let U be a subspace of $\mathbb{F}^{n\times 1}$.
    (a) Does there exist a matrix in $\mathbb{F}^{n\times n}$ such that $U = R(A)$?
    (b) Does there exist a matrix in $\mathbb{F}^{n\times n}$ such that $U = N(A)$?
12. Let $A \in \mathbb{F}^{m\times n}$. Let $\{u_1, \ldots, u_k\}$ be a basis for $N(A)$. Argue that there exist vectors $v_1, \ldots, v_{n-k}$ so that $\{u_1, \ldots, u_k, v_1, \ldots, v_{n-k}\}$ is a basis for $\mathbb{F}^{n\times 1}$. Show that $\{Av_1, \ldots, Av_{n-k}\}$ is a basis for $R(A)$. This will give an alternate proof of the rank nullity theorem.
13. Let $A \in \mathbb{F}^{m\times n}$. Let $P \in \mathbb{F}^{n\times n}$ and $Q \in \mathbb{F}^{m\times m}$ be invertible matrices. Show the following:
    (a) $N(QA) = N(A)$ (b) $R(AP) = R(A)$
14. Using rank nullity theorem, deduce Theorem 2.8.
15. Let $A, B \in \mathbb{F}^{n\times n}$ have the same rank. Is it true that $\text{rank}(A^2) = \text{rank}(B^2)$?
16. Let $A, B \in \mathbb{F}^{n\times n}$. Show that A is similar to B iff there exist matrices $P, Q \in \mathbb{F}^{n\times n}$, with P invertible, such that $A = PQ$ and $B = QP$.
17. Let A and B be similar matrices. Are the following pairs similar?
    (a) A^t, B^t (b) A^{-1}, B^{-1} (c) A^k, B^k for $k \in \mathbb{N}$ (d) $A - \alpha I, B - \alpha I$
18. If A and B are similar matrices, show that $\text{tr}(A) = \text{tr}(B)$.
19. Let $A, B \in \mathbb{F}^{n\times n}$ have the same trace. Are A and B
    (a) similar? (b) equivalent?
20. Let $A, B \in \mathbb{F}^{n\times n}$ have the same determinant. Are A and B
    (a) similar? (b) equivalent?
21. Is a rank factorization of a matrix unique?
22. Is a full rank factorization of a matrix unique?
23. Using rank factorization, show that for any $m \times k$ matrix A and $k \times n$ matrix B, $\text{rank}(AB) \le \min\{\text{rank}(A), \text{rank}(B)\}$.
24. Let A and B be $m \times n$ matrices. Using full rank factorization, show that $\text{rank}(A + B) \le \text{rank}(A) + \text{rank}(B)$.
25. Using the rank theorem, prove that the row rank of a matrix is equal to its column rank.

Let $A \in F^{n\times n}$. Then A is said to be an orthogonal projection if $A^* = A = A^2$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\langle x\, y\rangle = \sum x_i \overline{y_i} = y^* x$$

Let $V \subseteq F^{n\times 1}$ be a subspace, then there exists a unique orthogonal projection denoted by $P_V \in F^{n\times n}$, such that range$(P_V) = V$. Infact if $\{v_1, v_2 \cdots v_k\}$ be an orthonormal basis, then $P_V = v_1 v_1^* + v_2 v_2^* \cdots + v_k v_k^*$

This unique P_V is known as the orthogonal projection of $F^{n\times 1}$ onto V.

Let $\{v_1, v_2 \cdots v_k\}$ be a ONB for V ($k = \dim(V)$)

Let $P_V = v_1 v_1^* + v_2 v_2^* + \cdots$

$$P_V^* = v_1 v_1^* + v_2 v_2^* + \cdots = P_V$$

$$P_V^2 = (v_1 v_1^* + v_2 v_2^* + \cdots)(v_1 v_1^* + v_2 v_2^* + \cdots)$$

$$v_i v_i^* v_j v_j^* = \langle v_j v_i\rangle v_i v_j = \delta_{ji} v_i v_j$$

$$P_V^2 = v_1 v_1^* + v_2 v_2^* + \cdots = P_V \longrightarrow P_V = \text{OGP}$$

Let $x \in F^n$. Then $P_U x = \left(\sum_{j=1}^{n} v_j v_j^*\right) x = \sum_{j=1}^{n} \langle x, v_j \rangle v_j$

$R(P_U) = \{P_U x : x \in F^n\} \subseteq V$

T such that $V \subseteq R(P_U)$

Let $v \in V$. By fourier series expansion, we have

$$v = \sum_{j=1}^{k} \langle v_i, v_j \rangle v_j = \sum_{j=1}^{k} v_j v_j^* v_i = P_U v \in R(P_U)$$

$$\Rightarrow R(P_U) = V$$

Let $V \subseteq F^{n \times 1}$ be a subspace and $\{v_1, v_2 \ldots v_k\}$ be a ONB of V. Then given any $x \in F^n$ the vectors

$$y_i = \sum_{j=1}^{k} \langle x, v_j \rangle v_j \in V \text{ is the unique vector}$$

in V with the property that $(x - y) \perp v \ \forall \ v \in V$

Chapter 5
Eigenvalues and Eigenvectors



5.1 Invariant Line

Let $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. We view A as the linear transformation: $A : \mathbb{R}^{2\times 1} \to \mathbb{R}^{2\times 1}$. It transforms straight lines to straight lines or points. Does there exist a straight line which is transformed to itself? We see that

$$A \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} y \\ x \end{bmatrix}.$$

Thus, the line $\{[x, x]^t : x \in \mathbb{R}\}$ never moves. So also the line $\{[x, -x]^t : x \in \mathbb{R}\}$. Observe that

$$A \begin{bmatrix} x \\ x \end{bmatrix} = 1 \begin{bmatrix} x \\ x \end{bmatrix}, \quad A \begin{bmatrix} x \\ -x \end{bmatrix} = (-1) \begin{bmatrix} x \\ -x \end{bmatrix}.$$

Let $A \in \mathbb{F}^{n\times n}$. A scalar $\lambda \in \mathbb{F}$ is called an **eigenvalue** of A iff there exists a nonzero vector $v \in \mathbb{F}^{n\times 1}$ such that $Av = \lambda v$. Such a nonzero vector v is called an **eigenvector of** A **for** (associated with, corresponding to) the eigenvalue λ.

Example 5.1 Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$. We see that $A \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = 1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$. Therefore, A has an eigenvector $[1, 0, 0]^t$ associated with the eigenvalue 1. Is $[c, 0, 0]^t$ also an eigenvector associated with the same eigenvalue 1?

What are the other eigenvalues, and what are the corresponding eigenvectors of A? If $v = [a, b, c]^t \neq 0$ is an eigenvector for an eigenvalue λ, then the equation $Av = \lambda v$ implies that

$$a + b + c = \lambda a, \quad b + c = \lambda b, \quad c = \lambda c.$$

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The last equation says that either $\lambda = 1$ or $c = 0$. If $c = 0$, then the second equation implies that either $\lambda = 1$ or $b = 0$. Then the first equation yields either $\lambda = 1$ or $a = 0$. Now, $c = 0,\ b = 0,\ a = 0$ is not possible, since it would lead to $v = 0$. In any case, $\lambda = 1$. Then the equations are simplified to

$$a + b + c = a, \quad b + c = b, \quad c = c.$$

It implies that $b = c = 0$. Then $v = [a, 0, 0]^t$ for any nonzero a. That is, such a vector v is an eigenvector for the only eigenvalue 1. □

Example 5.2 Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{bmatrix}$. We see that $A \begin{bmatrix} a \\ b \\ c \end{bmatrix} = (a + b + c) \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$. Therefore, when $a + b + c = 0$, we have 0 as an eigenvalue with eigenvector $[a, b, c]$. Verify that the vectors $[1, 0, -1]^t$, $[0, 1, -1]^t$ are eigenvectors for the eigenvalue 0.

For $a = 1,\ b = 2,\ c = 3$, we see that an eigenvalue is $a + b + c = 6$ with a corresponding eigenvector as $[1, 2, 3]^t$.

Does A have eigenvalues other than 0 and 6? □

Corresponding to an eigenvalue, there are infinitely many eigenvectors.

Exercises for Sect. 5.1

1. Suppose $A \in \mathbb{F}^{n\times n}$, $\lambda \in \mathbb{F}$, and $b \in \mathbb{F}^{n\times 1}$ are such that $(A - \lambda I)x = b$ has a unique solution. Show that λ is not an eigenvalue of A.
2. Formulate a converse of the statement in Exercise 1 and prove it.
3. Let $A \in \mathbb{F}^{n\times n}$. Show that a scalar $\lambda \in \mathbb{F}$ is an eigenvalue of A iff the map $A - \lambda I : \mathbb{F}^{n\times 1} \to \mathbb{F}^{n\times 1}$ is not one-one.
4. Let $A \in \mathbb{F}^{n\times n}$. Show that A is invertible iff 0 is not an eigenvalue of A.
5. Let $A \in \mathbb{F}^{n\times n}$. If the entries in each column of A add up to a scalar λ, then find an eigenvector of A.

5.2 The Characteristic Polynomial

Eigenvalues of small matrices can be computed using the following theorem.

Theorem 5.1 *Let $A \in \mathbb{F}^{n\times n}$. A scalar $\lambda \in \mathbb{F}$ is an eigenvalue of A iff* $\det(A - \lambda I) = 0$.

Proof A scalar $\lambda \in \mathbb{F}$ is an eigenvalue of A
iff $Av = \lambda v$ for some $v \neq 0$
iff $(A - \lambda I)v = 0$ for some $v \neq 0$
iff the linear system $(A - \lambda I)x = 0$ has a nonzero solution
iff $\operatorname{rank}(A - \lambda I) < n$
iff $\det(A - \lambda I) = 0$. ■

The polynomial $(-1)^n \det(A - tI)$, which is also equal to $\det(tI - A)$, is called the **characteristic polynomial** of the matrix A; and it is denoted by $\chi_A(t)$. Each eigenvalue of A is a zero of the characteristic polynomial of A. Conversely, each zero of the characteristic polynomial is said to be a **complex eigenvalue** of A.

If A is a complex matrix of order n, then $\chi_A(t)$ is a polynomial of degree n in t. The *fundamental theorem of algebra* states that any polynomial of degree n, with complex coefficients, can be written as a product of n linear factors each with complex coefficients. Thus, $\chi_A(t)$ has exactly n, not necessarily distinct, zeros in $\mathbb{C}$. And these are the eigenvalues (complex eigenvalues) of the matrix A.

If A is a matrix with real entries, some of the zeros of $\chi_A(t)$ may turn out to be complex numbers with nonzero imaginary parts. Considering A as a linear transformation on $\mathbb{R}^{n\times 1}$, the scalars are now real numbers. Thus each zero of the characteristic polynomial may not be an eigenvalue; only the real zeros are.

However, if we regard a real matrix A as a matrix with complex entries, then A is a linear transformation on $\mathbb{C}^{n\times 1}$. Each complex eigenvalue, that is, a zero of the characteristic polynomial of A, is an eigenvalue of A.

Due to this obvious advantage, we consider a matrix in $\mathbb{R}^{n\times n}$ as one in $\mathbb{C}^{n\times n}$ so that each root of the characteristic polynomial of a matrix is considered an eigenvalue of the matrix. In this sense, an eigenvalue is taken as a complex eigenvalue, by default. Observe that when λ is an (a complex) eigenvalue of $A \in \mathbb{F}^{n\times n}$, a corresponding eigenvector x is a vector in $\mathbb{C}^{n\times 1}$, in general.

Example 5.3 Find the eigenvalues and corresponding eigenvectors of the matrix

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}.$$

Here, $\chi_A(t) = (-1)^3 \det(A - tI) = -\begin{vmatrix} 1-t & 0 & 0 \\ 1 & 1-t & 0 \\ 1 & 1 & 1-t \end{vmatrix} = (t-1)^3.$

The eigenvalues of A are its zeros, that is, $1,\ 1,\ 1$. To get an eigenvector, we solve $A[a, b, c]^t = [a, b, c]^t$ or that

$$a = a,\ a + b = b,\ a + b + c = c.$$

It gives $a = b = 0$, and $c \in \mathbb{F}$ can be arbitrary. Since an eigenvector is nonzero, all the eigenvectors are given by $(0, 0, c)^t$, for $c \neq 0$. □

Example 5.4 For the matrix $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \in \mathbb{R}^{2\times 2}$, the characteristic polynomial is $\chi_A(t) = (-1)^2 \det(A - tI) = t^2 + 1$. It has no real zeros. Thus, A has no real eigenvalues. However, i and $-i$ are its complex eigenvalues. That is, the same matrix $A \in \mathbb{C}^{2\times 2}$ has eigenvalues as i and $-i$. The corresponding eigenvectors are obtained by solving

$$A[a, b]^t = i[a, b]^t \quad \text{and} \quad A[a, b] = -i[a, b]^t.$$

For $\lambda = i$, we have $b = ia$, $-a = ib$. Thus, $[a,\ ia]^t$ is an eigenvector for $a \neq 0$. For the eigenvalue $-i$, the eigenvectors are $[a,\ \ -ia]$ for $a \neq 0$.

We consider A as a matrix with complex entries. With this convention, the matrix A has (complex) eigenvalues i and $-i$. □

A polynomial with the coefficient of the highest degree as 1 is called a *monic polynomial*. In the characteristic polynomial, the factor $(-1)^n$ is multiplied with the determinant to make the result a monic polynomial. We see that

$$\chi_A(t) = (-1)^n \det(A - tI) = \det(tI - A) = t^n + a_{n-1}t^{n-1} + \cdots + a_1 t + a_0$$

for some scalars $a_0, \ldots, a_{n-1}$. If $\lambda_1, \ldots, \lambda_n$ are the complex eigenvalues of A, counting multiplicities, then

$$\chi_A(t) = (t - \lambda_1) \cdots (t - \lambda_n).$$

Exercises for Sect. 5.2

1. Find all eigenvalues and corresponding eigenvectors of the following matrices:

   (a) $\begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & -2 \\ 0 & 0 & 2 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 \end{bmatrix}$

2. Find all eigenvalues and their corresponding eigenvectors of the $n \times n$ matrix A whose jth row has each entry j.
3. Let $p(t) = t^3 - \alpha t^2 - (\alpha + 3)t - 1$ for a real number α. Consider

$$A = \begin{bmatrix} \alpha & \alpha + 3 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \quad P = \begin{bmatrix} -1 & 2 & -\alpha - 3 \\ 1 & -1 & \alpha + 2 \\ -1 & 1 & -\alpha - 1 \end{bmatrix}.$$

   (a) Compute the characteristic polynomials of A and $P^{-1}AP$.
   (b) Show that all zeros of $p(t)$ are real regardless of the value of α.

5.3 The Spectrum

If the characteristic polynomial of a 3×3 matrix A is $\chi_A(t) = (t - 1)^2(t - 2)$, then its eigenvalues are 1 and 2. Thus, the set of eigenvalues $\{1, 2\}$ does not give information as to how many times an eigenvalue is a repeated zero of the characteristic polynomial. In this case, the notion of a multiset comes of help. The sets $\{1, 1, 2\}$ and $\{1, 2, 2\}$ are both equal to the set $\{1, 2\}$, but the multiset $\{1, 1, 2\}$ is different from the multiset $\{1, 2, 2\}$. Again, the multiset $\{1, 1, 2\}$ is same as the multiset $\{1, 2, 1\}$. That is, the number of times an element occurs in a multiset is significant whereas the ordering of elements in a multiset is ignored.

The multiset of all complex eigenvalues of a matrix $A \in \mathbb{F}^{n\times n}$ is called the **spectrum** of A; and we denote it by $\sigma(A)$. If $A \in \mathbb{F}^{n\times n}$ has the (complex) eigenvalues $\lambda_1, \ldots, \lambda_n$, counting multiplicities, then the spectrum of A is $\sigma(A) = \{\lambda_1, \ldots, \lambda_n\}$; and vice versa. Notice that the spectrum of $A \in \mathbb{F}^{n\times n}$ always has n elements; though they may not be distinct. The following theorem lists some important facts about eigenvalues, using the notion of spectrum.

Theorem 5.2 *Let $A \in \mathbb{F}^{n\times n}$. Then the following are true.*

1. $\sigma(A^t) = \sigma(A)$.
2. *If B is similar to A, then* $\sigma(B) = \sigma(A)$.
3. *If $A = [a_{ij}]$ is a diagonal or an upper triangular or a lower triangular matrix, then* $\sigma(A) = \{a_{11}, \ldots, a_{nn}\}$.
4. *If* $\sigma(A) = \{\lambda_1, \ldots, \lambda_n\}$, *then* $\det(A) = \lambda_1 \cdots \lambda_n$ *and* $\operatorname{tr}(A) = \lambda_1 + \cdots + \lambda_n$.

Proof (1) $\chi_{A^t}(t) = \det(A^t - tI) = \det((A - tI)^t) = \det(A - tI) = \chi_A(t)$.
(2) $\chi_{P^{-1}AP}(t) = \det(P^{-1}AP - tI) = \det(P^{-1}(A - tI)P) = \det(P^{-1})\det(A - tI)$ $\det(P) = \det(A - tI) = \chi_A(t)$.
(3) In all these cases, $\chi_A(t) = \det(A - tI) = (a_{11} - t)\cdots(a_{nn} - t)$.
(4) Let $\sigma(A) = \{\lambda_1, \ldots, \lambda_n\}$. Then

$$\chi_A(t) = \det(A - tI) = (\lambda_1 - t)\cdots(\lambda_n - t). \tag{5.1}$$

Substituting $t = 0$, we obtain $\det(A) = \lambda_1 \cdots \lambda_n$.
For the other equality, we compute the coefficient of t^{n-1} using (5.1). Expanding $\det(A - tI)$ in its first row, we find that the first term in the expansion is $(a_{11} - t)\det(A_{11})$, where A_{11} is the minor corresponding to the $(1, 1)$th entry. All other terms are polynomials of degree less than or equal to $n - 2$.

Next, we expand $\det(A_{11})$; and continue in a similar way to obtain the following:
Coefficient of t^{n-1} in $\det(A - tI)$
$=$ Coefficient of t^{n-1} in $(a_{11} - t) \times \det(A_{11})$
$= \cdots =$ Coefficient of t^{n-1} in $(a_{11} - t) \times (a_{22} - t) \times \cdots \times (a_{nn} - t)$
$= (-1)^{n-1}(a_{11} + \cdots + a_{nn})$.
$=$ Coefficient of t^{n-1} in $(\lambda_1 - t)\cdots(\lambda_n - t)$, using (5.1)
$= (-1)^{n-1}(\lambda_1 + \cdots + \lambda_n)$.
Therefore, $\lambda_1 + \cdots + \lambda_n = a_{11} + \cdots + a_{nn} = \operatorname{tr}(A)$. ■

Theorem 5.3 *Eigenvectors associated with distinct eigenvalues of a matrix are linearly independent.*

Proof Let $\lambda_1, \ldots, \lambda_k$ be the distinct eigenvalues of a matrix A, and let $v_1, \ldots, v_k$ be corresponding eigenvectors. We use induction on $j \in \{1, \ldots, k\}$.

For $j = 1$, since $v_1 \neq 0$, $\{v_1\}$ is linearly independent. Lay out the induction hypothesis (for $j = m$): the set $\{v_1, \ldots, v_m\}$ is linearly independent. To show that (for $j = m + 1$): the set $\{v_1, \ldots, v_m, v_{m+1}\}$ is linearly independent, let

$$\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_m v_m + \alpha_{m+1} v_{m+1} = 0. \tag{5.2}$$

Then, $A(\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_m v_m + \alpha_{m+1} v_{m+1}) = 0$ gives (since $Av_i = \lambda_i v_i$)

$$\alpha_1 \lambda_1 v_1 + \alpha_2 \lambda_2 v_2 + \cdots + \alpha_m \lambda_m v_m + \alpha_{m+1} \lambda_{m+1} v_{m+1} = 0.$$

Multiply (5.2) with λ_{m+1} and subtract from the last equation to get

$$\alpha_1(\lambda_1 - \lambda_{m+1})v_1 + \cdots + \alpha_m(\lambda_m - \lambda_{m+1})v_m = 0.$$

By the Induction Hypothesis, $\alpha_i(\lambda_i - \lambda_{m+1}) = 0$ for $i = 1, 2, \ldots, m$. Since $\lambda_i \neq \lambda_{m+1}$, $\alpha_i = 0$ for each such i. Then, (5.2) yields $\alpha_{m+1} v_{m+1} = 0$. Then $v_{m+1} \neq 0$ implies $\alpha_{m+1} = 0$. This completes the proof of linear independence of eigenvectors associated with distinct eigenvalues. ■

Exercises for Sect. 5.3

1. Let A be a 4×4 non-invertible matrix with all diagonal entries 1. If $2 + 3i$ is an eigenvalue of A, what are its other eigenvalues?
2. Show that if each eigenvalue of $A \in \mathbb{F}^{n \times n}$ has absolute value less than 1, then both $I - A$ and $I + A$ are invertible.
3. Show that if rank of an $n \times n$ matrix is 1, then its trace is one of its eigenvalues. What are its other eigenvalues?
4. Find the spectrum of a matrix where the sum of entries in each row is same.
5. Let $A, B, P \in \mathbb{C}^{n \times n}$ be such that $B = P^{-1}AP$. Let λ be an eigenvalue of A. Show that a vector v is an eigenvector of B corresponding to the eigenvalue λ iff Pv is an eigenvector of A corresponding to the same eigenvalue λ.
6. Do equivalent matrices have same eigenvalues?

5.4 Special Types of Matrices

A square matrix A is called **hermitian** iff $A^* = A$; **skew hermitian** iff $A^* = -A$; **symmetric** iff $A^t = A$; and **skew symmetric** iff $A^t = -A$.

Thus, a *real symmetric* matrix is real hermitian; and a *real skew symmetric* matrix is real skew hermitian. Hermitian matrices are also called *self-adjoint* matrices. In the following, B is symmetric, C is skew symmetric, D is hermitian, and E is skew hermitian; B is also hermitian and C is also skew hermitian.

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}, \; C = \begin{bmatrix} 0 & 2 & -3 \\ -2 & 0 & 4 \\ 3 & -4 & 0 \end{bmatrix}, \; D = \begin{bmatrix} 1 & -2i & 3 \\ 2i & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}, \; E = \begin{bmatrix} 0 & 2+i & 3 \\ 2-i & i & 4i \\ 3 & -4i & 0 \end{bmatrix}.$$

Notice that all diagonal entries of a hermitian matrix are real since $\overline{a}_{ii} = a_{ii}$. Similarly, each diagonal entry of a skew symmetric matrix must be zero, since $a_{ii} = -a_{ii}$. And each diagonal entry of a skew hermitian matrix must be 0 or purely imaginary, as $\overline{a}_{ii} = -a_{ii}$ implies $2\mathrm{Re}(a_{ii}) = 0$.

Let A be a square matrix. Since $A + A^t$ is symmetric and $A - A^t$ is skew symmetric, every square matrix can be written as a sum of a symmetric matrix and a skew symmetric matrix:

$$A = \tfrac{1}{2}(A + A^t) + \tfrac{1}{2}(A - A^t).$$

Similar rewriting is possible with hermitian and skew hermitian matrices:

$$A = \tfrac{1}{2}(A + A^*) + \tfrac{1}{2}(A - A^*).$$

A square matrix A is called **unitary** iff $A^*A = I = AA^*$. In addition, if A is real, then it is called an orthogonal matrix. That is, an **orthogonal matrix** is a matrix with real entries satisfying $A^tA = I = AA^t$.

Thus, a square matrix is unitary iff it is invertible and its inverse is equal to its adjoint. Similarly, a real matrix is orthogonal iff it is invertible and its inverse is its transpose. In the following, B is a unitary matrix of order 2, and C is an orthogonal matrix (also unitary) of order 3:

$$B = \frac{1}{2}\begin{bmatrix} 1+i & 1-i \\ 1-i & 1+i \end{bmatrix}, \quad C = \frac{1}{3}\begin{bmatrix} 2 & 1 & 2 \\ -2 & 2 & 1 \\ 1 & 2 & -2 \end{bmatrix}.$$

Example 5.5 For any $\theta \in \mathbb{R}$, the following are orthogonal matrices of order 2:

$$O_1 := \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}, \quad O_2 := \begin{bmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{bmatrix}.$$

Let u be the vector in the plane that starts at the origin and ends at the point (a, b). Writing the point (a, b) as a column vector $[a,\ b]^t$, we see that

$$O_1\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} a\cos\theta - b\sin\theta \\ a\sin\theta + b\cos\theta \end{bmatrix}, \quad O_2\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} a\cos\theta + b\sin\theta \\ a\sin\theta - b\cos\theta \end{bmatrix}.$$

Thus, $O_1[a,\ b]^t$ is the end-point of the vector obtained by rotating the vector u by an angle θ. Similarly, $O_2[a,\ b]^t$ is a point obtained by reflecting u along a straight line that makes an angle $\theta/2$ with the x-axis. Accordingly, O_1 is called a *rotation* by an angle θ, and O_2 is called a *reflection* along a line making an angle of $\theta/2$ with the x-axis.

If $A = [a_{ij}]$ is an orthogonal matrix of order 2, then $A^tA = I$ implies

$$a_{11}^2 + a_{21}^2 = 1 = a_{12}^2 + a_{22}^2, \quad a_{11}a_{12} + a_{21}a_{22} = 0.$$

Then there exist $\alpha, \beta \in \mathbb{R}$ such that $a_{11} = \cos\alpha,\ a_{21} = \sin\alpha,\ a_{12} = \cos\beta, a_{22} = \sin\beta$ and $\cos(\alpha - \beta) = 0$. Thus $\alpha - \beta = \pm(\pi/2)$. It then follows that A is in either

of the forms O_1 or O_2. That is, an orthogonal matrix of order 2 is either a rotation or a reflection. □

Theorem 5.4 *Let A be a unitary or an orthogonal matrix of order n. Then the following are true:*

1. *For each pair of vectors $x, y \in \mathbb{F}^{n\times 1}$, $\langle Ax, Ay\rangle = \langle x, y\rangle$.*
2. *For each $x \in \mathbb{F}^{n\times 1}$, $\|Ax\| = \|x\|$.*
3. *The columns of A are orthonormal.*
4. *The rows of A are orthonormal.*
5. $|\det(A)| = 1$.

Proof (1) $\langle Ax, Ay\rangle = (Ay)^*Ax = y^*A^*Ax = y^*x = \langle x, y\rangle$.
(2) Take $x = y$ in (1).
(3) Let the columns of A be $u_1, \ldots, u_n$ in that order. Then $\langle u_j, u_i\rangle = u_i^* u_j$ is the (i, j)th entry of A^*A. Since $A^*A = I$, $\langle u_j, u_i\rangle = \delta_{ij}$. Therefore, $\{u_1, \ldots, u_n\}$ is an orthonormal set.
(4) Considering $AA^* = I$, we get this result.
(5) Notice that $\det(A^*) = \det(\overline{A}) = \overline{\det(A)}$. Thus

$$\det(A^*A) = \det(A^*)\det(A) = \overline{\det(A)}\det(A) = |\det(A)|^2.$$

However, $\det(A^*A) = \det(I) = 1$. Therefore, $|\det(A)| = 1$. ■

Theorem 5.4(1)–(2) show that unitary and orthogonal matrices preserve the inner product and the norm. Thus they are also called *isometries*. The condition $A^*A = I$ is equivalent to the condition that the columns of A are orthonormal. Similarly, the rows of A are orthonormal iff $AA^* = I$.

Theorem 5.4 implies that the determinant of an orthogonal matrix is either 1 or -1. It follows that the product of all eigenvalues, counting multiplicities, of an orthogonal matrix is ± 1.

The determinant of a hermitian matrix is a real number since $A = A^*$ implies

$$\det(A) = \det(A^*) = \det(\overline{A}) = \overline{\det(A)}.$$

We prove some interesting facts about the eigenvalues and eigenvectors of these special types of matrices.

Theorem 5.5 *Let $A \in \mathbb{F}^{n\times n}$ and let λ be any eigenvalue of A.*

1. *If A is hermitian, then $\lambda \in \mathbb{R}$; and eigenvectors corresponding to distinct eigenvalues of A are orthogonal.*
2. *If A is real symmetric, then $\lambda \in \mathbb{R}$; and there exists a real eigenvector corresponding to λ.*
3. *If A is skew hermitian or real skew symmetric, then λ is purely imaginary or zero.*
4. *If A is unitary or orthogonal, then $|\lambda| = 1$.*

Proof (1) Let A be hermitian; that is, $A^* = A$. Let $v \in \mathbb{C}^{n\times 1}$ be an eigenvector corresponding to the eigenvalue λ of A. Then

$$\lambda v^* v = v^*(\lambda v) = v^* A v = v^* A^* v = (Av)^* v = (\lambda v)^* v = \overline{\lambda} v^* v.$$

Since $v \neq 0$, we have $v^* v \neq 0$; consequently, $\overline{\lambda} = \lambda$. Therefore, $\lambda \in \mathbb{R}$.

For the second conclusion, let $\lambda \neq \mu$ be two distinct eigenvalues of A with corresponding eigenvectors as x and y. Both $\lambda \in \mathbb{R}$ and $\mu \in \mathbb{R}$. Then

$$\lambda y^* x = y^* \lambda x = y^* A x = y^* A^* x = (Ay)^* x = (\mu y)^* x = \overline{\mu} y^* x = \mu y^* x.$$

Hence $(\lambda - \mu) y^* x = 0$. As $\lambda \neq \mu$, we conclude that $y^* x = 0$. That is, $x \perp y$.
(2) Let A be real symmetric. Then $\lambda \in \mathbb{R}$ follows from (1). For the second statement, let $v = x + iy \in \mathbb{C}^{n\times 1}$ be an eigenvector of A corresponding to λ, with $x, y \in \mathbb{R}^{n\times 1}$. Then

$$A(x + iy) = \lambda(x + iy).$$

Comparing the real and imaginary parts, we have

$$Ax = \lambda x, \quad Ay = \lambda y.$$

Since $x + iy \neq 0$, at least one of x or y is nonzero. One such nonzero vector is a real eigenvector corresponding to the (real) eigenvalue λ of A.
(3) Let A be skew hermitian. That is, $A^* = -A$. Let $v \in \mathbb{C}^{n\times 1}$ be an eigenvector corresponding to the eigenvalue λ of A. Then

$$\lambda v^* v = v^*(\lambda v) = v^* A v = -v^* A^* v = -(Av)^* v = -(\lambda v)^* v = -\overline{\lambda} v^* v.$$

Since $v^* v \neq 0$, $\lambda = -\overline{\lambda}$. That is, $2\mathrm{Re}(\lambda) = 0$. This shows that λ is purely imaginary or zero.

When A is real skew symmetric, we take transpose instead of adjoint everywhere in the above proof.
(4) Suppose A is unitary, i.e., $A^* A = I$. Let v be an eigenvector corresponding to the eigenvalue λ. Now, $Av = \lambda v$ implies $v^* A^* = (\lambda v)^* = \overline{\lambda} v^*$. Then

$$v^* v = v^* I v = v^* A^* A v = \overline{\lambda}\lambda v^* v = |\lambda|^2 v^* v.$$

Since $v^* v \neq 0$, $|\lambda| = 1$.

Replacing A^* with A^t in the above proof yields the same conclusion when A is an orthogonal matrix. ■

Existence of a real eigenvector corresponding to a hermitian matrix cannot be guaranteed, in general.

Example 5.6 Consider the matrix $A = \begin{bmatrix} 1 & 1-i \\ 1+i & -1 \end{bmatrix}$. It is a hermitian matrix. Its characteristic polynomial is $\chi_A(t) = t^2 - 3$. Its eigenvalues are $\pm\sqrt{3}$. To compute an eigenvector corresponding to the eigenvalue $\sqrt{3}$, we set up the linear system $(A - \sqrt{3}I)[a,\ b]^t = 0$. That is,

$$(1 - \sqrt{3})a + (1 - i)b = 0, \quad (1 + i)a - (1 + \sqrt{3})b = 0.$$

This equation has a nonzero solution $a = 1 - i,\ b = \sqrt{3} - 1$. So, the vector $v = [1 - i,\ \sqrt{3} - 1]^t$ is an eigenvector corresponding to the eigenvalue $\sqrt{3}$.

If both $a, b \in \mathbb{R}$, then comparing the imaginary parts in the equations, we get $-b = 0$ and $a = 0$. But that is impossible since $v \neq 0$. Hence, there does not exist a real eigenvector for the eigenvalue $\sqrt{3}$ of A. □

Similarly, a real skew symmetric matrix can have purely imaginary eigenvalues. The only real eigenvalue of a real skew symmetric matrix is 0. Also, an orthogonal matrix can have complex eigenvalues so that each eigenvalue need not be ± 1. For instance, the matrix

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

has eigenvalues $\pm i$. However, all real eigenvalues of an orthogonal matrix are ± 1; and any eigenvalue of an orthogonal matrix is of the form $e^{i\theta}$ for $\theta \in \mathbb{R}$.

Exercises for Sect. 5.4

1. Let $A \in \mathbb{F}^{n\times n}$. Show that for all $x, y \in \mathbb{F}^{n\times 1}$, $\langle Ax, Ay\rangle = \langle x, y\rangle$ iff for all $x \in \mathbb{F}^{n\times 1}$, $\|Ax\| = \|x\|$.
2. Construct a 3×3 hermitian matrix with no zero entries whose eigenvalues are $1,\ 2$ and 3.
3. Construct a 2×2 real skew symmetric matrix whose eigenvalues are purely imaginary.
4. Show that if an invertible matrix is real symmetric, then so is its inverse.
5. Show that if an invertible matrix is hermitian, then so is its inverse.
6. Construct an orthogonal 2×2 matrix whose determinant is 1.
7. Construct an orthogonal 2×2 matrix whose determinant is -1.

5.5 Problems

1. Let $a + bi$ be an eigenvalue of $A \in \mathbb{R}^{n\times n}$ with an associated eigenvector $x + iy$, where $a, b \in \mathbb{R}$ and $x, y \in \mathbb{R}^{n\times 1}$. Show the following:

   (a) The vectors $x + iy$ and $x - iy$ are linearly independent in $\mathbb{C}^{n\times 1}$.
   (b) The vectors x and y are linearly independent in $\mathbb{R}^{n\times 1}$.

2. Let $A = [a_{ij}] \in \mathbb{R}^{2\times 2}$, where $a_{12}a_{21} > 0$. Write $B = \operatorname{diag}\big(\sqrt{a_{21}/a_{12}},\ 1\big)$. Let $C = B^{-1}AB$. What do you conclude about the eigenvalues and eigenvectors of C; and those of A?
3. Show that if λ is a nonzero eigenvalue of an $n \times n$ matrix A, then $N(A - \lambda I) \subseteq R(A)$.
4. An $n \times n$ matrix A is said to be *idempotent* if $A^2 = A$. Show that the only possible eigenvalues of an idempotent matrix are 0 or 1.
5. An $n \times n$ matrix A is said to be *nilpotent* if $A^k = 0$ for some natural number k. Show that 0 is the only eigenvalue of a nilpotent matrix.
6. Let $A \in \mathbb{F}^{n\times n}$ have an eigenvalue λ with an associated eigenvector x. Suppose A^* has an eigenvalue μ associated with y. If $\lambda \neq \overline{\mu}$, then show that $x \perp y$.
7. Let $A, B \in \mathbb{F}^{n\times n}$. Show that

$$\begin{bmatrix} I & A \\ 0 & I \end{bmatrix}\begin{bmatrix} 0 & 0 \\ B & BA \end{bmatrix}\begin{bmatrix} I & -A \\ 0 & I \end{bmatrix} = \begin{bmatrix} AB & 0 \\ B & 0 \end{bmatrix}.$$

   Then deduce that $\chi_{AB}(t) = \chi_{BA}(t)$.
8. Let $a_0, a_1, \ldots, a_{n-1} \in \mathbb{F}$. Let $p(t) = t^n - a_{n-1}t^{n-1} - \cdots - a_0$ be a polynomial. Construct the matrix

$$C = \begin{bmatrix} a_{n-1} & a_{n-2} & \cdots & a_1 & a_0 \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ & & \ddots & & \\ 0 & 0 & \cdots & 1 & 0 \end{bmatrix}.$$

   The matrix C is called the *companion matrix* of the polynomial $p(t)$. Show the following:

   (a) $\chi_C(t) = p(t)$.
   (b) If $p(\lambda) = 0$ for some $\lambda \in \mathbb{F}$, then λ is an eigenvalue of the matrix C with an associated eigenvector $[\lambda^{n-1},\ \lambda^{n-2}, \cdots, \lambda, 1]^t$.

9. Let $A \in \mathbb{F}^{n\times n}$ and let $B = I - 2A + A^2$. Show the following:

   (a) If 1 is an eigenvalue of A, then B is not invertible.
   (b) If v is an eigenvector of A, then v is an eigenvector of B. Are the corresponding eigenvalues equal?

10. For which scalars α, are the $n \times n$ matrices A and $A + \alpha I$ similar?
11. Show that there do not exist $n \times n$ matrices A and B with $AB - BA = I$.
12. Let $A, B \in \mathbb{F}^{n\times n}$. Show that if 1 is not an eigenvalue of A, then the matrix equation $AX + B = X$ has a unique solution in $\mathbb{F}^{n\times n}$.
13. Let A and B be hermitian matrices. Show that AB is hermitian iff $AB = BA$.
14. Let A and B be hermitian matrices. Determine whether the matrices $A + B$, ABA, $AB + BA$, and $AB - BA$ are hermitian or not.

15. Let A be a non-hermitian matrix. Determine whether the matrices $A + A^*$, $A - A^*$, $A^*A - AA^*$, $(I + A)(A + A^*)$, and $(I + A)(I - A^*)$ are hermitian or not.
16. Show that a square matrix A is invertible iff $\det(A^*A) > 0$.
17. Let A be a skew hermitian matrix. Show that $I - A$ is invertible.
18. Let $A \in \mathbb{F}^{m\times n}$ have rank n. Let $P = A(A^*A)^{-1}A^*$. Show the following:

    (a) A^*A is invertible so that the construction of P is meaningful.
    (b) P is hermitian.
    (c) $Py = y$ for each $y \in R(A)$.
    (d) $P^k y = Py$ for each $y \in \mathbb{F}^{m\times 1}$ and for each $k \in \mathbb{N}$.
    (e) Let $y \in \mathbb{F}^{m\times 1}$ satisfy $y \perp Ax$ for each $x \in \mathbb{F}^{n\times 1}$. Then $Py = 0$.

19. Let $B \in \mathbb{F}^{n\times n}$. Suppose that $I + B$ is invertible. Let $C = (I + B)^{-1}(I - B)$. Show the following:

    (a) If B is skew hermitian, then C is unitary.
    (b) If B is unitary, then C is skew hermitian.

20. Show that in the plane,

    (a) a rotation following a rotation is a rotation.
    (b) a rotation following a reflection is a reflection.
    (c) a reflection following a rotation is a reflection.
    (d) a reflection following a reflection is a rotation.

21. Show that the eigenvalues of a reflection in the plane are ± 1. What could be the eigenvalues of a rotation?
22. Let $A \in \mathbb{R}^{2\times 2}$ be an orthogonal matrix. Suppose that A has a non-trivial *fixed point*; that is, there exists a nonzero vector $v \in \mathbb{R}^{2\times 1}$ such that $Av = v$. Show that with respect to any orthonormal basis B of $\mathbb{R}^{2\times 1}$, the coordinate matrix $[A]_{B,B}$ is in the form
$$\begin{bmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{bmatrix}.$$
23. Show that 1 is an eigenvalue of any 3×3 orthogonal matrix.
24. Using the previous problem, show that an orthogonal matrix of order 3 is similar to a matrix of the form $\begin{bmatrix} 1 & 0 \\ 0 & R \end{bmatrix}$, where R is either a reflection or a rotation in the plane.
25. Let A be an orthogonal matrix of order 3. Show that either $\sigma(A) = \{1, 1, 1\}$ or $\sigma(A) = \{1, e^{i\theta}, e^{-i\theta}\}$ for some $\theta \in \mathbb{R}$.
26. Show that the ith column of an orthogonal upper triangular matrix is either e_i or $-e_i$.
27. Let A be an orthogonal matrix with an eigenvalue 1 and a corresponding eigenvector v. Show that v is also an eigenvector of A^t.
28. Let $u_1, \ldots, u_n$ be the columns of an orthogonal matrix of order n. Show that $u_1u_1^t + \cdots + u_nu_n^t = I$.

29. Let $\{v_1, \ldots, v_n\}$ be an orthonormal set in $\mathbb{F}^{n\times 1}$. Let $\lambda_1, \ldots, \lambda_n \in \mathbb{R}$. Show that the matrix $A = \lambda_1 v_1 v_1^* + \cdots + \lambda_n v_n v_n^*$ is hermitian. Can you find eigenvalues and eigenvectors of A?
30. A *permutation matrix* of order n is an $n \times n$ matrix obtained from the identity matrix I_n by reordering its rows. Show the following:

    (a) A permutation matrix may be obtained by reordering the columns of the identity matrix.
    (b) A permutation matrix is a product of elementary matrices of the first type.
    (c) Let P be a permutation matrix of order m whose ith row is the k_ith row of I_m. Let A be an $m \times n$ matrix. Then PA is the $m \times n$ matrix whose ith row is the k_ith row of A.
    (d) Let Q be a permutation matrix of order n whose jth column is the k_jth column of I_n. Let A be an $m \times n$ matrix. Then AQ is the $m \times n$ matrix whose jth column is the k_jth column of A.
    (e) A permutation matrix is orthogonal.
    (f) If a permutation matrix P is symmetric, then $P^{2k} = I$ and $P^{2k+1} = P$ for $k = 0,\ 1,\ 2, \ldots$.

Chapter 6
Canonical Forms



6.1 Schur Triangularization

Eigenvalues and eigenvectors can be used to bring a matrix to nice forms using similarity transformations. A very general result in this direction is Schur's unitary triangularization. It says that using a suitable similarity transformation, we can represent a square matrix by an upper triangular matrix. Thus, the diagonal entries of the upper triangular matrix must be the eigenvalues of the given matrix.

Theorem 6.1 (Schur Triangularization) *Let $A \in \mathbb{C}^{n\times n}$. Then there exists a unitary matrix $P \in \mathbb{C}^{n\times n}$ such that P^*AP is upper triangular. Moreover, if $A \in \mathbb{R}^{n\times n}$ and all eigenvalues of A are real, then P can be chosen to be an orthogonal matrix.*

Proof We use induction on n. If $n = 1$, then clearly A is an upper triangular matrix, and we take $P = [1]$, the identity matrix with a single entry as 1, which is both unitary and orthogonal.

Assume that for all $B \in \mathbb{C}^{m\times m}$, $m \geq 1$, we have a unitary matrix $Q \in \mathbb{C}^{m\times m}$ such that Q^*BQ is upper triangular. Let $A \in \mathbb{C}^{(m+1)\times(m+1)}$. Let $\lambda \in \mathbb{C}$ be an eigenvalue of A with an associated eigenvector u. Recall that $\langle w, z\rangle = z^*w$ defines an inner product on $\mathbb{C}^{(m+1)\times 1}$. Then the unit vector $v = u/\|u\|$ is an eigenvector of A associated with the eigenvalue λ.

Extend the set $\{v\}$ to an orthonormal (ordered) basis $E = \{v, v_1, \ldots, v_m\}$ for $\mathbb{C}^{(m+1)\times 1}$. Here, you may have to use an extension of a basis, and then Gram–Schmidt orthonormalization process. Next, construct the matrix $R \in \mathbb{C}^{(m+1)\times(m+1)}$ by taking these basis vectors as its columns, in that order. That is, let

$$R = \begin{bmatrix} v & v_1 & \cdots & v_m \end{bmatrix}.$$

Since E is an orthonormal set, R is unitary. Consider the coordinate matrix of A with respect to the basis E. It is given by $[A]_{E,E} = R^{-1}AR = R^*AR$. The first column of R^*AR is

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$$R^*ARe_1 = R^*Av = R^{-1}\lambda v = \lambda R^{-1}v = \lambda R^{-1}Re_1 = \lambda e_1,$$

where $e_1 \in \mathbb{C}^{(m+1)\times 1}$ has first component as 1 and all other components 0. Then R^*AR can be written in the following block form:

$$R^*AR = \begin{bmatrix} \lambda & x \\ 0 & C \end{bmatrix},$$

where $0 \in \mathbb{C}^{m\times 1}$, $C \in \mathbb{C}^{m\times m}$, and $x = [v^*Av_1 \quad v^*Av_2 \quad \cdots \quad v^*Av_m] \in \mathbb{C}^{1\times m}$.

Notice that if $m = 1$, the proof is complete. For $m > 1$, by induction hypothesis, we have a matrix $S \in \mathbb{C}^{m\times m}$ such that S^*CS is upper triangular. Then take

$$P = R\begin{bmatrix} 1 & 0 \\ 0 & S \end{bmatrix}.$$

Since R and S are unitary, $P^*P = PP^* = I$;, that is, P is unitary. Moreover,

$$P^*AP = \begin{bmatrix} 1 & 0 \\ 0 & S \end{bmatrix}^* R^*AR \begin{bmatrix} 1 & 0 \\ 0 & S \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & S^* \end{bmatrix} \begin{bmatrix} \lambda & x \\ 0 & C \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & S \end{bmatrix} = \begin{bmatrix} \lambda & xS \\ 0 & S^*CS \end{bmatrix}.$$

Of course, $xS \in \mathbb{C}^{1\times m}$. Since S^*CS is upper triangular, the induction proof is complete.

When $A \in \mathbb{R}^{n\times n}$, and all eigenvalues of A are real, we use the transpose instead of the adjoint everywhere in the above proof. Thus, P can be chosen to be an orthogonal matrix. ■

To eradicate possible misunderstanding, we recall that A has only real eigenvalues means that when we consider this A as a matrix in $\mathbb{C}^{n\times n}$, all its complex eigenvalues turn out to be real numbers. This again means that all zeros of the characteristic polynomial of A are real.

We may unfold the inductive proof of Schur's triangularization as follows. Once we obtain a matrix similar to A in the form

$$\begin{bmatrix} \lambda & w \\ 0 & S^*CS \end{bmatrix},$$

we look for whether λ is still an eigenvalue of S^*CS. If so, we choose this eigenvalue over others for further reduction. In the next step, we obtain a matrix similar to A in the form

$$\begin{bmatrix} \lambda & w & y \\ 0 & \lambda & z \\ 0 & 0 & M \end{bmatrix}$$

where M is an $(n-2)\times(n-2)$ matrix. Continuing further this way, we see that a Schur triangularization of A exists, where on the diagonal of the final upper triangular matrix, equal eigenvalues occur together. This particular form

will be helpful later. In general, the eigenvalues can occur on the diagonal of the Schur form in any prescribed order, depending on our choice of eigenvalue in each step.

Example 6.1 Consider the matrix $A = \begin{bmatrix} 2 & 1 & 0 \\ 2 & 3 & 0 \\ -1 & -1 & 1 \end{bmatrix}$ for Schur triangularization.

We find that $\chi_A(t) = (t-1)^2(t-4)$. All eigenvalues of A are real; thus, there exists an orthogonal matrix P such that $P^t AP$ is upper triangular. To determine such a matrix P, we take one of the eigenvalues, say 1. An associated eigenvector of norm 1 is $v = [0, 0, 1]^t$. We extend $\{v\}$ to an orthonormal basis for $\mathbb{C}^{3\times 1}$. For convenience, we take the orthonormal basis as

$$\{[0, 0, 1]^t,\ [1, 0, 0]^t,\ [0, 1, 0]^t\}.$$

Taking the basis vectors as columns, we form the matrix R as

$$R = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}.$$

We then find that

$$R^t AR = \begin{bmatrix} 1 & -1 & -1 \\ 0 & 2 & 1 \\ 0 & 2 & 3 \end{bmatrix}.$$

Next, we triangularize the matrix $C = \begin{bmatrix} 2 & 1 \\ 2 & 3 \end{bmatrix}$. It has eigenvalues 1 and 4. The eigenvector of norm 1 associated with the eigenvalue 1 is $[1/\sqrt{2},\ -1/\sqrt{2}]^t$. We extend it to an orthonormal basis

$$\left\{[1/\sqrt{2},\ -1/\sqrt{2}]^t,\ [1/\sqrt{2},\ 1/\sqrt{2}]^t\right\}$$

for $\mathbb{C}^{2\times 1}$. Then we construct the matrix S by taking these basis vectors as its columns, that is,

$$S = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}.$$

We find that $S^t CS = \begin{bmatrix} 1 & -1 \\ 0 & 4 \end{bmatrix}$, which is an upper triangular matrix. Then

$$P = R \begin{bmatrix} 1 & 0 \\ 0 & S \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & -1/\sqrt{2} & 1/\sqrt{2} \\ 1 & 0 & 0 \end{bmatrix}.$$

Consequently, $P^tAP = \begin{bmatrix} 1 & 0 & -\sqrt{2} \\ 0 & 1 & -1 \\ 0 & 0 & 4 \end{bmatrix}$. □

Since $P^* = P^{-1}$, Schur triangularization is informally stated as follows:

Every square matrix is unitarily similar to an upper triangular matrix.

Further, there is nothing sacred about being *upper* triangular. For, given a matrix $A \in \mathbb{C}^{n\times n}$, consider using Schur triangularization of A^*. There exists a unitary matrix Q such that Q^*A^*Q is upper triangular. Then taking adjoint, we have Q^*AQ is lower triangular. Thus, the following holds:

Every square matrix is unitarily similar to a lower triangular matrix.

Analogously, a real square matrix having only real eigenvalues is also orthogonally similar to a lower triangular matrix. We remark that the lower triangular form of a matrix need not be the transpose or the adjoint of its upper triangular form.

Moreover, neither the unitary matrix P nor the upper triangular matrix P^*AP in Schur triangularization is unique. That is, there can be unitary matrices P and Q such that both P^*AP and Q^*AQ are upper triangular, and $P \neq Q$, $P^*AP \neq Q^*AQ$. The non-uniqueness stems from the choice of eigenvalues, their associated eigenvectors, and in extending those to an orthonormal basis. For instance, in Example 6.1, if you extend $\{[0, 0, 1]^t\}$ to the orthonormal basis $\{[0, 0, 1]^t, [0, 1, 0]^t, [1, 0, 0]^t\}$, then you end up with (Verify.)

$$P = \begin{bmatrix} 0 & -1/\sqrt{2} & 1/\sqrt{2} \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ 1 & 0 & 0 \end{bmatrix}, \quad P^tAP = \begin{bmatrix} 1 & 0 & -\sqrt{2} \\ 0 & 1 & 1 \\ 0 & 0 & 4 \end{bmatrix}.$$

Exercises for Sect. 6.1

1. Let $A = \begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 3 \\ 0 & 0 \end{bmatrix}$ and $C = \begin{bmatrix} 0 & 3 \\ 0 & -1 \end{bmatrix}$. Show that both B and C are Schur triangularizations of A.
2. Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 \\ 0 & 7 \end{bmatrix}$ and $C = \begin{bmatrix} 0 & 2 \\ 0 & 7 \end{bmatrix}$. Show that B is a Schur triangularization of A but C is not.
3. Determine a matrix A such that $A^*A = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 1 & i \\ 0 & -i & 1 \end{bmatrix}$.
4. Using Schur triangularization prove the *Spectral mapping theorem*: for any polynomial $p(t)$, $\sigma(p(A)) = p(\sigma(A))$. In other words, for any polynomial $p(t)$, if λ is an eigenvalue of A, then $p(\lambda)$ is an eigenvalue of $p(A)$; and all (complex) eigenvalues of $p(A)$ are of this form.

6.2 Annihilating Polynomials

Schur triangularization brings a square matrix to an upper triangular form by a similarity transformation. For an upper triangular matrix $A \in \mathbb{F}^{n\times n}$ with diagonal entries $d_1, \dots, d_n$, we wish to compute the product

$$(A - d_1 I)(A - d_2 I) \cdots (A - d_n I).$$

For example, let $A = \begin{bmatrix} d_1 & \star & \star \\ 0 & d_2 & \star \\ 0 & 0 & d_3 \end{bmatrix}$, where $\star$ stands for any entry possibly nonzero. We see that

$$\begin{aligned} A - d_1 I &= \begin{bmatrix} 0 & \star & \star \\ 0 & \star & \star \\ 0 & 0 & \star \end{bmatrix} \\ (A - d_1 I)(A - d_2 I) &= \begin{bmatrix} 0 & \star & \star \\ 0 & \star & \star \\ 0 & 0 & \star \end{bmatrix} \begin{bmatrix} \star & \star & \star \\ 0 & 0 & \star \\ 0 & 0 & \star \end{bmatrix} = \begin{bmatrix} 0 & 0 & \star \\ 0 & 0 & \star \\ 0 & 0 & \star \end{bmatrix} \\ (A - d_1 I)(A - d_2 I)(A - d_3 I) &= \begin{bmatrix} 0 & 0 & \star \\ 0 & 0 & \star \\ 0 & 0 & \star \end{bmatrix} \begin{bmatrix} \star & \star & \star \\ 0 & \star & \star \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \end{aligned}$$

What do you observe?

Theorem 6.2 *Let $A \in \mathbb{F}^{n\times n}$ be an upper triangular matrix with diagonal entries $d_1, \dots, d_n$, in that order. Let $1 \le k \le n$. Then the first k columns of the product $(A - d_1 I) \cdots (A - d_k I)$ are zero columns.*

Proof For $k = 1$, we see that the first column of $A - d_1 I$ is a zero column. It means $(A - d_1 I)e_1 = 0$, where $e_1, \dots, e_n$ are the standard basis vectors for $\mathbb{F}^{n\times 1}$. Assume that the result is true for $k = m < n$. That is,

$$(A - d_1 I) \cdots (A - d_m I)e_1 = 0, \dots, (A - d_1 I) \cdots (A - d_m I)e_m = 0.$$

Notice that

$$(A - d_1 I) \cdots (A - d_m I)(A - d_{m+1} I) = (A - d_{m+1} I)(A - d_1 I) \cdots (A - d_m I).$$

It then follows that for $1 \le j \le m$,

$$\begin{aligned} &(A - d_1 I) \cdots (A - d_m I)(A - d_{m+1} I)e_j \\ &\qquad = (A - d_{m+1} I)(A - d_1 I) \cdots (A - d_m I)e_j = 0. \end{aligned}$$

Next, $(A - d_{m+1}I)e_{m+1}$ is the $(m+1)$th column of $A - d_{m+1}I$, which has all zero entries beyond the first m entries. That is, there are scalars $\alpha_1, \ldots, \alpha_m$ such that $(A - d_{m+1}I)e_{m+1} = \alpha_1 e_1 + \cdots + \alpha_m e_m$. Then

$$(A - d_1 I) \cdots (A - d_m I)(A - d_{m+1}I)e_{m+1} \\ = (A - d_1 I) \cdots (A - d_m I)(\alpha_1 e_1 + \cdots + \alpha_m e_m) = 0.$$

Therefore, the first $m+1$ columns of $(A - d_1 I) \cdots (A - d_m I)(A - d_{m+1}I)$ are zero columns. By induction, the proof is complete. ■

In what follows, we will be substituting the variable t by a square matrix in a polynomial $q(t)$. In such a substitution, we will interpret the constant term a_0 as $a_0 I$. For instance,

$$\text{if } A \in \mathbb{F}^{n\times n} \text{ and } q(t) = 2 + t + 5t^2, \text{ then } q(A) = 2I_n + A + 5A^2.$$

If $q(A)$ turns out to be the zero matrix, then we say that the polynomial $q(t)$ **annihilates** the matrix A; we also say that $q(t)$ is an **annihilating polynomial** of A; and that A is **annihilated by** $q(t)$. The zero polynomial annihilates every matrix. Does every matrix have an annihilating nonzero polynomial?

Theorem 6.3 (Cayley–Hamilton) *Each square matrix is annihilated by its characteristic polynomial.*

Proof Let $A \in \mathbb{F}^{n\times n}$. We consider A as a matrix in $\mathbb{C}^{n\times n}$. By Schur triangularization, there exist a unitary matrix $P \in \mathbb{C}^{n\times n}$ and an upper triangular matrix $U \in \mathbb{C}^{n\times n}$ such that $P^*AP = U$. That is, $A = PUP^*$. Then, for all scalars α and β, we have

$$\alpha I + \beta A = \alpha I + \beta PUP^* = P(\alpha I)P^* + P(\beta U)P^* = P(\alpha I + \beta U)P^*.$$

And $A^2 = PUP^*PUP^* = PU^2P^*$. By induction, it follows that for any polynomial $q(t)$, $q(A) = P\,q(U)\,P^*$. In particular,

$$\chi_A(A) = P\,\chi_A(U)\,P^*.$$

Suppose the diagonal entries of U are $\lambda_1, \ldots, \lambda_n$. Since A and U are similar matrices,

$$\chi_A(t) = \chi_U(t) = (t - \lambda_1) \cdots (t - \lambda_n).$$

Theorem 6.2 implies that the n columns of $(U - \lambda_1 I) \cdots (U - \lambda_n I)$ are zero columns. That is, $\chi_A(U) = 0$. Therefore, $\chi_A(A) = P\,\chi_A(U)\,P^* = 0$. ■

Cayley–Hamilton theorem helps us in computing powers of matrices and also the inverse of a matrix if it exists. Suppose $A \in \mathbb{F}^{n\times n}$ has the characteristic polynomial

$$\chi_A(t) = t^n + a_{n-1}t^{n-1} + \cdots + a_1 t + a_0.$$

By Cayley–Hamilton theorem, $A^n + a_{n-1}A^{n-1} + \cdots + a_1 A + a_0 I = 0$. Then

$$A^n = -(a_0 I + a_1 A + \cdots + a_{n-1}A^{n-1}).$$

Thereby, computation of A^n, $A^{n+1}, \ldots$ can be reduced to that of $A, \ldots, A^{n-1}$.

Next, suppose that A is invertible. Then $\det(A) \neq 0$. Since $\det(A)$ is the product of all eigenvalues of A, $\lambda = 0$ is not an eigenvalue of A. It implies that t, which is equal to $(t - \lambda)$ for $\lambda = 0$, is not a factor of the characteristic polynomial of A. Therefore, the constant term a_0 in the characteristic polynomial of A is nonzero. By Cayley–Hamilton theorem,

$$a_0 I = -(a_1 A + a_2 A^2 + \cdots + a_{n-1}A^{n-1} + A^n).$$

Multiplying A^{-1} and simplifying, we obtain

$$A^{-1} = -\frac{1}{a_0}\big(a_1 I + a_2 A + \cdots + a_{n-1}A^{n-2} + A^{n-1}\big).$$

This way, A^{-1} can also be computed from A, $A^2, \ldots, A^{n-1}$.

Example 6.2 Let $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$. Its characteristic polynomial is $\chi_A(t) = (t-1)^3$. Cayley–Hamilton theorem says that $(A - I)^3 = 0$. By direct computation, we find that

$$(A - I)^2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}^2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Thus, it is not necessary that the degree of an annihilating polynomial of A has to be the order of A. □

Let $A \in \mathbb{F}^{n\times n}$. A monic polynomial of least degree that annihilates A is called the **minimal polynomial** of A.

In Example 6.2, no polynomial of degree 1 annihilates A. But the monic polynomial $(t-1)^2$ annihilates it. Therefore, $(t-1)^2$ is the minimal polynomial of A.

Observe that the degree of a minimal polynomial is unique. Moreover, if $q(t)$ is a minimal polynomial of A, and $q(t)$ has degree k, then no monic polynomial of degree less than k annihilates A. Therefore, no nonzero polynomial of degree less than k annihilates A. In other words,

> if a polynomial of degree less than the degree of the minimal polynomial of A annihilates A, then this polynomial must be the zero polynomial.

Cayley–Hamilton theorem implies that the degree of the minimal polynomial does not exceed the order of the matrix. However, any multiple of an annihilating polynomial also annihilates the given matrix. For instance, the polynomials

$10(t-1)^2$, $t(t-1)^2$ and $(1+2t+t^2+3t^3)(t-1)^2$ annihilate the matrix A in Example 6.2. Is the characteristic polynomial one among the multiples of the minimal polynomial?

Theorem 6.4 *The minimal polynomial of a square matrix is unique; and it divides all annihilating polynomials of the matrix.*

Proof Let $A \in \mathbb{F}^{n\times n}$. Since the degree of a minimal polynomial is unique, let $p(t) = t^k + p_1(t)$ and $q(t) = t^k + q_1(t)$ be two minimal polynomials of A, where $p_1(t)$ and $q_1(t)$ are polynomials of degree less than k. Then $p(A) - q(A) = p_1(A) - q_1(A) = 0$.

The polynomial $r(t) = p_1(t) - q_1(t)$ has degree less than k and it annihilates A. Thus, $r(t)$ is the zero polynomial. In that case, $p(t) = q(t)$. This proves that the minimal polynomial of A is unique.

For the second statement, let $q(t)$ be the minimal polynomial of A. Let $p(t)$ be an annihilating polynomial of A. The degree of $p(t)$ is at least as large as that of $q(t)$. So, suppose the degree of $q(t)$ is k and the degree of $p(t)$ is $m \geq k$. By the division algorithm, we have $p(t) = s(t)q(t) + r(t)$, where $s(t)$ is a polynomial of degree $m-k$, and $r(t)$ is a polynomial of degree less than k. Then

$$0 = p(A) = s(A)\,q(A) + r(A) = 0 + r(A) = r(A).$$

That is, $r(t)$ is a polynomial of degree less than k that annihilates A. Hence, $r(t)$ is the zero polynomial. Therefore, $q(t)$ divides $p(t)$. ■

Theorem 6.3 implies that

the minimal polynomial of a matrix divides its characteristic polynomial.

This fact is sometimes mentioned as the Cayley–Hamilton theorem.

Exercises for Sect. 6.2

1. Compute A^{-1} and A^{10} where A is the matrix of Example 6.2.
2. Use the formula $(A - tI)\,\mathrm{adj}(A - tI) = \det(A - tI)I$ to give another proof of Cayley–Hamilton theorem. [Hint: For matrices $B_0, \ldots, B_{n-1}$, write $\mathrm{adj}(A - tI) = B_0 + B_1 t + \cdots + B_{n-1}t^{n-1}$.]
3. Let $A \in \mathbb{F}^{n\times n}$. Let $p(t)$ be a monic polynomial of degree n. If $p(A) = 0$, then does it follow that $p(t)$ is the characteristic polynomial of A?
4. Basing on Theorem 6.4, describe an algorithm for computing the minimal polynomial of a matrix.

6.3 Diagonalizability

Schur triangularization implies that each square matrix with complex entries is similar to an upper triangular matrix. Moreover, a square matrix with real entries is similar to an upper triangular real matrix provided all zeros of its characteristic polynomial

are real. The upper triangular matrix similar to a given square matrix takes a better form when the matrix is hermitian.

Theorem 6.5 (Spectral theorem for hermitian matrices) *Each hermitian matrix is unitarily similar to a real diagonal matrix. And, each real symmetric matrix is orthogonally similar to a real diagonal matrix.*

Proof Let $A \in \mathbb{C}^{n\times n}$ be a hermitian matrix. Due to Schur triangularization, we have a unitary matrix P such that $D = P^*AP$ is upper triangular. Now,

$$D^* = P^*A^*P = P^*AP = D.$$

Since D is upper triangular and $D^* = D$, we see that D is a diagonal matrix. Comparing the diagonal entries in $D^* = D$, we see that all diagonal entries in D are real.

Further, if A is real symmetric, then all its eigenvalues are real. Due to Schur triangularization, the matrix P above can be chosen to be an orthogonal matrix so that $D = P^tAP$ is upper triangular. By a similar argument as above, it follows that D is a diagonal matrix with real entries. ■

A matrix $A \in \mathbb{F}^{n\times n}$ is called **diagonalizable** if there exists an invertible matrix $P \in \mathbb{F}^{n\times n}$ such that $P^{-1}AP$ is a diagonal matrix. When $P^{-1}AP$ is a diagonal matrix, we say that A is **diagonalized by** P. In this language, the spectral theorem for hermitian matrices may be stated as follows:

> Every hermitian matrix is unitarily diagonalizable; and every real symmetric matrix is orthogonally diagonalizable.

To see how the eigenvalues and eigenvectors are involved in the diagonalization process, let $A, P, D \in \mathbb{F}^{n\times n}$ be matrices, where

$$P = [v_1 \quad \cdots \quad v_n], \quad D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n).$$

Then $Pe_j = v_j$ and $De_j = \lambda_j e_j$ for each $1 \le j \le n$. Suppose $AP = PD$. Then

$$Av_j = APe_j = PDe_j = P(\lambda_j e_j) = \lambda_j(Pe_j) = \lambda_j v_j \quad \text{for } 1 \le j \le n.$$

Conversely, suppose $Av_j = \lambda_j v_j$ for $1 \le j \le n$. Then

$$APe_j = Av_j = \lambda_j v_j = \lambda_j\, Pe_j = P\,(\lambda_j e_j) = PDe_j \quad \text{for } 1 \le j \le n.$$

That is, $AP = PD$. We summarize this as follows.

Observation 6.1 Let $A, P, D \in \mathbb{F}^{n\times n}$. For $j = 1, \ldots, n$, let v_j be the jth column of P. Let $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. Then, $AP = PD$ iff $Av_j = \lambda_j v_j$ for each $j = 1, \ldots, n$.

For diagonalization of a matrix $A \in \mathbb{F}^{n\times n}$, we find the eigenvalues of A and the associated eigenvectors.

If $\mathbb{F} = \mathbb{R}$ and there exists an eigenvalue with nonzero imaginary part, then A cannot be diagonalized by a matrix with real entries.

Else, suppose we have n number of eigenvalues in $\mathbb{F}$ counting multiplicities.

If n linearly independent eigenvectors cannot be found, then A cannot be diagonalized.

Otherwise, we put the eigenvectors together as columns to form the matrix P; and $P^{-1}AP$ is a diagonalization of A. Further, by taking orthonormal eigenvectors, the matrix P can be made unitary or orthogonal.

Example 6.3 The matrix $A = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{bmatrix}$ is real symmetric. It has eigenvalues $-1,\ 2$ and 2, with associated orthonormal eigenvectors as

$$\begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}, \quad \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}, \quad \begin{bmatrix} -1/\sqrt{6} \\ -1/\sqrt{6} \\ 2/\sqrt{6} \end{bmatrix}.$$

Thus, the diagonalizing orthogonal matrix is given by

$$P = \begin{bmatrix} 1/\sqrt{3} & -1/\sqrt{2} & -1/\sqrt{6} \\ 1/\sqrt{3} & 1/\sqrt{2} & -1/\sqrt{6} \\ 1/\sqrt{3} & 0 & 2/\sqrt{6} \end{bmatrix}.$$

We see that $P^{-1} = P^t$ and $P^{-1}AP = P^tAP = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$. □

In fact, the spectral theorem holds for a bigger class of matrices. A matrix $A \in \mathbb{C}^{n\times n}$ is called a **normal matrix** iff $A^*A = AA^*$. All unitary matrices and all hermitian matrices are normal matrices. All diagonal matrices are normal matrices. In addition, a week converse to the last statement holds.

Theorem 6.6 *Each upper triangular normal matrix is diagonal.*

Proof Let $U \in \mathbb{C}^{n\times n}$ be an upper triangular matrix. If $n = 1$, then clearly U is a diagonal matrix. Suppose that each upper triangular normal matrix of order k is diagonal. Let U be an upper triangular normal matrix of order $k + 1$. Write U in a partitioned form as in the following:

$$U = \begin{bmatrix} R & u \\ 0 & a \end{bmatrix}$$

where $R \in \mathbb{C}^{k\times k}$, $u \in \mathbb{C}^{k\times 1}$, 0 is the zero row vector in $\mathbb{C}^{1\times k}$, and $a \in \mathbb{C}$. Since U is normal,

$$U^*U = \begin{bmatrix} R^*R & R^*u \\ u^*R & u^*u + |a|^2 \end{bmatrix} = UU^* = \begin{bmatrix} RR^* + uu^* & \bar{a}\,u \\ a\,u^* & |a|^2 \end{bmatrix}.$$

It implies that $u^*u + |a|^2 = |a|^2$. That is, $u = 0$. Plugging $u = 0$ in the above equation, we see that $R^*R = RR^*$. Since R is upper triangular, by the induction hypothesis, R is a diagonal matrix. Then with $u = 0$, U is also a diagonal matrix. The proof is complete by induction. ■

Using this result on upper triangular normal matrices, we can generalize the spectral theorem to normal matrices.

Theorem 6.7 (Spectral theorem for normal matrices) *A square matrix is unitarily diagonalizable iff it is a normal matrix.*

Proof Let $A \in \mathbb{C}^{n\times n}$. Let A be unitarily diagonalizable. Then there exist a unitary matrix P and a matrix $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ such that $A = PDP^*$. Then $A^*A = PD^*DP^*$ and $AA^* = PDD^*P^*$. However,

$$D^*D = \operatorname{diag}\big(|\lambda_1|^2, \ldots, |\lambda_n|^2\big) = DD^*.$$

Therefore, $A^*A = AA^*$. So, A is a normal matrix.

Conversely let A be a normal matrix; so $A^*A = AA^*$. Due to Schur triangularization, let Q be a unitary matrix such that $Q^*AQ = U$, an upper triangular matrix. Since $Q^* = Q^{-1}$, the condition $A^*A = AA^*$ implies that $U^*U = UU^*$. By Theorem 6.6, U is a diagonal matrix. ■

There can be non-normal matrices which are diagonalizable. For example, with

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 3 & -2 \\ 2 & 1 & 0 \end{bmatrix}, \quad P = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 5 & 2 \\ 1 & 3 & 1 \end{bmatrix}$$

we see that $A^*A \neq AA^*$, $P^*P \neq I$ but

$$P^{-1}AP = \begin{bmatrix} 1 & -1 & 2 \\ -1 & 0 & 0 \\ 2 & 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 4 & 3 & -2 \\ 2 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 \\ 1 & 5 & 2 \\ 1 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

Observe that in such a case, the diagonalizing matrix P is non-unitary.

In general, we have the following characterization of diagonalizability.

Theorem 6.8 *A matrix $A \in \mathbb{F}^{n\times n}$ is diagonalizable iff there exists a basis of $\mathbb{F}^{n\times 1}$ consisting of eigenvectors of A.*

Proof Let $A \in \mathbb{F}^{n\times n}$. Suppose A is diagonalizable. Then there exist an invertible matrix P and a diagonal matrix D such that $AP = PD$. Let v_j be the jth column of

P. By Observation 6.1, $Av_j = \lambda_j v_j$. As P is invertible, its columns $v_1, \ldots, v_n$ are nonzero; thus they are eigenvectors of A, and they form a basis for $\mathbb{F}^{n\times 1}$.

Conversely, suppose that $\{v_1, \ldots, v_n\}$ is a basis of $\mathbb{F}^{n\times 1}$ and that for each $j = 1, \ldots, n$, v_j is an eigenvector of A. Then there exists $\lambda_j \in \mathbb{F}$ such that $Av_j = \lambda_j v_j$. Construct $n \times n$ matrices

$$P = \begin{bmatrix} v_1 & \cdots & v_n \end{bmatrix}, \quad D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n).$$

Observation 6.1 implies that $AP = PD$. Since the columns of P form a basis for $\mathbb{F}^{n\times 1}$, P is invertible. Therefore, A is diagonalizable. ■

In case, we have a basis $B = \{v_1, \ldots, v_n\}$ for $\mathbb{F}^{n\times 1}$ with $Av_j = \lambda_j v_j$, due to Theorems 3.9 and 3.10, the co-ordinate matrix $[A]_{B,B}$ is the diagonal matrix $\operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. The question is, when are there n linearly independent eigenvectors of A? The spectral theorem for normal matrices provides a partial answer. Another partial answer on diagonalizability is as follows.

Theorem 6.9 *If an $n \times n$ matrix has n distinct eigenvalues, then it is diagonalizable.*

Proof Suppose $A \in \mathbb{C}^{n\times n}$ has n distinct eigenvalues $\lambda_1, \ldots, \lambda_n$ with corresponding eigenvectors $v_1, \ldots, v_n$. By Theorem 5.3, the vectors $v_1, \ldots, v_n$ are linearly independent, and thus form a basis for $\mathbb{C}^{n\times 1}$. Therefore, A is diagonalizable.

More directly, take $P = [v_1 \quad \cdots \quad v_n]$. Then P is invertible. By Observation 6.1, $P^{-1}AP = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. ■

If λ is an eigenvalue of a matrix A with an associated eigenvector u, then $Au = \lambda u$; that is, $u \in N(A - \lambda I)$. The number of linearly independent solution vectors of $Au = \lambda u$ is $\dim(N(A - \lambda I))$, the nullity of $A - \lambda I$. This number has certain relation with diagonalizability of T.

Let λ be an eigenvalue of a matrix $A \in \mathbb{F}^{n\times n}$. The **geometric multiplicity** of λ is $\dim(N(A - \lambda I))$; and the **algebraic multiplicity** of λ is the largest natural number k such that $(t - \lambda)^k$ divides $\chi_A(t)$.

Observe that if λ is an eigenvalue of A, then its geometric multiplicity is the maximum number of linearly independent eigenvectors associated with λ; and its algebraic multiplicity is the number of times λ is a zero of the characteristic polynomial. Thus the algebraic multiplicity of an eigenvalue is often called its *multiplicity*.

Example 6.4 Let $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and let $B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$.

The characteristic polynomials of both A and B are equal to $(t - 1)^2$. The eigenvalue $\lambda = 1$ has algebraic multiplicity 2 for both A and B.

For geometric multiplicities, we solve $Ax = x$ and $By = y$.

Now, $Ax = x$ gives $x = x$, which is satisfied by any vector in $\mathbb{F}^{2\times 1}$. Thus, $N(A - I) = \mathbb{F}^{2\times 1}$; consequently, the geometric multiplicity of the only eigenvalue 1 of A is $\dim(N(A - I)) = 2$.

For the matrix B, take $y = [a, b]^t$. Then $By = y$ gives $a + b = a$ and $b = b$. That is, $a = 0$; and b can be any scalar. For example, $[0, 1]^t$ is a solution. Then the geometric multiplicity of the eigenvalue 1 of B is $\dim(N(B - I)) = 1$. □

Theorem 6.10 *The geometric multiplicity of an eigenvalue of a matrix is less than or equal to the algebraic multiplicity of that eigenvalue.*

Proof Let λ be an eigenvalue of a matrix A. Let ℓ be the geometric multiplicity and let k be the algebraic multiplicity of the eigenvalue λ. We have ℓ number of linearly independent eigenvectors of A associated with the eigenvalue λ, and no more. Extend the set of these eigenvectors to an ordered basis B of V. Let P be the matrix whose columns are the vectors in B. Due to Theorems 3.9 and 3.10, the matrix $M = P^{-1}AP = [A]_{B,B}$ may be written as

$$M = \begin{bmatrix} \lambda I_\ell & C \\ 0 & D \end{bmatrix}$$

for some matrices $C \in \mathbb{C}^{\ell \times (n-\ell)}$ and $D \in \mathbb{C}^{(n-\ell)\times(n-\ell)}$. Since A and M are similar, they have the same characteristic polynomial $\chi(t) = (\lambda - t)^\ell p(t)$ for some polynomial $p(t)$ of degree $n - \ell$.

But the zero λ of $\chi(t)$ is repeated k times. That is, $\chi(t) = (\lambda - t)^k q(t)$ for some polynomial $q(t)$ of which $(\lambda - t)$ is not a factor.

Notice that $\lambda - t$ may or may not be a factor of $p(t)$. In any case, $\ell \le k$. ■

Theorem 6.11 *An $n \times n$ matrix A is diagonalizable iff the geometric multiplicity of each eigenvalue of A is equal to its algebraic multiplicity iff the sum of geometric multiplicities of all eigenvalues of A is n.*

Proof Let λ be an eigenvalue of an $n \times n$ matrix A with geometric multiplicity ℓ and algebraic multiplicity k. Let A be diagonalizable. Then we have a basis E of $\mathbb{F}^{n\times 1}$ which consists of eigenvectors of A, with respect to which the matrix of A is diagonal. In this diagonal matrix, there are exactly k number of entries equal to λ. In the basis E, there are k number of eigenvectors associated with λ. These eigenvectors corresponding to λ are linearly independent. There may be more number of linearly independent eigenvectors associated with λ, but no less. So, $\ell \ge k$. Then Theorem 6.10 implies that $\ell = k$.

Conversely, suppose that the geometric multiplicity of each eigenvalue is equal to its algebraic multiplicity. Then corresponding to each eigenvalue λ, we have exactly that many linearly independent eigenvectors as its algebraic multiplicity. Moreover, eigenvectors corresponding to distinct eigenvalues are linearly independent. Thus, collecting together the eigenvectors associated with all eigenvalues, we get n linearly independent eigenvectors which form a basis for $\mathbb{F}^{n\times 1}$. (See Problem 7.) Therefore, A is diagonalizable.

The second 'iff' statement follows since geometric multiplicity of each eigenvalue is at most its algebraic multiplicity. ■

In Example 6.4, we see that the geometric multiplicity of each (the only) eigenvalue of the matrix A is equal to its algebraic multiplicity; both are 2. So, A is diagonalizable; in fact, it is already diagonal. But the geometric multiplicity of the (only) eigenvalue 1 of B is 1 while the algebraic multiplicity is 2. Therefore, B is not diagonalizable.

Exercises for Sect. 6.3

1. Diagonalize the given matrix, and then compute its fifth power:

(a) $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 7 & -2 & 0 \\ -2 & 6 & -2 \\ 0 & -2 & 5 \end{bmatrix}$ (c) $\begin{bmatrix} 7 & -5 & 15 \\ 6 & -4 & 15 \\ 0 & 0 & 1 \end{bmatrix}$

2. Show that the following matrices are diagonalized by matrices in $\mathbb{R}^{3\times 3}$.

(a) $\begin{bmatrix} 3/2 & -1/2 & 0 \\ -1/2 & 3/2 & 0 \\ 1/2 & -1/2 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 3 & -1/2 & -3/2 \\ 1 & 3/2 & 3/2 \\ -1 & 1/2 & 5/2 \end{bmatrix}$ (c) $\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & 0 \\ 2 & 2 & 3 \end{bmatrix}$

3. If possible, diagonalize the following matrices:

(a) $\begin{bmatrix} 2 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 2 & -1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & -2 & 1 \\ 0 & 1 & 1 \\ 0 & 3 & -1 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ -1 & -2 & -3 \end{bmatrix}$

4. Are the following matrices diagonalizable?

(a) $\begin{bmatrix} 2 & 3 \\ 6 & -1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & -10 & 0 \\ -1 & 3 & 1 \\ -1 & 0 & 4 \end{bmatrix}$ (c) $\begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix}$

5. Check whether each of the following matrices is diagonalizable. If diagonalizable, find a basis of eigenvectors for $\mathbb{C}^{3\times 1}$:

(a) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$

6. Find orthogonal or unitary diagonalizing matrices for the following:

(a) $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 3+i \\ 3-i & 4 \end{bmatrix}$ (c) $\begin{bmatrix} -4 & -2 & 2 \\ -2 & -1 & 1 \\ 2 & 1 & -1 \end{bmatrix}$

7. Determine A^5 and a matrix B such that $B^2 = A$, where

(a) $A = \begin{bmatrix} 2 & 1 \\ -2 & -1 \end{bmatrix}$ (b) $A = \begin{bmatrix} 9 & -5 & 3 \\ 0 & 4 & 3 \\ 0 & 0 & 1 \end{bmatrix}$

8. If A is a normal matrix, then show that A^*, $I + A$ and A^2 are also normal.

6.4 Jordan Form

All matrices cannot be diagonalized since corresponding to an eigenvalue, there may not be sufficient number of linearly independent eigenvectors. Non-diagonalizability of a matrix $A \in \mathbb{F}^{n\times n}$ means that we cannot have a basis consisting of vectors v_j for $\mathbb{F}^{n\times 1}$ so that $Av_j = \lambda_j v_j$ for scalars λ_j. In that case, we would like to have a basis which would bring the matrix to a nearly diagonal form. Specifically, if possible, we would try to construct a basis $\{v_1, \ldots, v_n\}$ such that

$$Av_j = \lambda_j v_j \quad \text{or} \quad Av_j = \lambda_j v_j + v_{j-1} \quad \text{for each } j.$$

Notice that the matrix similar to A with respect to such a basis would have λ_js on the diagonal, and possibly nonzero entries on the super diagonal (entries above the diagonal); all other entries being 0.

We will show that it is possible, by proving that there exists an invertible matrix P such that

$$P^{-1}AP = \operatorname{diag}(J_1,\ J_2, \ldots, J_k),$$

where each J_i is a block diagonal matrix of the form

$$J_i = \operatorname{diag}(\tilde{J}_1(\lambda_i),\ \tilde{J}_2(\lambda_i), \ldots, \tilde{J}_{s_i}(\lambda_i)),$$

for some s_i. Each matrix $\tilde{J}_j(\lambda_i)$ of order j here has the form

$$\tilde{J}_j(\lambda_i) = \begin{bmatrix} \lambda_i & 1 & & & \\ & \lambda_i & 1 & & \\ & & \ddots & \ddots & \\ & & & & 1 \\ & & & & \lambda_i \end{bmatrix}.$$

The missing entries are all 0. Such a matrix $\tilde{J}_j(\lambda_i)$ is called a **Jordan block** with diagonal entries λ_i. The order of the Jordan block is its order as a square matrix. Any matrix which is in the block diagonal form $\operatorname{diag}(J_1,\ J_2, \ldots, J_k)$ is said to be in **Jordan form**.

In writing Jordan blocks and Jordan forms, we do not show the zero entries for improving legibility. For instance, the following are possible Jordan form matrices of order 3 with all diagonal entries as 1:

$$\begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 1 & \\ & 1 & \\ & & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & & \\ & 1 & 1 \\ & & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 1 & \\ & 1 & 1 \\ & & 1 \end{bmatrix}$$

Example 6.5 The following matrix is in Jordan form:

$$\begin{bmatrix} 1 & & & & & & \\ & 1 & & & & & \\ & & 1 & 1 & & & \\ & & & 1 & & & \\ & & & & 2 & 1 & \\ & & & & & 2 & 1 \\ & & & & & & 2 \end{bmatrix}.$$

It has three Jordan blocks for the eigenvalue 1 of which two are of order 1 and one of order 2; and it has one block of order 3 for the eigenvalue 2.

The eigenvalue 1 has geometric multiplicity 3, algebraic multiplicity 4, and the eigenvalue 2 has geometric multiplicity 1 and algebraic multiplicity 3. □

In what follows, we will be using similarity transformations resulting out of elementary matrices. A similarity transformation that uses an elementary matrix $E[i, j]$ on a matrix A transforms A to $(E[i, j])^{-1} A\, E[i, j]$. Since $(E[i, j])^{-1} = (E[i, j])^t = E[i, j]$, the net effect of this transformation is described as follows:

> $E[i, j]^{-1} A\, E[i, j] = E[i, j]\, A\, E[i, j]$ exchanges the ith and jth rows, and then exchanges the ith and the jth columns of A.

We will refer to this type of similarity transformations by the name **permutation similarity**.

Using the second type of elementary matrices, we have a similarity transformation $(E_\alpha[i])^{-1} A\, E_\alpha[i]$ for $\alpha \neq 0$. Since $(E_\alpha[i])^{-1} = E_{1/\alpha}[i]$ and $(E_\alpha[i])^t = E_\alpha[i]$, this similarity transformation has the following effect:

> $(E_\alpha[i])^{-1} A\, E_\alpha[i] = E_{1/\alpha}[i]\, A\, E_\alpha[i]$ multiplies all entries in the ith row with $1/\alpha$, and then multiplies all entries in the ith column with α; thus keeping (i, i)th entry intact.

We will refer to this type of similarity transformation as **dilation similarity**. In particular, if A is such a matrix that its ith row has all entries 0 except the (i, i)th entry, and there is another entry on the ith column which is $\alpha \neq 0$, then $(E_\alpha[i])^{-1} A E_\alpha[i]$ is the matrix in which this α changes to 1 and all other entries are as in A.

The third type of similarity transformation applied on A yields the matrix $(E_\alpha[i, j])^{-1} A E_\alpha[i, j]$. Notice that $(E_\alpha[i, j])^{-1} = E_{-\alpha}[i, j]$ and $(E_\alpha[j, i])^t = E_\alpha[i, j]$. This similarity transformation changes a matrix A as described below:

> $(E_\alpha[i, j])^{-1} A E_\alpha[i, j] = E_{-\alpha}[i, j]\, A\, E_\alpha[i, j]$ is obtained from A by subtracting α times the jth row from the ith row, and then adding α times the ith column to the jth column.

We name this type of similarity as a **combination similarity**.

In the formula for m_k, below we use the convention that for any matrix B of order n, B^0 is the identity matrix of order n.

Theorem 6.12 (Jordan form) *Each matrix $A \in \mathbb{C}^{n\times n}$ is similar to a matrix in Jordan form J, where the diagonal entries are the eigenvalues of A. For $1 \le k \le n$, if $m_k(\lambda)$ is the number of Jordan blocks of order k with diagonal entry λ, in J, then*

$$m_k(\lambda) = \operatorname{rank}((A-\lambda I)^{k-1}) - 2\operatorname{rank}((A-\lambda I)^k) + \operatorname{rank}((A-\lambda I))^{k+1}.$$

The Jordan form of A is unique up to a permutation of the blocks.

Proof First, we will show the existence of a Jordan form, and then we will come back to the formula m_k, which will show the uniqueness of a Jordan form up to a permutation of Jordan blocks.

Due to Schur triangularization, we assume that A is an upper triangular matrix, where the eigenvalues of A occur on the diagonal, and equal eigenvalues occur together. If $\lambda_1, \ldots, \lambda_k$ are the distinct eigenvalues of A, then our assumption means that A is an upper triangular matrix with diagonal entries, read from top left to bottom right, appear as

$$\lambda_1, \ldots, \lambda_1; \lambda_2, \ldots, \lambda_2; \ldots; \lambda_k, \ldots, \lambda_k.$$

Let n_i denote the number of times λ_i occurs in this list. First, we show that by way of a similarity transformation, A can be brought to the form

$$\operatorname{diag}(A_1, A_2, \ldots, A_k),$$

where each A_i is an upper triangular matrix of size $n_i \times n_i$ and each diagonal entry of A_i is λ_i. Our requirement is shown schematically as follows, where each such element marked x that is not inside the blocks A_i needs to be zeroed-out by a similarity transformation.

$$\begin{bmatrix} \boxed{A_1} & & x & \\ & \boxed{A_2} & & \\ & & \ddots & \\ & & & \boxed{A_k} \end{bmatrix} \longrightarrow \begin{bmatrix} \boxed{A_1} & & 0 & \\ & \boxed{A_2} & & \\ & & \ddots & \\ & & & \boxed{A_k} \end{bmatrix}$$

If such an x occurs as the (r, s)th entry in A, then $r < s$. Moreover, the corresponding diagonal entries a_{rr} and a_{ss} are eigenvalues of A occurring in different blocks A_i and A_j. Thus $a_{rr} \neq a_{ss}$. Further, all entries below the diagonals of A_i and of A_j are 0. We use a combination similarity to obtain

$$E_{-\alpha}[r, s]\, A\, E_{\alpha}[r, s] \quad \text{with} \quad \alpha = \frac{-x}{a_{rr} - a_{ss}}.$$

This similarity transformation subtracts α times the sth row from the rth row and then adds α times the rth column to the sth column. Since $r < s$, it changes the entries of A in the rth row to the right of the sth column, and the entries in the sth column above the rth row. Thus, the upper triangular nature of the matrix does not change. Further, it replaces the (r, s)th entry x with

$$a_{rs} + \alpha(a_{rr} - a_{ss}) = x + \frac{-x}{a_{rr} - a_{ss}}(a_{rr} - a_{ss}) = 0.$$

We use a sequence of such similarity transformations starting from the last row of A_{k-1} with smallest column index and ending in the first row with largest column index. Observe that an entry beyond the blocks, which was 0 previously can become nonzero after a single such similarity transformation. Such an entry will eventually be zeroed-out. Finally, each position which is not inside any of the k blocks $A_1, \ldots, A_k$ contains only 0. On completion of this stage, we end up with a matrix

$$\operatorname{diag}(A_1,\ A_2, \ldots, A_k).$$

In the second stage, we focus on bringing each block A_i to the Jordan form. For notational convenience, write λ_i as a. If $n_i = 1$, then such an A_i is already in Jordan form. We use induction on the order n_i of A_i. Lay out the induction hypothesis that each such matrix of order $m-1$ has a Jordan form. Suppose A_i has order m. Look at A_i in the following partitioned form:

$$A_i = \left[\begin{array}{c|c} B & u \\ \hline 0 & a \end{array}\right],$$

where B is the first $(m-1) \times (m-1)$ block, 0 is the zero row vector in $\mathbb{C}^{1\times(m-1)}$, and u is a column vector in $\mathbb{C}^{(m-1)\times 1}$. By the induction hypothesis, there exists an invertible matrix Q such that $Q^{-1}BQ$ is in Jordan form; it looks like

$$Q^{-1}BQ = \begin{bmatrix} \boxed{B_1} & & & \\ & \boxed{B_2} & & \\ & & \ddots & \\ & & & \boxed{B_\ell} \end{bmatrix} \text{ where each } B_j = \begin{bmatrix} a & 1 & & & \\ & a & 1 & & \\ & & \ddots & \ddots & \\ & & & & 1 \\ & & & & a \end{bmatrix}.$$

Then

$$\begin{bmatrix} Q & 0 \\ 0 & 1 \end{bmatrix}^{-1} A_i \begin{bmatrix} Q & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} Q^{-1}BQ & Q^{-1}u \\ 0 & a \end{bmatrix} = \begin{bmatrix} a & * & & & & b_1 \\ & a & * & & & b_2 \\ & & \ddots & \ddots & & \\ & & & & * & b_{m-2} \\ & & & & a & b_{m-1} \\ & & & & & a \end{bmatrix}.$$

Call the above matrix as C. In the matrix C, the sequence of $*$'s on the super-diagonal, read from top left to right bottom, comprise a block of 1s followed by a 0, and then a block of 1s followed by a 0, and so on. The number of 1s depends on the sizes of B_1, B_2, etc. That is, when B_1 is over, and B_2 starts, we have a 0. Also, we have shown $Q^{-1}u$ as $[b_1 \ \cdots \ b_{m-1}]^t$. Our goal is to zero-out all b_js except b_{m-1} which may be made a 0 or 1.

In the next sub-stage, call it the third stage, we apply similarity transformations to zero-out (all or except one of) the entries $b_1, \ldots, b_{m-2}$. In any row of C, the entry above the diagonal (the $*$ there) is either 0 or 1. The $*$ is a 0 at the last row of each block B_j. We leave all such b's right now; they are to be tackled separately. So, suppose in the rth row, $b_r \neq 0$ and the $(r, r+1)$th entry (the $*$ above the diagonal entry) is a 1. We wish to zero-out each such b_r which is in the (r, m) position. For this purpose, we use a combination similarity to transform C to

$$E_{b_r}[r+1, m]\, C\, (E_{b_r}[r+1, m])^{-1} = E_{b_r}[r+1, m]\, C\, E_{-b_r}[r+1, m].$$

Observe that this matrix is obtained from C by adding b_r times the last row to the $(r+1)$th row, and then subtracting b_r times the $(r+1)$th column from the last column. Its net result is replacing the (r, m)th entry by 0, and keeping all other entries intact. Continuing this process of applying a suitable combination similarity transformation, each nonzero b_i with a corresponding 1 on the super-diagonal on the same row is reduced to 0. We then obtain a matrix, where all entries in the last column of C have been zeroed-out, without touching the entries at the last row of any of the blocks B_j. Write such entries as $c_1, \ldots, c_\ell$. Thus, at the end of third stage, A_i has been brought to the following from by similarity transformations:

$$F := \begin{bmatrix} \boxed{B_1} & & & & c_1 \\ & \boxed{B_2} & & & c_2 \\ & & \ddots & & \\ & & & \boxed{B_\ell} & c_\ell \\ & & & & a \end{bmatrix}$$

Notice that if B_j is a 1×1 block, then the corresponding entry c_j on the last column is already 0. In the next sub-stage, call it the fourth stage, we keep the nonzero c corresponding to the last block (the c entry with highest column index), and zero-out all other c's. Let B_q be the last block so that its corresponding c entry is $c_q \neq 0$ in the sth row. (It may not be c_ℓ; in that case, all of $c_{q+1}, \ldots, c_\ell$ are already 0.) We first make c_q a 1 by using a dilation similarity:

$$G := E_{1/c_q}[s]\, F\, E_{c_q}[s].$$

In G, the earlier c_q at (s, m)th position is now 1. Let B_p be any block other than B_q with $c_p \neq 0$ in the rth row. Our goal in this sub-stage, call it the fifth stage, is to zero-out c_p. We use two combination similarity transformations as shown below:

$$H := E_{-c_p}[r-1, s-1]\, E_{-c_p}[r, s]\, G\, E_{c_p}[r, s]\, E_{c_p}[r-1, s-1].$$

This similarity transformation brings c_p to 0 and keeps other entries intact. We do this for each such c_p. Thus in the mth column of H, we have only one nonzero entry

1 at (s, m)th position. If this happens to be at the last row, then we have obtained a Jordan form. Otherwise, in this sub-stage (call it the seventh stage), we move this 1 to the $(s, s+1)$th position by the following sequence of permutation similarities:

$$E[m-1,m] \cdots E[s+2,m]E[s+1,m]\, H\, E[s+1,m]E[s+2,m] \cdots E[m-1,m].$$

This transformation exchanges the rows and columns beyond the sth so that the 1 in (s, m)th position moves to $(s, s+1)$th position making up a block; and other entries remain as they were earlier.

Here ends the proof by induction that each block A_i can be brought to a Jordan form by similarity transformations. From a similarity transformation for A_i, a similarity transformation can be constructed for the block diagonal matrix

$$\tilde{A} := \operatorname{diag}(A_1,\ A_2, \ldots, A_k)$$

by putting identity matrices of suitable order and the similarity transformation for A_i in a block form. As these transformations do not affect any other rows and columns of $\tilde{A}$, a sequence of such transformations brings $\tilde{A}$ to its Jordan form, proving the existence part in the theorem.

Toward the formula for m_k, let λ be an eigenvalue of A, and let $1 \le k \le n$. Observe that $A - \lambda I$ is similar to $J - \lambda I$. Thus,

$$\operatorname{rank}((A-\lambda I)^i) = \operatorname{rank}((J-\lambda I)^i) \quad \text{for each } i.$$

Therefore, it is enough to prove the formula for J instead of A.

We use induction on n. In the basis case, $J = [\lambda]$. Here, $k = 1$ and $m_k = m_1 = 1$. On the right hand side,

$$(J-\lambda I)^{k-1} = I = [1],\ (J-\lambda I)^k = [0]^1 = [0],\ (J-\lambda I)^{k+1} = [0]^2 = [0].$$

So, the formula holds for $n = 1$.

Lay out the induction hypothesis that for all matrices in Jordan form of order less than n, the formula holds. Let J be a matrix of order n, which is in Jordan form. We consider two cases.

Case 1: Let J have a single Jordan block corresponding to λ. That is,

$$J = \begin{bmatrix} \lambda & 1 & & & \\ & \lambda & 1 & & \\ & & \ddots & \ddots & \\ & & & & 1 \\ & & & & \lambda \end{bmatrix}, \quad J - \lambda I = \begin{bmatrix} 0 & 1 & & & \\ & 0 & 1 & & \\ & & \ddots & \ddots & \\ & & & & 1 \\ & & & & 0 \end{bmatrix}.$$

Here $m_1 = 0,\ m_2 = 0, \ldots, m_{n-1} = 0$ and $m_n = 1$. We see that $(J - \lambda I)^2$ has 1s on the super-super-diagonal, and 0 elsewhere. Proceeding similarly for higher powers of $J - \lambda I$, we see that their ranks are given by

$$\operatorname{rank}(J-\lambda I) = n-1, \operatorname{rank}\big((J-\lambda I)^2\big) = n-2, \ldots, \operatorname{rank}\big((J-\lambda I)^i\big) = n-i,$$
$$\operatorname{rank}\big((J-\lambda I)^n\big) = 0, \operatorname{rank}\big((J-\lambda I)^{n+1}\big) = 0, \ldots$$

Then for $k < n$, $\operatorname{rank}\big((J-\lambda I)^{k-1}\big) - 2\operatorname{rank}\big((J-\lambda I)^k\big) + \operatorname{rank}\big((J-\lambda I)^{k+1}\big)$
$$= (n-(k-1)) - 2(n-k) + (n-k-1) = 0.$$

And for $k = n$, $\operatorname{rank}\big((J-\lambda I)^{k-1}\big) - 2\operatorname{rank}\big((J-\lambda I)^k\big) + \operatorname{rank}\big((J-\lambda I)^{k+1}\big)$
$$= (n-(n-1)) - 2\times 0 + 0 = 1 = m_n.$$

Case 2: Suppose J has more than one Jordan block corresponding to λ. The first Jordan block in J corresponds to λ and has order r for some $r < n$. Then $J - \lambda I$ can be written in block form as

$$J - \lambda I = \begin{bmatrix} C & 0 \\ 0 & D \end{bmatrix},$$

where C is the Jordan block of order r with diagonal entries as 0, and D is the matrix of order $n-r$ in Jordan form consisting of other blocks of $J - \lambda I$. Then, for any j,

$$(J-\lambda I)^j = \begin{bmatrix} C^j & 0 \\ 0 & D^j \end{bmatrix}.$$

Therefore, $\operatorname{rank}(J-\lambda I)^j = \operatorname{rank}(C^j) + \operatorname{rank}(D^j)$. Write $m_k(C)$ and $m_k(D)$ for the number of Jordan blocks of order k for the eigenvalue λ that appear in C, and in D, respectively. Then

$$m_k = m_k(C) + m_k(D).$$

By the induction hypothesis,

$$m_k(C) = \operatorname{rank}(C^{k-1}) - 2\operatorname{rank}(C^k) + \operatorname{rank}(C)^{k+1},$$
$$m_k(D) = \operatorname{rank}(D^{k-1}) - 2\operatorname{rank}(D^k) + \operatorname{rank}(D)^{k+1}.$$

It then follows that

$$m_k = \operatorname{rank}((J-\lambda I)^{k-1}) - 2\operatorname{rank}((J-\lambda I)^k) + \operatorname{rank}((J-\lambda I))^{k+1}.$$

Since the number of Jordan blocks of order k corresponding to each eigenvalue of A is uniquely determined, the Jordan form of A is also uniquely determined up to a permutation of blocks. ■

To obtain a Jordan form of a given matrix, we may use the construction of similarity transformations as used in the proof of Theorem 6.12, or we may use the formula for m_k as given there. We illustrate these methods in the following examples.

Example 6.6 Let $A = \begin{bmatrix} 2 & 1 & 0 & 0 & 0 & 1 & 0 & \textcircled{2} & 0 \\ & 2 & 0 & 0 & 0 & 3 & 0 & 0 & \textcircled{1} \\ & & 2 & 1 & 0 & 0 & \textcircled{2} & 0 & 0 \\ & & & 2 & 0 & 2 & 0 & 0 & 0 \\ & & & & 2 & 0 & 0 & 0 & 0 \\ & & & & & 2 & 0 & 0 & 0 \\ & & & & & & 3 & 1 & 1 \\ & & & & & & & 3 & 1 \\ & & & & & & & & 3 \end{bmatrix}$.

This is an upper triangular matrix. Following the proof of Theorem 6.12, we first zero-out the circled entries, starting from the entry on the third row. Here, the row index is $r = 3$, the column index is $s = 7$, the eigenvalues are $a_{rr} = 2$, $a_{ss} = 3$, and the entry to be zeroed-out is $x = 2$. Thus, $\alpha = -2/(2-3) = 2$. We use an appropriate combination similarity to obtain

$$M_1 = E_{-2}[3, 7]\, A\, E_2[3, 7].$$

That is, in A, we replace $row(3)$ with $row(3) - 2 \times row(7)$ and then replace $col(7)$ with $col(7) + 2 \times col(3)$. It leads to

$$M_1 = \begin{bmatrix} 2 & 1 & 0 & 0 & 0 & 1 & 0 & \textcircled{2} & 0 \\ & 2 & 0 & 0 & 0 & 3 & 0 & 0 & \textcircled{1} \\ & & 2 & 1 & 0 & 0 & 0 & -2 & 0 \\ & & & 2 & 0 & 2 & 0 & 0 & 0 \\ & & & & 2 & 0 & 0 & 0 & 0 \\ & & & & & 2 & 0 & 0 & 0 \\ & & & & & & 3 & 1 & 1 \\ & & & & & & & 3 & 1 \\ & & & & & & & & 3 \end{bmatrix}.$$

Notice that the similarity transformation brought in a new nonzero entry such as -2 in $(3, 8)$ position. But its column index has increased. Looking at the matrix afresh, we must zero-out this entry first. The suitable combination similarity yields

$$M_2 = E_2[3, 8]\, M_1\, E_{-2}[3, 8]$$

which replaces $row(3)$ with $row(3) + 2 \times row(8)$ and then replaces $col(8)$ with $col(8) - 2 \times col(3)$. Verify that it zeroes-out the entry -2 but introduces 2 at $(3, 9)$ position. Once more, we use a combination similarity to obtain

$$M_3 = E_{-2}[3, 9]\, M_2\, E_2[3, 9]$$

replacing $row(3)$ with $row(3) - 2 \times row(9)$ and then replacing $col(9)$ with $col(9) + 2 \times col(3)$. Now,

$$M_3 = \begin{bmatrix} 2 & 1 & 0 & 0 & 0 & 1 & 0 & \textcircled{2} & 0 \\ & 2 & 0 & 0 & 0 & 3 & 0 & 0 & \textcircled{1} \\ & & 2 & 1 & 0 & 0 & 0 & 0 & 0 \\ & & & 2 & 0 & 2 & 0 & 0 & 0 \\ & & & & 2 & 0 & 0 & 0 & 0 \\ & & & & & 2 & 0 & 0 & 0 \\ & & & & & & 3 & 1 & 1 \\ & & & & & & & 3 & 1 \\ & & & & & & & & 3 \end{bmatrix}.$$

Similar to the above, we use the combination similarities to reduce M_3 to M_4, where

$$M_4 = E_{-1}[2, 9]\, M_3\, E_1[2, 9].$$

To zero-out the encircled 2, we use a suitable combination similarity, and get

$$M_5 = E_{-2}[1, 8]\, M_4\, E_2[1, 8].$$

It zeroes-out the encircled 2 but introduces -2 at $(1, 9)$ position. Once more, we use a suitable combination similarity to obtain

$$M_6 = E_2[1, 9]\, M_5\, E_{-2}[1, 9] = \begin{bmatrix} 2 & 1 & 0 & 0 & 0 & \textcircled{1} & & & \\ & 2 & 0 & 0 & 0 & 3 & & & \\ & & 2 & 1 & 0 & 0 & & & \\ & & & 2 & 0 & 2 & & & \\ & & & & 2 & 0 & & & \\ & & & & & 2 & & & \\ & & & & & & 3 & 1 & 1 \\ & & & & & & & 3 & 1 \\ & & & & & & & & 3 \end{bmatrix}.$$

Now, the matrix M_6 is in block diagonal form. We focus on each of the blocks, though we will be working with the whole matrix. We consider the block corresponding to the eigenvalue 2 first. Since this step is inductive we scan this block from the top left corner. The 2×2 principal sub-matrix of this block is already in Jordan form. The 3×3 principal sub-matrix is also in Jordan form. We see that the principal sub-matrix of size 4×4 and 5×5 is also in Jordan form, but the 6×6 sub-matrix, which is the block itself is not in Jordan form.

We wish to bring the sixth column to its proper shape. Recall that our strategy is to zero out all those entries on the sixth column which are opposite to a 1 on the super-diagonal of this block. There is only one such entry, which is encircled in M_6 above.

The row index of this entry is $r = 1$, its column index is $m = 6$, and the entry itself is $b_r = 1$. We use a combination similarity to obtain

$$M_7 = E_1[2,6]\, M_6\, E_{-1}[2,6] = \begin{bmatrix} 2 & 1 & 0 & 0 & 0 & 0 & & & \\ & 2 & 0 & 0 & 0 & 5 & & & \\ & & 2 & 1 & 0 & 0 & & & \\ & & & 2 & 0 & 2 & & & \\ & & & & 2 & 0 & & & \\ & & & & & 2 & & & \\ & & & & & & 3 & 1 & 1 \\ & & & & & & & 3 & 1 \\ & & & & & & & & 3 \end{bmatrix}.$$

Next, among the nonzero entries 5 and 2 at the positions $(2,6)$ and $(4,6)$, we wish to zero-out the 5 and keep 2 as the row index of 2 is higher. First, we use a dilation similarity to make this entry 1 as in the following:

$$M_8 = E_{1/2}[4]\, M_7\, E_2[4].$$

It replaces $row(4)$ with $1/2$ times itself and then replaces $col(4)$ with 2 times itself, thus making $(4,6)$th entry 1 and keeping all other entries intact. Next, we zero-out the 5 on $(2,4)$ position by using the two combination similarities. Here, $c_p = 5,\ r = 2,\ s = 4$; thus

$$M_9 = E_{-5}[1,3]\, E_{-5}[2,4]\, M_8\, E_5[2,4]\, E_5[1,3] = \begin{bmatrix} 2 & 1 & 0 & 0 & 0 & 0 & & & \\ & 2 & 0 & 0 & 0 & 0 & & & \\ & & 2 & 1 & 0 & 0 & & & \\ & & & 2 & 0 & \textcircled{1} & & & \\ & & & & 2 & 0 & & & \\ & & & & & 2 & & & \\ & & & & & & 3 & 1 & 1 \\ & & & & & & & 3 & 1 \\ & & & & & & & & 3 \end{bmatrix}.$$

Here, M_9 has been obtained from M_8 by replacing $row(2)$ with $row(2) - 5 \times row(4)$, $col(4)$ with $col(4) + 5 \times col(2)$, $row(1)$ with $row(1) - 5 \times row(3)$, and then $col(3)$ with $col(3) + 5 \times col(1)$.

Next, we move this encircled 1 to $(4,5)$ position by similarity. Here, $s = 4$, $m = 6$. Thus, the sequence of permutation similarities boils down to only one, i.e., exchanging $row(5)$ with $row(6)$ and then exchanging $col(6)$ with $col(5)$. Observe that we would have to use more number of permutation similarities if the difference between m and s is more than 2. It gives

$$M_{10} = E[5,6]\, M_9\, E[5,6] = \begin{bmatrix} 2 & 1 & 0 & 0 & 0 & 0 & & & \\ & 2 & 0 & 0 & 0 & 0 & & & \\ & & 2 & 1 & 0 & 0 & & & \\ & & & 2 & 1 & 0 & & & \\ & & & & 2 & 0 & & & \\ & & & & & 2 & & & \\ & & & & & & 3 & 1 & 1 \\ & & & & & & & 3 & 1 \\ & & & & & & & & 3 \end{bmatrix}.$$

Now, the diagonal block corresponding to the eigenvalue 2 is in Jordan form. We focus on the other block corresponding to 3. Here, $(7, 9)$th entry which contains a 1 is to be zeroed-out. This entry is opposite to a 1 on the super-diagonal. We use a combination similarity. Here, the row index is $r = 7$, the column index $m = 9$, and the entry is $b_r = 1$. Thus, we have Jordan form as

$$M_{11} = E_1[8,9]\, M_{10}\, E_{-1}[8,9] = \begin{bmatrix} 2 & 1 & & & & & & & \\ & 2 & & & & & & & \\ & & 2 & 1 & & & & & \\ & & & 2 & 1 & & & & \\ & & & & 2 & & & & \\ & & & & & 2 & & & \\ & & & & & & 3 & 1 & \\ & & & & & & & 3 & 1 \\ & & & & & & & & 3 \end{bmatrix}. \qquad \square$$

Example 6.7 Consider the matrix A of Example 6.6. Here, we compute the number m_k of Jordan blocks of size k corresponding to each eigenvalue. For this purpose, we require the ranks of the matrices $(A - \lambda I)^k$ for successive k, and for each eigenvalue λ of A. We see that A has two eigenvalues 2 and 3.

For the eigenvalue 2,

$$\operatorname{rank}(A - 2I)^0 = \operatorname{rank}(I) = 9,\ \operatorname{rank}(A - 2I) = 6,\ \operatorname{rank}(A - 2I)^2 = 4,$$
$$\operatorname{rank}(A - 2I)^{3+k} = 3 \quad \text{for } k = 0,\ 1,\ 2, \ldots$$

For the eigenvalue 3,

$$\operatorname{rank}(A - 3I)^0 = \operatorname{rank}(I) = 9,\ \operatorname{rank}(A - 3I) = 8,\ \operatorname{rank}(A - 3I)^2 = 7,$$
$$\operatorname{rank}(A - 3I)^{3+k} = 6 \quad \text{for } k = 0,\ 1,\ 2, \ldots$$

Using the formula for $m_k(\lambda)$, we obtain

$$m_1(2) = 9 - 2 \times 6 + 4 = 1, \quad m_2(2) = 6 - 2 \times 4 + 3 = 1,$$
$$m_3(2) = 4 - 2 \times 3 + 3 = 1, \quad m_{3+k}(2) = 3 - 2 \times 3 + 3 = 0.$$
$$m_1(3) = 9 - 2 \times 8 + 7 = 0, \quad m_2(3) = 8 - 2 \times 7 + 6 = 0,$$
$$m_3(3) = 7 - 2 \times 6 + 6 = 1, \quad m_{3+k}(3) = 6 - 2 \times 6 + 6 = 0.$$

Therefore, in the Jordan form of A, there is one Jordan block of size 1, one of size 2 and one of size 3 with eigenvalue 2, and one block of size 3 with eigenvalue 3. From this information, we see that the Jordan form of A is uniquely determined up to any rearrangement of the blocks. Check that M_{11} as obtained in Example 6.6 is one such Jordan form of A. □

Suppose that a matrix $A \in \mathbb{C}^{n \times n}$ has a Jordan form $J = P^{-1}AP$, in which the first Jordan block is of size k with diagonal entries as λ. If $P = [v_1 \quad \cdots \quad v_n]$, then $AP = PJ$ implies that

$$A(v_1) = \lambda v_1, \ A(v_2) = v_1 + \lambda v_2, \ldots, A(v_k) = v_{k-1} + \lambda v_k.$$

If the next Jordan block in J has diagonal entries as μ (which may or may not be equal to λ), then we have $Av_{k+1} = \mu v_{k+1}$, $Av_{k+2} = v_{k+1} + \mu v_{k+2}, \ldots$, and so on.

The list of vectors $v_1, \ldots, v_k$ above is called a *Jordan string that starts with* v_1 *and ends with* v_k. The number k is called the *length* of the Jordan string. In such a Jordan string, we see that

$$v_1 \in N(A - \lambda I), \ v_2 \in N(A - \lambda I)^2, \ldots, v_k \in N(A - \lambda I)^k.$$

Any vector in $N((A - \lambda I)^j)$, for some j, is called a **generalized eigenvector** corresponding to the eigenvalue λ of A.

The columns of P are all generalized eigenvectors of A corresponding to the eigenvalues of A. These generalized eigenvectors form a basis for $\mathbb{F}^{n \times 1}$. Such a basis is called a *Jordan basis*. The coordinate matrix of A with respect to the Jordan basis is the Jordan form J of A.

The Jordan basis consists of Jordan strings. Each Jordan string starts with an eigenvector of A, such as v_1 above. If λ is an eigenvalue of A having geometric multiplicity γ, then there are exactly γ number of Jordan strings in the Jordan basis corresponding to the eigenvalue λ. Thus, there are exactly γ number of Jordan blocks in J with diagonal entries as λ. The size of any such block is equal to the length of the corresponding Jordan string.

The uniqueness of a Jordan form can be made exact by first ordering the eigenvalues of A and then arranging the blocks corresponding to each eigenvalue (which now appear together on the diagonal) in some order, say in ascending order of their size. In doing so, the Jordan form of any matrix becomes unique. Such a Jordan form is called the **Jordan canonical form** of a matrix. It then follows that if two matrices are similar, then they have the same Jordan canonical form. Moreover, uniqueness

also implies that two dissimilar matrices will have different Jordan canonical forms. Therefore, Jordan form characterizes similarity of matrices.

As an application of Jordan form, we will show that each matrix is similar to its transpose. Suppose $J = P^{-1}AP$. Now, $J^t = P^tA^t(P^{-1})^t = P^tA^t(P^t)^{-1}$. That is, A^t is similar to J^t. Thus, it is enough to show that J^t is similar to J. First, let us see it for a single Jordan block. For a Jordan block J_λ, consider the matrix Q of the same order as in the following:

$$J_\lambda = \begin{bmatrix} \lambda & 1 & & & \\ & \lambda & 1 & & \\ & & \ddots & \ddots & \\ & & & & 1 \\ & & & & \lambda \end{bmatrix}, \quad Q = \begin{bmatrix} & & & & 1 \\ & & & 1 & \\ & & \iddots & & \\ & 1 & & & \\ 1 & & & & \end{bmatrix}.$$

In the matrix Q, the entries on the *anti-diagonal* are all 1 and all other entries are 0. We see that $Q^2 = I$. Thus, $Q^{-1} = Q$. Further,

$$Q^{-1}J_\lambda\, Q = Q\, J_\lambda\, Q = (J_\lambda)^t.$$

Therefore, each Jordan block is similar to its transpose. Now, construct a matrix R by putting matrices such as Q as its blocks matching the orders of each Jordan block in J. Then it follows that $R^{-1}J\,R = J^t$.

Jordan form guarantees that one can always choose m linearly independent generalized eigenvectors corresponding to the eigenvalue λ, where m is the algebraic multiplicity of λ. Moreover, the following is guaranteed:

> If the linear system $(A - \lambda I)^k x = 0$ has $r < m$ number of linearly independent solutions, then $(A - \lambda I)^{k+1} = 0$ has at least $r + 1$ number of linearly independent solutions.

This result is more useful in computing the exponential of a square matrix rather than using the Jordan form explicitly. See Sect. 7.6 for details.

Exercises for Sect. 6.4

1. Determine the Jordan forms of the following matrices:
   (a) $\begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 2 & 1 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ (c) $\begin{bmatrix} -2 & -1 & -3 \\ 4 & 3 & 3 \\ -2 & 1 & -1 \end{bmatrix}$
2. Determine the matrix $P \in \mathbb{C}^{3\times 3}$ such that $P^{-1}AP$ is in Jordan form, where A is the matrix in Exercise 1(c).
3. Let A be a 7×7 matrix with characteristic polynomial $(t-2)^4(3-t)^3$. It is known that in the Jordan form of A, the largest blocks for both the eigenvalues are of order 2. Show that there are only two possible Jordan forms for A; and determine those Jordan forms.
4. Let A be a 5×5 matrix whose first and second rows are, respectively, $[0, 1, 1, 0, 1]$ and $[0, 0, 1, 1, 1]$; and all other rows are zero rows. What is the Jordan form of A?

5. Let A be an $n \times n$ a lower triangular matrix with each diagonal entry as 1, each sub-diagonal entry (just below each diagonal entry) as 1, and all other entries being 0. What is the Jordan form of A?
6. Let A be an $n \times n$ matrix, where the diagonal entries are $1, 2, \ldots, n$, from top left to right bottom, the super-diagonal entries are all 1, and all other entries are 0. What is the Jordan form of A?
7. What is the Jordan form of the $n \times n$ matrix whose each row is equal to $[1,\ 2, \cdots, n]$?
8. Let λ be an eigenvalue of the $n \times n$ matrix A. Suppose for each $k \in \mathbb{N}$, we know the number m_k. Show that for each j, both $\text{rank}(A - \lambda I)^j$ and $\text{null}(A - \lambda I)^j$ are uniquely determined.
9. Prove that two matrices $A, B \in \mathbb{C}^{n\times n}$ are similar iff they have the same eigenvalues, and for each eigenvalue λ, $\text{rank}(A - \lambda I)^k = \text{rank}(B - \lambda I)^k$ for each $k \in \mathbb{N}$.
10. Show that two matrices $A, B \in \mathbb{C}^{n\times n}$ are similar iff they have the same eigenvalues, and for each eigenvalue λ, $\text{null}(A - \lambda I)^k = \text{null}(B - \lambda I)^k$ for each $k \in \mathbb{N}$.

6.5 Singular Value Decomposition

Given an $m \times n$ matrix A with complex entries, there are two hermitian matrices that can be constructed naturally from it, namely A^*A and AA^*. We wish to study the eigenvalues and eigenvectors of these matrices and their relations to certain parameters associated with A. We will see that these concerns yield a factorization of A.

All eigenvalues of the hermitian matrix $A^*A \in \mathbb{C}^{n\times n}$ are real. If $\lambda \in \mathbb{R}$ is such an eigenvalue with an associated eigenvector $v \in \mathbb{C}^{n\times 1}$, then $A^*Av = \lambda v$ implies that

$$\lambda \|v\|^2 = \lambda v^*v = v^*(\lambda v) = v^*A^*Av = (Av)^*(Av) = \|Av\|^2.$$

Since $\|v\| > 0$, we see that $\lambda \geq 0$. The eigenvalues of A^*A can thus be arranged in a decreasing list

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_r > 0 = \lambda_{r+1} = \cdots = \lambda_n$$

for some r with $0 \leq r \leq n$. Notice that $\lambda_1, \ldots, \lambda_r$ are all positive and the rest are all equal to 0. In the following we relate this r with $\text{rank}(A)$. Of course, we could have considered AA^* instead of A^*A.

Let $A \in \mathbb{C}^{m\times n}$. Let $\lambda_1 \geq \cdots \geq \lambda_n \geq 0$ be the n eigenvalues of A^*A. The non-negative square roots of these real numbers are called the **singular values** of A. Conventionally, we denote the singular values of A by s_i. The eigenvalues of A^*A are then denoted by $s_1^2 \geq \cdots \geq s_n^2 \geq 0$.

Theorem 6.13 *Let* $A \in \mathbb{C}^{m\times n}$. *Then* $\text{rank}(A) = \text{rank}(A^*A) = \text{rank}(AA^*) = \text{rank}(A^*) =$ *the number of positive singular values of* A.

Proof Let $v \in \mathbb{C}^{n\times 1}$. If $Av = 0$, then $A^*Av = 0$. Conversely, if $A^*Av = 0$, then $v^*A^*Av = 0$. It implies that $\|Av\|^2 = 0$, giving $Av = 0$. Therefore, $N(A^*A) = N(A)$. It follows that $\text{null}(A^*A) = \text{null}(A)$. By the rank nullity theorem, we conclude that $\text{rank}(A^*A) = \text{rank}(A)$.

Consider A^* instead of A to obtain $\text{rank}(AA^*) = \text{rank}((A^*)^*A^*) = \text{rank}(A^*)$.

Since $\text{rank}(A^t) = \text{rank}(A) = \text{rank}(\overline{A})$, $\text{rank}(A^*) = \text{rank}(A)$.

Next, let $s_1 \geq s_2 \geq \cdots \geq s_r > 0 = s_{r+1} = \cdots = s_n$ be the singular values of A. That is, there are exactly r number of positive singular values of A. By the spectral theorem, the hermitian matrix A^*A is unitarily diagonalizable. So, there exists a unitary matrix Q such that

$$Q^*(A^*A)Q = \text{diag}(s_1^2, \ldots, s_r^2, 0, \ldots, 0).$$

Therefore, $\text{rank}(A^*A)$ is equal to the rank of the above diagonal matrix; and that is equal to r. This completes the proof. ■

We remark that the number of nonzero eigenvalues of a matrix, counting multiplicities, need not be same as the rank of the matrix. For instance, the matrix

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

has no nonzero eigenvalues, but its rank is 1.

Suppose $\lambda > 0$ is an eigenvalue of A^*A with an associated eigenvector v. Then $A^*Av = \lambda v$ implies $(AA^*)(Av) = \lambda(Av)$. Since $\lambda v \neq 0$, $Av \neq 0$. Thus, λ is also an eigenvalue of AA^* with an associated eigenvector Av.

Similarly, if $\lambda > 0$ is an eigenvalue of AA^*, then it follows that the same λ is an eigenvalue of A^*A. That is,

a positive real number is an eigenvalue of A^*A iff it is an eigenvalue of AA^*.

From Theorem 6.13, it follows that A and A^* have the same r number of positive singular values, where $r = \text{rank}(A) = \text{rank}(A^*)$. Further, A has $n - r$ number of zero singular values, whereas A^* has $m - r$ number of zero singular values. In addition, if $A \in \mathbb{C}^{n\times n}$ is hermitian and has eigenvalues $\lambda_1, \ldots, \lambda_n$, then its singular values are $|\lambda_1|, \ldots, |\lambda_n|$.

Analogous to the factorization of A^*A, we have one for A itself.

Theorem 6.14 (SVD) *Let $A \in \mathbb{C}^{m\times n}$ be of rank r. Let $s_1 \geq \ldots \geq s_r$ be the positive singular values of A. Write $S = \text{diag}(s_1, \ldots, s_r) \in \mathbb{C}^{r\times r}$ and $\Sigma = \begin{bmatrix} S & 0 \\ 0 & 0 \end{bmatrix} \in \mathbb{C}^{m\times n}$. Then there exist unitary matrices $P = [u_1 \ \cdots \ u_m] \in \mathbb{C}^{m\times m}$ and $Q = [v_1 \ \cdots \ v_n] \in \mathbb{C}^{n\times n}$ such that the following are true:*

(1) $A = P\,\Sigma\,Q^* = s_1u_1v_1^* + \cdots + s_ru_rv_r^*$.

(2) (a) *For* $1 \leq i \leq r$, $AA^*u_i = s_i^2AA^*u_i$; *and for* $r < i \leq m$, $AA^*u_i = 0$.

(b) *For* $1 \leq j \leq r$, $A^*Av_j = s_j^2v_j$; *and for* $r < j \leq n$, $A^*Av_j = 0$.

(3) *For* $1 \le i \le r$, $u_i = (s_i)^{-1} A v_i$ *and* $v_i = (s_i)^{-1} A^* u_i$.
(4) (a) $\{u_1, \ldots, u_r\}$ *is an orthonormal basis of* $R(A)$.
(b) $\{v_1, \ldots, v_r\}$ *is an orthonormal basis of* $R(A^*)$.
(c) $\{v_{r+1}, \ldots, v_n\}$ *is an orthonormal basis of* $N(A)$.
(d) $\{u_{r+1}, \ldots, u_m\}$ *is an orthonormal basis of* $N(A^*)$.

Proof All positive singular values of A are $s_1 \ge \cdots \ge s_r$. Thus, the eigenvalues of A^*A are $s_1^2 \ge \cdots \ge s_r^2$, and 0 repeated $n - r$ times. Since $A^*A \in \mathbb{C}^{n\times n}$ is hermitian, there exists a unitary matrix $Q \in \mathbb{C}^{n\times n}$ such that

$$Q^* A^* A Q = \operatorname{diag}(s_1^2, \ldots, s_r^2,\ 0, \ldots, 0) = \begin{bmatrix} S^2 & 0 \\ 0 & 0 \end{bmatrix},$$

where the columns of Q are eigenvectors of A^*A. Define

$$D = \operatorname{diag}\big(s_1^{-1}, \ldots, s_r^{-1},\ 1, \ldots, 1\big) = \begin{bmatrix} S^{-1} & 0 \\ 0 & I_{n-r} \end{bmatrix}.$$

Then $D^* = D$, and

$$(AQD)^*(AQD) = D^*(Q^*A^*AQ)D = \begin{bmatrix} S^{-1} & 0 \\ 0 & I_{n-r} \end{bmatrix} \begin{bmatrix} S^2 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} S^{-1} & 0 \\ 0 & I_{n-r} \end{bmatrix} = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}.$$

Therefore, the first r columns, say, $u_1, \ldots, u_r$, of the $m \times n$ matrix AQD form an orthonormal set in $\mathbb{C}^{m\times 1}$; and the other columns of AQD are zero columns. Extend this orthonormal set $\{u_1, \ldots, u_r\}$ to an orthonormal basis $\{u_1, \ldots, u_r, u_{r+1}, \ldots, u_m\}$ for $\mathbb{C}^{m\times 1}$. Construct the matrices

$$P_1 = \begin{bmatrix} u_1 & \cdots & u_r \end{bmatrix}, \quad P_2 = \begin{bmatrix} u_{r+1} & \cdots & u_m \end{bmatrix}, \quad P = \begin{bmatrix} P_1 & P_2 \end{bmatrix}.$$

(1) We find that $P \in \mathbb{C}^{m\times m}$ is unitary, and $AQD = \begin{bmatrix} P_1 & 0 \end{bmatrix}$. We already have the unitary matrix $Q \in \mathbb{C}^{n\times n}$. Hence

$$A = (AQD)D^{-1}Q^* = \begin{bmatrix} P_1 & 0 \end{bmatrix} \begin{bmatrix} S & 0 \\ 0 & I_{n-r} \end{bmatrix} Q^* = \begin{bmatrix} P_1 & P_2 \end{bmatrix} \begin{bmatrix} S & 0 \\ 0 & 0 \end{bmatrix} Q^* = P\,\Sigma\,Q^*.$$

Also, we see that

$$P\,\Sigma\,Q = \begin{bmatrix} u_1 & \cdots & u_m \end{bmatrix} \begin{bmatrix} s_1 & & & & \\ & s_2 & & & \\ & & \ddots & & \\ & & & s_r & \\ & & & & 0 \end{bmatrix} \begin{bmatrix} v_1^* \\ \vdots \\ v_n^* \end{bmatrix} = s_1 u_1 v_1^* + \cdots + s_r u_r v_r^*.$$

(2) (a) $AA^*P = P\,\Sigma\,Q^*\,Q\,\Sigma^*\,P^*P = \Sigma\,\Sigma^*P$. The matrix $\Sigma\,\Sigma^*$ is a diagonal matrix with diagonal entries as $s_1^2, \ldots, s_r^2, 0, \ldots, 0$. Therefore, $AA^*u_i = s_i^2 u_i$ for $1 \le i \le r$; and $AA^*u_i = 0$ for $i > r$.

(b) As in (a), $Q^*A^*AQ = \text{diag}(s_1^2, \ldots, s_r^2,\ 0, \ldots, 0)$ proves analogous facts about the vectors v_j.

(3) Let $1 \le i \le r$. The vector u_i is the ith column of AQD. Thus

$$u_i = AQDe_i = AQ(s_i)^{-1}e_i = (s_i)^{-1}AQe_i = (s_i)^{-1}Av_i.$$

Using this and (2a), we obtain

$$(s_i)^{-1}A^*u_i = (s_i)^{-2}A^*Av_i = (s_i)^{-2}(s_i)^2 v_i = v_i.$$

(4) (a) For $1 \le i \le r$, $u_i = (s_i)^{-1}Av_i$ implies that $u_i \in R(A)$. The vectors $u_1, \ldots, u_r$ are orthonormal and $\dim(R(A)) = r$. Therefore, $\{u_1, \ldots, u_r\}$ is an orthonormal basis of $R(A)$.

(b) As in (a), $\{v_1, \ldots, v_r\}$ is an orthonormal basis of $R(A^*)$.

(c) Let $r < j \le n$. Now, $A^*Av_j = 0$ implies $v_j^*A^*Av_j = 0$. So, $\|Av_j\|^2 = 0$; or that $Av_j = 0$. Then $v_j \in N(A)$. But $\dim(N(A)) = n - r$. Therefore, the $n - r$ orthonormal vectors $v_{r+1}, \ldots, v_n$ form an orthonormal basis for $N(A)$.

(d) As in (c), $\{u_{r+1}, \ldots, u_m\}$ is an orthonormal basis for $N(A^*)$. ■

Theorem 6.14 (2) and (4) imply that the columns of P are eigenvectors of AA^*, and the columns of Q are eigenvectors of A^*A. Accordingly, the columns of P are called the *left singular vectors* of A; and the columns of Q are called the *right singular vectors* of A. Notice that computing both sets of left and right singular vectors independently will not serve the purpose since they may not satisfy the equations in Theorem 6.14 (3).

Example 6.8 To determine the SVD of the matrix $A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$, we compute

$$AA^* = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \quad A^*A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}.$$

Since AA^* is smaller, we compute its eigenvalues and eigenvectors. The eigenvalues of AA^* with multiplicities are $2,\ 2$; thus the eigenvalues of A^*A are $2,\ 2,\ 0,\ 0$. Choosing simpler eigenvectors of AA^*, we have

$$s_1^2 = 2,\ u_1 = e_1; \quad s_2^2 = 2,\ u_2 = e_2.$$

Here, u_1 and u_2 are the left singular vectors. The corresponding right singular vectors are:

$$v_1 = \frac{1}{s_1}A^*u_1 = \tfrac{1}{\sqrt{2}}\begin{bmatrix}1&0\\0&1\\1&0\\0&1\end{bmatrix}\begin{bmatrix}1&0\end{bmatrix} = \tfrac{1}{\sqrt{2}}\begin{bmatrix}1\\0\\1\\0\end{bmatrix}, \quad v_2 = \frac{1}{s_1}A^*u_2 = \tfrac{1}{\sqrt{2}}\begin{bmatrix}0\\1\\0\\1\end{bmatrix}.$$

These eigenvectors are associated with the eigenvalues 2 and 2 of A^*A. We need the eigenvectors v_3 and v_4 associated with the remaining eigenvalues 0 and 0, which should also form an orthonormal set along with v_1 and v_2. Thus, we solve $A^*Ax = 0$. With $x = [a, b, c, d]^t$, the equations are

$$a + c = 0 = b + d.$$

Two linearly independent solutions of these equations are obtained by setting $a = 1,\ b = 0,\ c = -1,\ d = 0$ and $a = 0,\ b = 1,\ c = 0,\ d = -1$. The corresponding vectors are $w_3 = [1, 0 - 1, 0]^t$ and $w_4 = [0, 1, 0, -1]^t$. We find that $\{v_1, v_2, w_3, w_4\}$ is an orthogonal set. We then orthonormalize w_3 and w_4 to obtain

$$v_3 = \frac{w_3}{\|w_3\|} = \tfrac{1}{\sqrt{2}}\begin{bmatrix}1\\0\\-1\\0\end{bmatrix}, \quad v_4 = \frac{w_4}{\|w_4\|} = \tfrac{1}{\sqrt{2}}\begin{bmatrix}0\\1\\0\\-1\end{bmatrix}.$$

We set $P = [u_1 \quad u_2]$, $\quad Q = [v_1 \quad v_2 \quad v_3 \quad v_4]$, and Σ as the 2×4 matrix with singular values $\sqrt{2}, \sqrt{2}$ on the 2×2 first block, and then other entries 0. We see that

$$A = \begin{bmatrix}1&0&1&0\\0&1&0&1\end{bmatrix} = \begin{bmatrix}1&0\\0&1\end{bmatrix}\begin{bmatrix}\sqrt{2}&0&0&0\\0&\sqrt{2}&0&0\end{bmatrix}\tfrac{1}{\sqrt{2}}\begin{bmatrix}1&0&1&0\\0&1&0&1\\1&0&-1&0\\0&1&0&-1\end{bmatrix} = P\,\Sigma\,Q^*. \ \square$$

In the product $P\,\Sigma\,Q^*$, there are possibly many zero rows that do not contribute to the end result. Thus some simplifications can be done in the SVD. Let $A \in \mathbb{C}^{m\times n}$ where $m \le n$. Suppose $A = P\,\Sigma\,Q^*$ is an SVD of A. Let the ith column of Q be denoted by $v_i \in \mathbb{C}^{n\times 1}$. Write

$$P_1 = P,\ Q_1 = \begin{bmatrix}v_1 & \cdots & v_m\end{bmatrix} \in \mathbb{C}^{m\times n},\ \Sigma_1 = \operatorname{diag}\big(s_1, \ldots, s_r,\ 0, \ldots, 0\big) \in \mathbb{C}^{m\times m}.$$

Notice that P_1 is unitary and the m columns of Q_1 are orthonormal. In block form, we have

$$Q = \begin{bmatrix}Q_1 & Q_3\end{bmatrix}, \quad \Sigma = \begin{bmatrix}\Sigma_1 & 0\end{bmatrix}, \quad Q_3 = \begin{bmatrix}v_{m+1} & \cdots & v_n\end{bmatrix}.$$

Then

$$A = P\,\Sigma\,Q^* = P_1\begin{bmatrix}\Sigma_1 & 0\end{bmatrix}\begin{bmatrix}Q_1^*\\Q_3^*\end{bmatrix} = P_1\Sigma_1 Q_1^* \quad \text{for } m \le n. \tag{6.1}$$

Similarly, when $m \ge n$, we may curtail P accordingly. That is, suppose the ith column of P is denoted by $u_i \in \mathbb{C}^{m\times 1}$. Write

$$P_2 = \begin{bmatrix} u_1 & \cdots & u_n \end{bmatrix} \in \mathbb{C}^{m\times n},\ \Sigma_2 = \operatorname{diag}(s_1, \ldots, s_r, 0, \ldots, 0) \in \mathbb{C}^{n\times n},\ Q_2 = Q.$$

Here, the n columns of P_2 are orthonormal, and Q_2 is unitary. We write

$$P = \begin{bmatrix} P_2 & P_3 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \Sigma_2 \\ 0 \end{bmatrix}, \quad P_3 = \begin{bmatrix} u_{n+1} & \cdots & u_m \end{bmatrix}.$$

Then

$$A = P\,\Sigma\,Q^* = \begin{bmatrix} P_2 & P_3 \end{bmatrix} \begin{bmatrix} \Sigma_2 \\ 0 \end{bmatrix} Q_2^* = P_2 \Sigma_2 Q_2^* \quad \text{for } m \ge n. \tag{6.2}$$

The two forms of SVD in (6.1)–(6.2), one for $m \le n$ and the other for $m \ge n$ are called the **thin SVD** of A. Of course, for $m = n$, both the thin SVDs coincide with the SVD. For a unified approach to the thin SVDs, take $k = \min\{m, n\}$. Then a matrix $A \in \mathbb{C}^{m\times n}$ of rank r can be written as the product

$$A = P\,\Sigma\,Q^*$$

where $P \in \mathbb{C}^{m\times k}$ and $Q \in \mathbb{C}^{n\times k}$ have orthonormal columns, and $\Sigma \in \mathbb{C}^{k\times k}$ is the diagonal matrix $\operatorname{diag}(s_1, \ldots, s_r, 0, \ldots, 0)$ with $s_1, \ldots, s_r$ being the positive singular values of A.

It is possible to simplify the thin SVDs further by deleting the zero rows. Observe that in the product P^*AQ, the first r columns of P and the first r columns of Q produce S; the other columns of P and of Q give the zero blocks. Thus taking

$$\tilde{P} = \begin{bmatrix} u_1 & \cdots & u_r \end{bmatrix}, \quad \tilde{Q} = \begin{bmatrix} v_1 & \cdots & v_r \end{bmatrix},$$

a simplified decomposition of A is given by

$$A = \tilde{P}\,S\,\tilde{Q}^*,$$

where $\tilde{P} \in \mathbb{C}^{m\times r}$ and $\tilde{Q} \in \mathbb{C}^{r\times n}$ are having orthonormal columns. Such a decomposition is called the **tight SVD** of the matrix A. In the tight SVD, Each of the matrices A, $\tilde{P}$, S and $\tilde{Q}^*$ is of rank r.

Write $B = \tilde{P}S$ and $C = S\tilde{Q}^*$ to obtain

$$A = B\,\tilde{Q}^* = \tilde{P}\,C,$$

where both $B \in \mathbb{C}^{m\times r}$ and $C \in \mathbb{C}^{r\times n}$ are of rank r. It shows that each $m \times n$ matrix of rank r can be written as a product of an $m \times r$ matrix of rank r and an $r \times n$ matrix, also of rank r. We recognize it as the *full rank factorization* of A.

Example 6.9 Obtain SVD, tight SVD, and a full rank factorization of

$$A = \begin{bmatrix} 2 & -1 \\ -2 & 1 \\ 4 & -2 \end{bmatrix}.$$

Here, $A^*A = \begin{bmatrix} 24 & -12 \\ -12 & 6 \end{bmatrix}$. It has eigenvalues 30 and 0. Thus $s_1 = \sqrt{30}$. Notice that AA^* is a 3×3 matrix with eigenvalues 30, 0 and 0. We see that $r = \text{rank}(A) =$ the number of positive singular values of $A = 1$.

For the eigenvalue 30, we solve the equation $A^*A[a, b]^t = 30[a, b]^t$, that is,

$$24a - 12b = 30a, \quad -12a + 6b = 30b.$$

It has the solution $a = -2,\ b = 1$. So, a unit eigenvector of A^*A corresponding to the eigenvalue 30 is $v_1 = \frac{1}{\sqrt{5}}[-2,\ 1]^t$.

For the eigenvalue 0, the equations are

$$24a - 12b = 0, \quad -12a + 6b = 0.$$

Thus a unit eigenvector orthogonal to v_1 is $v_2 = \frac{1}{\sqrt{5}}[1,\ 2]^t$. Then,

$$u_1 = \tfrac{1}{\sqrt{30}} A v_1 = \tfrac{1}{\sqrt{30}} \begin{bmatrix} 2 & -1 \\ -2 & 1 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} -2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix} = \tfrac{1}{\sqrt{6}} \begin{bmatrix} -1 \\ 1 \\ -2 \end{bmatrix}.$$

Notice that $\|u_1\| = 1$. We extend $\{u_1\}$ to an orthonormal basis of $\mathbb{C}^{3\times 1}$. It is

$$\left\{ u_1 = \tfrac{1}{\sqrt{6}} \begin{bmatrix} -1 \\ 1 \\ -2 \end{bmatrix}, \quad u_2 = \tfrac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad u_3 = \tfrac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}.$$

Next, we take u_1, u_2, u_3 as the columns of P and v_1, v_2 as the columns of Q to obtain the SVD of A as

$$\begin{bmatrix} 2 & -1 \\ -2 & 1 \\ 4 & -2 \end{bmatrix} = P \Sigma Q^* = \begin{bmatrix} -1/\sqrt{6} & 1/\sqrt{2} & 1/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{2} & -1/\sqrt{3} \\ -2/\sqrt{6} & 0 & 1/\sqrt{3} \end{bmatrix} \begin{bmatrix} \sqrt{30} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} -2/\sqrt{5} & 1/\sqrt{5} \\ 1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix}^*.$$

For the tight SVD, we construct $\tilde{P}$ with its r columns as the the first r columns of P, $\tilde{Q}$ with its r columns as the first r columns of Q, and S as the $r \times r$ block consisting of first r singular values of A as the diagonal entries. With $r = 1$, we thus have the tight SVD as

$$\begin{bmatrix} 2 & -1 \\ -2 & 1 \\ 4 & -2 \end{bmatrix} = \tilde{P}S\tilde{Q}^* = \begin{bmatrix} -1/\sqrt{6} \\ 1/\sqrt{6} \\ -2/\sqrt{6} \end{bmatrix} \begin{bmatrix} \sqrt{30} \end{bmatrix} \begin{bmatrix} -2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix}^*.$$

In the tight SVD, using associativity of matrix product, we get the full rank factorizations as

$$\begin{bmatrix} 2 & -1 \\ -2 & 1 \\ 4 & -2 \end{bmatrix} = \begin{bmatrix} -\sqrt{5} \\ \sqrt{5} \\ -2\sqrt{5} \end{bmatrix} \begin{bmatrix} -2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix}^* = \begin{bmatrix} -1/\sqrt{6} \\ 1/\sqrt{6} \\ -2/\sqrt{6} \end{bmatrix} \begin{bmatrix} -2/\sqrt{6} \\ \sqrt{6} \end{bmatrix}^*.$$

It may be checked that the columns of Q are eigenvectors of AA^*. □

A singular value decomposition of a matrix is not unique. For, orthonormal bases that comprise the columns of P and Q can always be chosen differently. For instance, by multiplying ± 1 to an already constructed orthonormal basis, we may obtain another.

Also, it is easy to see that when $A \in \mathbb{R}^{m\times n}$, the matrices P and Q can be chosen to have real entries.

Singular value decomposition is the most important result for scientists and engineers, perhaps, next to the theory of linear equations. It shows clearly the power of eigenvalues and eigenvectors. The SVD in the summation form, in Theorem 6.14 (1), looks like

$$A = s_1 u_1 v_1^* + \cdots + s_r u_r v_r^*.$$

Each matrix $u_i v_i^*$ is of rank 1. This means that if we know the first r singular values of A and we know their corresponding left and right singular vectors, we know A completely. This is particularly useful when A is a very large matrix of low rank. No wonder, SVD is used in image processing, various compression algorithms, and in principal components analysis.

Let $A = P\,\Sigma\,Q^*$ be an SVD of $A \in \mathbb{C}^{m\times n}$, and let $x \in \mathbb{C}^{n\times 1}$ be a unit vector. Let $s_1 \geq \cdots \geq s_r$ be the positive singular values of A. Write $y = Q^*x = [\alpha_1, \ldots, \alpha_n]^t$. Then

$$\begin{aligned} \|y\|^2 &= \sum_{i=1}^{n} |\alpha_i|^2 = \|Q^*x\|^2 = x^*QQ^*x = x^*x = \|x\|^2 = 1. \\ \|Ax\|^2 &= \|P\,\Sigma\,Q^*x\|^2 = x^*Q\,\Sigma^*P^*P\,\Sigma\,Q^*x = x^*Q\,\Sigma^2Q^*x \\ &= y^*\,\Sigma^2 y = \sum_{j=1}^{r} s_j^2|\alpha_j|^2 \leq s_1^2 \sum_{j=1}^{r} |\alpha_j|^2 \leq s_1^2. \end{aligned}$$

Also, for $x = v_1$, $\|Av_1\|^2 = s_1^2\|u_1\|^2 = s_1^2$. Therefore, we conclude that

$$s_1 = \max\{\|Ax\| : x \in \mathbb{C}^{n\times 1},\ \|x\| = 1\}.$$

That is, the first singular value s_1 gives the maximum magnification that a vector experiences under the linear transformation A. Similarly, from above workout it follows that

$$\|Ax\|^2 = \sum_{j=1}^{r} s_j^2 |\alpha_j|^2 \geq s_r^2 \sum_{j=1}^{r} |\alpha_j|^2 \geq s_r^2.$$

With $x = v_r$, we have $\|Av_r\|^2 = s_r^2 \|u_r\|^2 = s_r^2$. Hence,

$$s_r = \min\{\|Ax\| : x \in \mathbb{C}^{n\times 1},\ \|x\| = 1\}.$$

That is, the minimum positive magnification is given by the smallest positive singular value s_r.

Notice that if x is a unit vector, then $s_r \leq \|Ax\| \leq s_1$.

Exercises for Sect. 6.5

1. Let $A \in \mathbb{C}^{m\times n}$. Show that the positive singular values of A are also the positive singular values of A^*.
2. Compute the singular value decompositions of the following matrices:

   (a) $\begin{bmatrix} 2 & -2 \\ 1 & -1 \\ -1 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 2 & 1 & 2 \\ -2 & 1 & 2 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 2 & 2 \\ 2 & 0 & -5 \\ 3 & 0 & 0 \end{bmatrix}$
3. Show that the matrices $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $\begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}$ are similar but they have different singular values.
4. Prove that if $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of an $n \times n$ hermitian matrix, then its singular values are $|\lambda_1|, \ldots, |\lambda_n|$.
5. Let $A \in \mathbb{C}^{n\times n}$ and let $\lambda \in \mathbb{C}$. Show that $A - \lambda I$ is invertible iff $A^* - \overline{\lambda} I$ is invertible.
6. Let $A \in \mathbb{F}^{n\times n}$. If A has eigenvalues $\lambda_1, \ldots, \lambda_n$ and singular values $s_1, \ldots, s_n$, then show that $|\lambda_1 \cdots \lambda_n| = s_1 \cdots s_n$.

6.6 Polar Decomposition

Square matrices behave like complex numbers in many ways. One such example is a powerful representation of square matrices using a stretch and a rotation. This mimics the polar representation of a complex number as $z = re^{i\theta}$, where r is a non-negative real number representing the stretch; and $e^{i\theta}$ is a rotation. Similarly, a square matrix can be written as a product of a positive semi-definite matrix, representing the stretch, and a unitary matrix representing the rotation. We slightly generalize it to any $m \times n$ matrix.

A hermitian matrix $P \in \mathbb{F}^{n\times n}$ is called **positive semidefinite** iff $x^*Px \geq 0$ for each $x \in \mathbb{F}^{n\times 1}$. We use such a matrix in the following matrix factorization.

Theorem 6.15 (Polar decomposition) *Let $A \in \mathbb{C}^{m\times n}$. Then there exist positive semi-definite matrices $P \in \mathbb{C}^{m\times m}$, $Q \in \mathbb{C}^{n\times n}$, and a matrix $U \in \mathbb{C}^{m\times n}$ such that*

$$A = PU = UQ,$$

where $P^2 = AA^*$, $Q^2 = A^*A$, and U satisfies the following:

(1) If $m = n$, then the $n \times n$ matrix U is unitary.
(2) If $m < n$, then the rows of U are orthonormal.
(3) If $m > n$, then the columns of U are orthonormal.

Proof Let $A \in \mathbb{C}^{m\times n}$ be a matrix of rank r with positive singular values $s_1 \geq \cdots \geq s_r$. Write $k = \min\{m, n\}$. Let $A = B\,\Sigma\,E^*$ be the thin SVD of A, where $B \in \mathbb{C}^{m\times k}$, $E \in \mathbb{C}^{n\times k}$ have orthonormal columns, and $\Sigma \in \mathbb{C}^{k\times k}$ has first r diagonal entries as $s_1, \ldots, s_r$, all other entries 0. Since $B^*B = E^*E = I$, we have

$$A = B\,\Sigma\,E^* = (B\,\Sigma\,B^*)(BE^*) = (BE^*)(E\,\Sigma\,E^*).$$

With $U = BE^*$, $P = B\,\Sigma\,B^*$ and $Q = E\,\Sigma\,E^*$, we obtain the polar decompositions $A = PU = UQ$. Moreover,

$$\begin{aligned} P^2 &= B\,\Sigma\,B^*B\,\Sigma\,B^* = B\,\Sigma\,E^*E\,\Sigma\,B^* = AA^*, \\ Q^2 &= E\,\Sigma\,E^*E\,\Sigma\,E^* = E\,\Sigma\,B^*B\,\Sigma\,E^* = A^*A. \end{aligned}$$

Notice that $\Sigma^* = \Sigma$. Thus $P^* = (B\,\Sigma\,B^*)^* = B\,\Sigma^*B^* = P$. That is, P is hermitian. Write $D = \mathrm{diag}(\sqrt{s_1}, \ldots, \sqrt{s_r}, 0, \ldots, 0) \in \mathbb{C}^{k\times k}$. Then $D^*D = \Sigma$. For each $x \in \mathbb{C}^{m\times 1}$,

$$x^*Px = x^*B\,\Sigma\,B^*x = x^*BD^*DB^*x = \|DB^*x\|^2 \geq 0.$$

Therefore, P is positive semi-definite. Similarly, Q is shown to be positive semi-definite.

(1) If $m = n$, then both B and E are unitary, i.e., $B^*B = BB^* = I = E^*E = EE^*$. Then $U^*U = (BE^*)^*BE^* = EB^*BE^* = EE^* = I$, and $UU^* = BE^*(BE^*)^* = BE^*EB^* = BB^* = I$. So, U is unitary.

(2) If $m < n$, then $k = m$; B is a square matrix with orthonormal columns, thus unitary; and $E^*E = I$. We have $UU^* = BE^*EB^* = BB^* = I$. Thus U has orthonormal rows.

(3) If $m > n$, then $k = n$. We see that E is unitary, and $B^*B = I$. Therefore, $U^*U = EB^*BE^* = EE^* = I$. That is, U has orthonormal columns. ■

Recall that a thin SVD is obtained from an SVD of $A = P\,\Sigma\,Q^*$ by keeping the first k columns of the larger of the two matrices P and Q, and then restricting Σ to

its first $k \times k$ block, where $k = \min\{m, n\}$. Thus, the polar decomposition of A may be constructed directly from the SVD. It is as follows:

If $A \in \mathbb{C}^{m\times n}$ has SVD as $A = BDE^*$, then $A = PU = UQ$, where

$$\begin{array}{lll} m = n: & U = BE^*, & P = BDB^*, \quad Q = EDE^*. \\ m < n: & U = BE_1^*, & P = BD_1B^*, \quad Q = E_1D_1E_1^*. \\ m > n: & U = B_1E^*, & P = B_1D_2B_1^*, \quad Q = ED_2E^*. \end{array}$$

Here, E_1 is constructed from E by taking its first m columns; D_1 is constructed from D by taking its first m columns; B_1 is constructed from B by taking its first n columns; and D_2 is constructed from D by taking its first n rows.

Example 6.10 Consider the matrix $A = \begin{bmatrix} 2 & -1 \\ -2 & 1 \\ 4 & -2 \end{bmatrix}$ of Example 6.9. We had obtained its SVD as $A = BDE^*$, where

$$B = \begin{bmatrix} -1/\sqrt{6} & 1/\sqrt{2} & 1/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{2} & -1/\sqrt{3} \\ -2/\sqrt{6} & 0 & 1/\sqrt{3} \end{bmatrix}, \quad D = \begin{bmatrix} \sqrt{30} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad E = \begin{bmatrix} -2/\sqrt{5} & 1/\sqrt{5} \\ 1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix}.$$

Here, $A \in \mathbb{C}^{3\times 2}$. Thus, Theorem 6.15 (3) is applicable; see the discussion following the proof of the theorem. We construct the matrices B_1 by taking first two columns of B, and D_2 by taking first two rows of D, as in the following:

$$B_1 = \begin{bmatrix} -1/\sqrt{6} & 1/\sqrt{2} \\ 1/\sqrt{6} & 1/\sqrt{2} \\ -2/\sqrt{6} & 0 \end{bmatrix}, \quad D_2 = \begin{bmatrix} \sqrt{30} & 0 \\ 0 & 0 \end{bmatrix}.$$

Then

$$U = B_1E^* = \tfrac{1}{\sqrt{6}}\begin{bmatrix} -1 & \sqrt{3} \\ 1 & \sqrt{3} \\ -2 & 0 \end{bmatrix} \tfrac{1}{\sqrt{5}}\begin{bmatrix} -2 & 1 \\ 1 & 2 \end{bmatrix} = \tfrac{1}{\sqrt{30}}\begin{bmatrix} 2+\sqrt{3} & -1+2\sqrt{3} \\ -2+\sqrt{3} & 1+2\sqrt{3} \\ 4 & -2 \end{bmatrix},$$

$$P = B_1D_2B_1^* = \sqrt{5}\begin{bmatrix} -1 & 0 \\ 1 & 0 \\ -2 & 0 \end{bmatrix} \tfrac{1}{\sqrt{6}}\begin{bmatrix} -1 & 1 & -2 \\ \sqrt{3} & \sqrt{3} & 0 \end{bmatrix} = \tfrac{\sqrt{5}}{\sqrt{6}}\begin{bmatrix} 1 & -1 & 2 \\ -1 & 1 & -2 \\ 2 & -2 & 4 \end{bmatrix},$$

$$Q = ED_2E^* = \sqrt{6}\begin{bmatrix} -2 & 0 \\ 1 & 0 \end{bmatrix} \tfrac{1}{\sqrt{5}}\begin{bmatrix} -2 & 1 \\ 1 & 2 \end{bmatrix} = \tfrac{\sqrt{6}}{\sqrt{5}}\begin{bmatrix} 4 & -2 \\ -2 & 1 \end{bmatrix}.$$

As expected we find that

$$PU = \tfrac{\sqrt{5}}{\sqrt{6}}\begin{bmatrix} 1 & -1 & 2 \\ -1 & 1 & -2 \\ 2 & -2 & 4 \end{bmatrix} \tfrac{1}{\sqrt{30}}\begin{bmatrix} 2+\sqrt{3} & -1+2\sqrt{3} \\ -2+\sqrt{3} & 1+2\sqrt{3} \\ 4 & -2 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -2 & 1 \\ 4 & -2 \end{bmatrix} = A.$$

$$UQ = \tfrac{1}{\sqrt{30}}\begin{bmatrix} 2+\sqrt{3} & -1+2\sqrt{3} \\ -2+\sqrt{3} & 1+2\sqrt{3} \\ 4 & -2 \end{bmatrix} \tfrac{\sqrt{6}}{\sqrt{5}}\begin{bmatrix} 4 & -2 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -2 & 1 \\ 4 & -2 \end{bmatrix} = A. \quad \square$$

In $A = PU = UQ$, the matrices P and Q satisfy $P^2 = AA^*$ and $Q^2 = A^*A$. If $A \in \mathbb{C}^{m\times n}$, then $AA^* \in \mathbb{C}^{m\times m}$ and $A^*A \in \mathbb{C}^{n\times n}$ are hermitian matrices with eigenvalues as $s_1^2, \ldots, s_r^2, 0, \ldots, 0$. The diagonalization of AA^* yields m linearly independent eigenvectors associated with these eigenvalues. If these eigenvectors are taken as the columns (in that order) of a matrix C, then

$$AA^* = C\,\mathrm{diag}(s_1^2, \ldots, s_r^2, 0, \ldots, 0)\,C^*, \quad P = C\,\mathrm{diag}(s_1, \ldots, s_r, 0, \ldots, 0)\,C^*.$$

Similarly, the matrix Q is equal to $M\,\mathrm{diag}(s_1, \ldots, s_r, 0, \ldots, 0)\,M^*$, where M consists of orthonormal eigenvectors of A^*A corresponding to the eigenvalues $s_1^2, \ldots, s_r^2, 0, \ldots, 0$.

Now, using the matrix P as computed above, we can determine U so that $A = PU$. Similarly, the matrix U in $A = UQ$ can also be determined. In this approach, SVD is not used; however, the two instances of U may differ since they depend on the choices of orthonormal eigenvectors of AA^* and A^*A. If A is invertible, you would end up with the same U.

Exercises for Sect. 6.6

1. Let A be an upper triangular matrix with distinct diagonal entries. Show that there exists an upper triangular matrix that diagonalizes A.
2. Let $A = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$. Determine whether A and A^2 are positive definite.
3. Determine the polar decompositions of the matrix A of Example 6.10 by diagonalizing AA^* and A^*A as mentioned in the text.
4. Let $A \in \mathbb{C}^{m\times n}$ with $m < n$. Prove that A can be written as $A = PU$, where $P \in \mathbb{C}^{m\times n}$, and $U \in \mathbb{C}^{n\times n}$ is unitary.
5. Let $A \in \mathbb{C}^{m\times n}$ with $m > n$. Prove that A can be written as $A = UQ$, where $U \in \mathbb{C}^{m\times m}$ is a unitary matrix and $Q \in \mathbb{C}^{m\times n}$.

6.7 Problems

1. Let P^*AP be the upper triangular matrix in Schur triangularization of $A \in \mathbb{F}^{n\times n}$. If A has n distinct eigenvalues, then show that there exists an upper triangular matrix Q such that PQ diagonalizes A.
2. If $A \in \mathbb{F}^{n\times n}$ is both diagonalizable and invertible, then how do you compute its inverse from its diagonalization?
3. Let $A \in \mathbb{F}^{n\times n}$ be diagonalizable with eigenvalues ± 1. Is $A^{-1} = A$?
4. How to diagonalize A^* if diagonalization of $A \in \mathbb{F}^{n\times n}$ is known?
5. Let $A \in \mathbb{R}^{n\times n}$ have real eigenvalues $\lambda_1 > \lambda_2 > \cdots > \lambda_n$ with corresponding eigenvectors $x_1, \ldots, x_n$. Let $x = \sum_{i=1}^n \alpha_i x_i$ for real numbers $\alpha_1, \ldots, \alpha_n$. Show the following:

   (a) $A^m x = \sum_{i=1}^n \alpha_i \lambda_i^m x_i$ (b) If $\lambda_1 = 1$, then $\lim_{m\to\infty} A^m x = \alpha_1 x_1$.

6. Prove that if a normal matrix has only real eigenvalues, then it is hermitian. Conclude that if a real normal matrix has only real eigenvalues, then it is real symmetric.
7. Let $A \in \mathbb{F}^{n\times n}$ have distinct eigenvalues $\lambda_1, \ldots, \lambda_k$. For $1 \le j \le k$, let the linearly independent eigenvectors associated with λ_j be $v_j^1, \ldots, v_j^{i_j}$. Prove that the set $\{v_1^1, \ldots, v_1^{i_1}, \ldots, v_k^1, \ldots, v_k^{i_k}\}$ is linearly independent. [Hint: See the proof of Theorem 5.3.]
8. Let $A = [a_{ij}] \in \mathbb{F}^{3\times 3}$ be such that $a_{31} = 0$ and $a_{32} \neq 0$. Show that each eigenvalue of A has geometric multiplicity 1.
9. Suppose $A \in \mathbb{F}^{4\times 4}$ has an eigenvalue λ with algebraic multiplicity 3, and $\operatorname{rank}(A - \lambda I) = 1$. Is A diagonalizable?
10. Show that there exists only one $n \times n$ diagonalizable matrix with an eigenvalue λ of algebraic multiplicity n.
11. Show that a nonzero nilpotent matrix is never diagonalizable.
[Hint: $A \neq 0$ but $A^m = 0$ for some $m \ge 2$.]
12. Let $A \in \mathbb{F}^{n\times n}$ and let $P^{-1}AP$ be a diagonal matrix. Show that the columns of P that are eigenvectors associated with nonzero eigenvalues of A form a basis for $R(A)$.
13. Let A be a diagonalizable matrix. Show that the number of nonzero eigenvalues of A is equal to $\operatorname{rank}(A)$.
14. Construct non-diagonalizable matrices A and B satisfying

    (a) $\operatorname{rank}(A)$ is equal to the number of nonzero eigenvalues of A;
    (b) $\operatorname{rank}(B)$ is not equal to the number of nonzero eigenvalues of B.

15. Let $x, y \in \mathbb{F}^{n\times 1}$ and let $A = xy^*$. Show the following:

    (a) A has an eigenvalue y^*x with an associated eigenvector x.
    (b) 0 is an eigenvalue of A with geometric multiplicity at least $n - 1$.
    (c) If $y^*x \neq 0$, then A is diagonalizable.

16. Using the Jordan form of a matrix show that a matrix A is diagonalizable iff for each eigenvalue of A, its geometric multiplicity is equal to its algebraic multiplicity.
17. Let $A \in \mathbb{C}^{n\times n}$ have an eigenvalue λ with algebraic multiplicity m. Prove that $\operatorname{null}((A - \lambda I)^m) = m$.
18. Let λ be an eigenvalue of a matrix $A \in \mathbb{C}^{n\times n}$ having algebraic multiplicity m. Prove that for each $k \in \mathbb{N}$, if $\operatorname{null}((A - \lambda I)^k) < m$, then $\operatorname{null}((A - \lambda I))^k < \operatorname{null}((A - \lambda I)^{k+1}$.
[Hint: Show that $N(A - \lambda I)^i \subseteq N(A - \lambda I)^{i+1}$. Then use Exercise 17.]
19. Let λ be an eigenvalue of a matrix A and let J be the Jordan form of A. Prove that the number of Jordan blocks with diagonal entry λ in J is the geometric multiplicity of λ.
20. Let A be a hermitian $n \times n$ matrix with eigenvalues $\lambda_1, \ldots, \lambda_n$. Show that there exist an orthonormal set $\{x_1, \ldots, x_n\}$ in $\mathbb{F}^{n\times 1}$ such that $x^*Ax = \sum_{i=1}^n \lambda_i |x^*x_i|^2$ for each $x \in \mathbb{F}^{n\times 1}$.

21. Let $A \in \mathbb{R}^{n\times n}$. Show that AA^t and A^tA are similar matrices.
22. Let A and B be hermitian matrices of the same order. Are the following statements true?

    (a) All eigenvalues of AB are real.
    (b) All eigenvalues of ABA are real.

23. Let $n > 1$ and let $u \in \mathbb{F}^{n\times 1}$ be a unit vector. Let $H = I - 2uu^*$. Show the following:

    (a) H is both hermitian and unitary; thus $H^{-1} = H$.
    (b) If λ and μ are two distinct eigenvalues of H, then $|\lambda - \mu| = 2$.
    (c) $Hu = -u$.
    (d) The trace of H is $n - 2$.
    (e) If $v \neq 0$ and $v \perp u$, then $Hv = v$.
    (f) The eigenvalue 1 has algebraic multiplicity $n - 1$.

24. Let $A \in \mathbb{C}^{m\times n}$ be a matrix of rank r. Let $s_1 \geq \cdots \geq s_r$ be the positive singular values of A. Let $A = P\,\Sigma_1 Q^*$ be a singular value decomposition of A, with $S = \operatorname{diag}(s_1, \ldots, s_r)$, and $\Sigma_1 = \begin{bmatrix} S & 0 \\ 0 & 0 \end{bmatrix} \in \mathbb{C}^{m\times n}$. Define matrices $\Sigma_2 = \begin{bmatrix} S^{-1} & 0 \\ 0 & 0 \end{bmatrix} \in \mathbb{C}^{n\times m}$ and $A^\dagger = Q\,\Sigma_2 P^*$. Prove the following:

    (a) The matrix $A^\dagger$ satisfies the following properties:

    $$(AA^\dagger)^* = AA^\dagger, \quad (A^\dagger A)^* = A^\dagger A, \quad AA^\dagger A = A, \quad A^\dagger AA^\dagger = A^\dagger.$$

    (b) There exists a unique matrix $A^\dagger \in \mathbb{F}^{n\times m}$ satisfying the four equations mentioned in (a). The matrix $A^\dagger$ is called the *generalized inverse* of A.
    (c) For any $b \in \mathbb{F}^{m\times 1}$, $A^\dagger b$ is the least squares solution of $Ax = b$.

25. Show that if s is a singular value of a matrix A, then there exists a nonzero vector x such that $\|Ax\| = s\|x\|$.
26. Let $A = P\Sigma Q^t$ be the SVD of a real $n \times n$ matrix. Let u_i be the ith column of P, and let v_i be the ith column of Q. Define the matrix B and the vectors x_i, y_i as follows:

    $$B = \begin{bmatrix} 0 & A^t \\ A & 0 \end{bmatrix}, \quad x_i = \begin{bmatrix} v_i \\ u_i \end{bmatrix}, \quad y_i = \begin{bmatrix} -v_i \\ u_i \end{bmatrix} \quad \text{for } 1 \leq i \leq n.$$

    Show that x_i and y_i are eigenvectors of B. How are the eigenvalues of B related to the singular values of A?
27. Derive the polar decomposition from the SVD. Also, derive singular value decomposition from the polar decomposition.
28. A *positive definite* matrix is a hermitian matrix such that for each $x \neq 0$, $x^*Ax > 0$. Show that a hermitian matrix is positive definite iff all its eigenvalues are positive.

29. Show that the square of a real symmetric invertible matrix is positive definite.
30. Show that if A is positive definite, then so is A^{-1}. Give an example of a 2×2 invertible matrix which is not positive definite.
31. Show that A^*A is positive semi-definite for any $A \in \mathbb{F}^{m\times n}$. Give an example of a matrix A where A^*A is not positive definite.
32. Show that if Q is unitary and A is positive definite, then QAQ^* is positive definite.
33. For a matrix $A \in \mathbb{F}^{n\times n}$, the *principal submatrices* are obtained by deleting its last r rows and last r columns for $r = 0, 1, \ldots, n-1$. Show that all principal submatrices of a positive definite matrix are positive definite. Further, verify that all principal submatrices of $A = \begin{bmatrix} 1 & 1 & -3 \\ 1 & 1 & -3 \\ -3 & -3 & 5 \end{bmatrix}$ have non-negative determinants but A is not positive semi-definite.
34. Let A be a real symmetric matrix. Show that the following are equivalent:

    (a) A is positive definite.
    (b) All principal submatrices of A have positive determinant.
    (c) A can be reduced to an upper triangular form using only elementary row operations of Type3, where all pivots are positive.
    (d) $A = U^tU$, where U is upper triangular with positive diagonal entries.
    (e) $A = B^tB$ for some invertible matrix B.

35. Let A be a real symmetric positive definite $n \times n$ matrix. Show the following:

    (a) All diagonal entries of A are positive.
    (b) For any invertible $n \times n$ matrix P, P^tAP is positive definite.
    (c) There exists an $n \times n$ orthogonal matrix Q such that $A = Q^tQ$.
    (d) There exist unique $n \times n$ matrices U and D where U is upper triangular with all diagonal entries 1, and D is a diagonal matrix with positive entries on the diagonal such that $A = U^tDU$.
    (e) *Cholesky factorization*: There exists a unique upper triangular matrix with positive diagonal entries such that $A = U^tU$.